

EXPLORING THE DESIGN OPPORTUNITIES FOR MEDIATED SPATIAL INTERACTIONS LEARNING FROM THE KITCHEN

by

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* This research will be presented as a Collection of Papers. The format requires combining several published conference and journal papers under a research question and hypothesis. The dissertation will be formed by presenting published work with an extensive introduction and conclusion.

To
Uzay,
and my family
whose love and support made this work possible.

Abstract

Exploring The Design Opportunities For Mediated Spatial Interactions Learning From the Kitchen

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Kitchen spaces occupy a unique and contested position in HCI research, simultaneously conceptualized as rational task environments and as emotionally charged social hubs. The dominant smart-kitchen framing prioritizes efficiency and functional performance, overlooking the kitchen's social dimensions and role in interaction, memory, and bonding. This dissertation conceptualizes the domestic kitchen as a hybrid environment supporting both productive doing and meaningful relating, and asks how Extended Reality (XR) can support mediated spatial interaction in domestic kitchens (RQ0). Two questions follow: in what ways XR can address the interaction challenges that emerge in domestic kitchen environments, particularly during hybrid or remote engagements (RQ1), and how designers can employ XR for envisioning, prototyping, and evaluating domestic experiences in kitchen contexts (RQ2). The work is grounded in a Design-oriented Research paradigm and shaped by a Programmatic Design Research framework, unfolding as a research program of four interrelated design experiments conducted with kitchen inhabitants and design professionals. Four artifacts anchor the program as epistemic instruments generating intermediate-level knowledge: KitchXR, Portal, MagMemo, and Ambient Cooking. Addressing RQ1, the dissertation identifies six recurring interactional frictions. From these it articulates four design suggestions, including tunable presence, arguing that presence be regulated along axes of space, time, and identity rather than as binary. Addressing RQ2, it identifies two constraints on how designers prototype and study domestic interaction, and positions XR as an intermediate living testbed, reflecting on where this stance held and where it strained. It reframes the kitchen as a spatial connection hub and cluster of active workzones. The frictions are grounded in inhabitant fieldwork; the later studies elaborate them with design practitioners.

Keywords: Human-computer interaction; Mediated Spatial Interaction; Extended reality; Smart kitchen;

Özetçe

Aracılı Mekânsal Etkileşimlere Yönelik Tasarım Olanaklarının

Mutfaktan Öğrenerek Keşfedilmesi

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Mekan olarak Mutfak, İnsan-Bilgisayar Etkileşimi (HCI) araştırmalarında hem rasyonel bir vaka alanı, hem de kullanıcıları bir araya getiren sosyal bir merkez biçiminde kavramsallaştırılabilindiği için kendine özgü ve çalışmaya açık bir konumda bulunmaktadır. Günümüzün yaygın Akıllı Mutfak yaklaşımı verimliliği ve işlevsel performansı önceliklendirmekte, mutfağın sosyal boyutlarını ve etkileşim, hafıza ve bağ kurmadaki rolünü göz ardı edebilmektedir. Bu tez, ev mutfağını üretken eylemi ve anlamlı ilişkilenebilirliği birlikte destekleyen hibrit bir ortam olarak kavramsallaştırmakta ve Genişletilmiş Gerçeklik (XR) ev mutfaklarında teknoloji aracılı mekânsal etkileşimi nasıl destekleyebileceğini incelemektedir (RQ0). Bunu iki alt soru ile incelemekte: XR, özellikle hibrit ya da uzaktan etkileşimler sırasında ev mutfağı ortamlarında ortaya çıkan etkileşim sorunlarını hangi biçimlerde ele alabilir?(RQ1) ve tasarımcılar, mutfak bağlamında evsel deneyimleri öngörme, prototipleme ve değerlendirme amacıyla XR'ı nasıl kullanabilir?(RQ2) Çalışma, Tasarım Odaklı Araştırma paradigmasına dayanmakta ve Programatik Tasarım Araştırması çerçevesiyle biçimlenmekte; ev sakinleri ve tasarım profesyonelleriyle yürütülen, birbiriyle ilişkili dört tasarım deneyinden oluşan bir araştırma programı olarak ilerlemektedir. Ara düzey bilgi üreten epistemik araçlar olarak dört artefakt bu programı temellendirilmektedir: “KitchXR”, “Portal”, “MagMemo” ve “Ambient Cooking”. Tez, RQ1 kapsamında, altı yinelenen etkileşimsel kullanım gerginliği tanımlamaktadır. Bunlardan hareketle, var olma hali(presence) ikili bir durum olarak değil mekân, zaman ve kimlik eksenlerinde düzenlenmesini savunan “ayarlanabilir var olma hali” başta olmak üzere dört tasarım önerisi geliştirmektedir. RQ2 kapsamında, tasarımcıların evsel etkileşimi prototipleme ve inceleme biçimlerine ilişkin iki kısıtlama belirlenmiş ve XR'ı ara düzey yaşayan bir test ortamı olarak konumlandırmakta; bu duruşun nerede işlediğini ve nerede zorlandığını değerlendirmektedir. Mutfağı uzamsal bir bağlantı merkezi ve etkin çalışma bölgeleri kümesi olarak yeniden çerçevelemektedir. Kullanım gerginlikleri, ev sakinleriyle yapılan saha çalışmasına dayanmakta, sonraki çalışmalar bunları tasarımcılarla geliştirmektedir.

Anahtar Kelimeler: İnsan-bilgisayar etkileşimi; Aracılı uzamsal etkileşim; Genişletilmiş gerçeklik; Akıllı mutfak

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Abbreviations

XR Extended Reality

VR Virtual Reality

AV Augmented Virtuality

AR Augmented Reality

MR Mixed Reality

HCI Human–Computer Interaction

AI Artificial Intelligence

IoT Internet of Things

UbiComp Ubiquitous Computing

CFKM Continuous Fuzzy Kano Model

Glossary

Ambient Computing An experiential orientation of ubiquitous computing in which computational systems operate in a context-aware, adaptive, and unobtrusive manner, shaping interaction through peripheral awareness rather than explicit user engagement.

Artificial Intelligence (AI) Computational systems and techniques that enable machines to perform tasks typically associated with human intelligence, such as perception, learning, decision-making, and adaptation.

Augmented Reality (AR) A form of extended reality in which digital content is overlaid onto the physical environment, enhancing real-world perception while preserving awareness of the surrounding physical space.

Augmented Virtuality (AV) A configuration along the reality–virtuality continuum in which elements of the physical world are integrated into a predominantly virtual environment.

Awareness The state of knowing about others’ presence, states, or activities within a shared environment.

Cohesion The group-level bonding that develops among participants over time.

Coordination The alignment of joint action among participants toward a shared goal.

Embodied Presence The sense that one’s body, or its mediated representation, is the locus from which the environment is perceived and acted upon [1].

Embodiment A sense of physical presence within an extended reality environment.

Extended Reality (XR) An umbrella term covering virtual, augmented, and mixed reality technologies, referring to systems that extend, modify, or blend physical and digital experiences.

Human–Computer Interaction (HCI) An interdisciplinary research field concerned with the design, evaluation, and use of interactive computing systems, focusing on the relationships between people, technologies, and contexts of use.

Internet of Things (IoT) A network of interconnected physical devices embedded with sensors, software, and connectivity, enabling data exchange, automation, and coordination across environments.

Mixed Reality (MR) A category of extended reality that blends physical and digital elements into a shared interactive space, allowing virtual content to coexist with and respond to the physical environment.

Phygital A hybrid of physical and digital interactions.

Presence The sense of “being there” in, or “being together” through, a mediated environment. [2]

Smart Home A domestic environment equipped with networked digital devices and automated systems designed to support household tasks, energy management, safety, and comfort.

Social Presence The sense of being with, and mutually aware of, another person; in its co-located form, co-presence, the mutual availability of participants to one another [3].

Spatial Computing A computing paradigm in which digital content and interactions are situated in and responsive to three-dimensional physical space, enabling computation to be anchored to spatial context, spatial relationships, and embodied user interaction.

Spatial Interaction Interaction based on three-dimensional spatial relationships between users, digital content, and the physical environment.

Spatial Presence The sense of being located within the mediated space rather than at one’s physical location, closely related to the place illusion and to presence-as-transportation [2, 4].

Telepresence The experience of remote “being there” through teleoperation [5].

Togetherness The felt affective sense of belonging and relatedness between people; a central concept for evaluating communication and awareness systems [6].

Tunable Presence A mediated social presence the user governs rather than the system fixes, graduated rather than binary, and adjustable across identity, time, and space.

Ubiquitous Computing (UbiComp) A computing paradigm in which digital technologies are distributed throughout everyday environments and activities, enabling continuous computational support beyond traditional desktop interaction.

Virtual Reality (VR) An immersive form of extended reality in which users experience a fully computer-generated environment, typically through head-mounted displays and embodied interaction techniques.

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Chapter 1

Introduction

Throughout history, the kitchen has functioned as a key site of development and progression, reflecting broader economic, technological, and socio-cultural transformations. In the mid-twentieth century, the widespread adoption of electric refrigerators, gas ovens, and microwaves reshaped how people cook and manage household tasks, automating domestic chores and altering everyday culinary routines. More recently, digital and networked technologies have extended this evolution: smart kitchens equipped with interactive devices, from touchscreen ovens to voice-assisted recipe systems, blend computing with cooking and blur the boundary between traditional utensils and informational interfaces [7–9]. As a result, the design of kitchen tools now extends beyond functionality to encompass interaction, experience, and engagement. Modern appliances feature interfaces ranging from simple tactile controls to complex IoT-connected displays, all of which demand intuitive, human-centered design to be effective [10]. Yet, the adoption and everyday use of these technologies are neither uniform nor inevitable; they are shaped by domestic routines, cultural norms, and household power relations that influence how, and whether, interactive systems become meaningfully integrated into kitchen life. The kitchen thus stands out as a significant site for human–computer interaction and innovation, a space where technological development and HCI design directly shape everyday experience, continuously adapting the age-old act of cooking to new tools and paradigms [11].

“Being with others occurred within the flow of kitchen life – a boy doing his homework or watching television in the kitchen whilst his mother cooked; a woman preparing food for consumption later whilst her husband prepared his own breakfast; a man who lived alone speaking to his deceased wife’s photograph whilst he chopped vegetables; or a younger man with a phone pressed to his ear whilst continuing with other tasks in the kitchen. Such encounters were important in shaping

the meaning of the kitchen and remind us that homes are socially constructed – what people say and do in and about their kitchen is shaped by their interactions and feelings about those other people they care about or spend time with (Murcott 2000; Wright-St Clair et al. 2005); these findings also show that people can display their family relationships (Finch 2007) through their kitchen lives” [12]

The quote from the study *Kitchen Life* captures the lived dynamics of the domestic kitchen. Unlike professional kitchen environments, it is not defined solely by food preparation or task efficiency. Instead, it accommodates a wide range of practices that play a central role in everyday household life across cultures. In the same study, Wills et al. describe the kitchen not as a collection of discrete actions, such as cooking, cleaning, or feeding pets, but as a complex entanglement of practices embedded in everyday life [12]. These practices evolve with shifting social and economic conditions, continually reshaping what the kitchen means and how it is experienced.

The dominant smart-kitchen framing in HCI prioritizes efficiency and functional performance, often overlooking the domestic kitchen’s social dimensions and its role in interaction, memory, and bonding. Building on these contrasting perspectives, this thesis approaches the kitchen as both a productive and relational environment, and explores how emerging technologies, particularly Extended Reality (XR) and mediated spatial interaction, might enrich not only what people do in kitchens but also how they share presence, communication, and togetherness within them.

1.1 Relevance, Significance and Motivation

Kitchen spaces occupy a unique and contested position in HCI research, simultaneously conceptualized as rational task environments and as emotionally charged social hubs. On one hand, a substantial body of work treats kitchens as sites of efficiency, automation, and information delivery, designing smart appliances, recipe displays, and process guidance systems that optimize workflows and reduce cognitive load [13, 14]. This efficiency paradigm, rooted in industrial-era visions of the “smart home”, frames cooking as a sequence of discrete tasks requiring technical support and error prevention [15].

A parallel research tradition has documented the kitchen as a deeply social and culturally significant domestic space. Ethnographic and critical HCI scholarship has shown that cooking practices are embedded in cultural identity, social relationships, and shared meaning [16]; that

kitchens function not as collections of discrete actions but as complex entanglements of practices woven into everyday life [12]; and that family cooking involves rich gestural coordination, embodied teaching, and emotional connection that extend far beyond task completion [17]. Other work has demonstrated how remote cooking sustains intimacy and co-presence for geographically distributed families [18], while Grimes and Harper [9, 19] have explicitly called for HCI to move beyond nutritional monitoring toward celebratory and socially meaningful food technologies. In my own Kitchen Talks study, I observed these dynamics directly: cooking together functions as a fundamental vehicle for teaching, learning, and cultural transmission, and the kitchen consistently emerges as a space of collaboration, creativity, and interpersonal connection.

Despite describing the same physical space, these two research traditions have largely developed in parallel rather than in dialogue. The disconnect stems from divergent disciplinary origins: task-oriented kitchen research inherits from engineering and ubiquitous computing, valuing measurable performance, controlled evaluations, and deployable prototypes; social and ethnographic work inherits from sociology, anthropology, and critical HCI, foregrounding meaning-making, care, and cultural reproduction through interpretive analysis. Each tradition tends to treat the other’s concerns as background rather than as a first-order design problem. This imbalance was diagnosed early in the Kitchen Manifesto by Bell and Kaye [15]. They argued that smart kitchen research had conflated the domestic with the industrial, calling for design that prioritizes experience and the rituals of domesticity over efficiency alone. As Ferdous [20] has further shown, the separation is empirically misleading: cooking routinely involves simultaneous task execution and social interaction, with activities fluidly shifting between instrumental and relational modes.

This thesis addresses that disconnect by reframing social interaction not as a competing concern to task performance, but as itself a core design task, that demands the same intentional design attention that HCI has long devoted to efficiency and automation. The table in Appendix 6.7 illustrates, existing studies cluster predominantly along this task/social axis, foregrounding one dimension while neglecting the other; the contribution of this thesis is to work across both, using mediated spatial interaction as the lens through which their integration becomes designable.

Kitchens are among the most dynamic domestic spaces in terms of technological integration. Over time, they have continually absorbed and adapted to new tools and appliances from

refrigerators, microwave ovens, and dishwashers to air fryers, smart assistants, and broader smart home ecosystems. This ongoing evolution reflects not only a high degree of technological acceptance but also demonstrates how kitchen spaces and associated practices evolve in response to innovation. While these technological advancements have primarily supported the task-oriented dimension of the kitchen, this thesis does not position such systems as inherently limited or reductive. Rather, it observes that their design has largely foregrounded efficiency-driven and instrumental concerns, leaving the social and relational dimensions of kitchen life comparatively underexplored. In this context, emerging paradigms such as Ambient Computing and Extended Reality (XR) create opportunities to extend the scope of supported interactions by foregrounding presence, awareness, and mediated togetherness alongside task performance.

XR, an umbrella term covering the reality-virtuality spectrum introduced by Milgram [21], provides contemporary technological means to explore these possibilities. It allows investigation of natural, embodied interaction with spatially aware systems while drawing on advances in motion tracking, image processing, displays, wearables, and real-time rendering. Rather than being treated as an end in itself, XR is approached here as a powerful enabler for envisioning how future ubiquitous and spatial computing technologies might reshape communication, collaboration, and shared experience in the home. These technological trajectories support forms of interaction that emphasize presence, awareness, and mediated togetherness, enabling more relational modes of engagement alongside task performance. Although their conceptual roots stretch back decades, their domestic adoption has accelerated only in the past ten years, making this an important moment to understand how such technologies might become meaningfully integrated into everyday environments such as the kitchen.

Despite increasing attention to smart homes and immersive technologies, important blind spots remain. There is still limited understanding of how interaction challenges unfold in dynamic domestic spaces such as kitchens, where movement, multitasking, collaboration, and social exchange constantly intersect. Much existing work continues to privilege automation and efficiency, leaving social, emotional, and relational dimensions underexplored. At the same time, designers lack clear guidance on how immersive and spatial computing tools can meaningfully support the creation of domestic experiences rather than merely adding new interfaces. This thesis responds to these gaps through a Research-through-Design approach that brings together contextual inquiry, speculative prototyping, and designer-focused evaluation to explore what mediated spatial interaction in the kitchen could become.

The motivation of this thesis is therefore to bridge task-oriented HCI studies of cooking with the social dimensions of the domestic kitchen. This arises from gaps in both directions: the limited consideration of social interaction within task-focused kitchen studies and the lack of task-aware, practically grounded implementations in research that frames the kitchen primarily as a social hub. Accordingly, this thesis conceptualizes the domestic kitchen as a hybrid environment that simultaneously supports productive doing and meaningful relating. The inquiry is framed around mediated spatial interaction, referring to digitally enhanced ways of communicating and co-experiencing presence across space through immersive and spatially aware technologies. Building on this, it advances a three-channel communication model as ambient, synchronous, and asynchronous, alongside forms of presence (social, spatial, embodied), and operationalized through speculative phygital XR artifacts: Ambient Cooking, Portal, and MagMemo. In this framing, designing for social interaction is not in competition with task support; rather, the XR-mediated social channels become the means through which the embodied, spatially constrained needs of the active kitchen are addressed holistically. Deep technical implementation issues, such as hardware engineering or system optimization, are considered outside the primary scope.

Finally, this work also centers on designers as key stakeholders, recognizing the need for effective tools and workflows to support design exploration in domestic contexts. While end-users contribute by articulating needs, practices, expectations, and experiential challenges, the research primarily foregrounds designer-facing methods for exploring, prototyping, and evaluating mediated spatial interactions. This positioning situates the work within Human–Computer Interaction (HCI), design research, and speculative prototyping, with a focus on how artifacts and immersive design processes function as research instruments rather than solely as deployable products.

Why XR in the Domestic Kitchen? The empirical findings reported across this thesis, do not arise from XR technologies themselves. Rather, they describe a set of recurring breakdowns and unmet needs in mediated kitchen interaction (without an XR-only focus): visibility constraints, audibility disruptions, spatial awareness deficits, hands-occupied conditions, and the absence of shared spatial reference between co-located and remote participants. The majority of these needs are fundamentally visual and spatial in nature. They concern where information lives, how it is anchored to objects, surfaces, and zones, and how presence and awareness can be distributed across a materially complex environment while attention remains tethered to active

tasks.

My focus on XR as a possible solution direction comes afterwards. Among the emerging technological paradigms available to address the breakdowns identified above, XR and spatial computing is one of the particularly well-suited technologies: it combines spatial anchoring, embodied interaction, and support for hands-occupied use in ways that align with the rhythms and material constraints of everyday kitchen activity.

This thesis therefore approaches XR not as an outcome to be argued for, but as the design and methodological lens best suited to address visual and spatial breakdowns of this kind. XR, encompassing the reality–virtuality spectrum from Real Reality through AR and MR to fully immersive VR [21], offers a coherent set of affordances that map onto the three modalities central to this work: spatial anchoring for ambient cues, embodied co-presence for synchronous communication, and persistent in-place traces for asynchronous exchange.

The contributions developed in this thesis are not framed as the only viable design response to the frictions identified. The findings, design qualities, and modality combinations described here are deliberately portable: they characterize interaction concerns that can inform the design of voice-based, robotic, and other emerging domestic systems, and the speculative artifacts proposed in later chapters may be supported, extended, or complemented by such technologies in future work.

Finally, XR plays a second and analytically distinct role in this thesis, which is methodological rather than technological. By enabling embodied, situated prototyping of interactions that are not yet feasible in everyday homes, XR functions as a speculative instrument, which is a means of making kitchen hybridity designable rather than only observable. In this sense, the use of XR is justified not solely by what it may become as a deployed technology, but by what it permits designers and researchers to investigate now: the spatial and relational dimensions of kitchen interaction can be enacted, perturbed, and refined under ecologically grounded yet controlled conditions in ways that alternative modalities currently do not afford.

1.2 Scope

This thesis is situated at the intersection of Interaction Design, Human-Computer Interaction (HCI), and Extended Reality (XR) studies. *Its core scope is the investigation of mediated spatial interaction in domestic kitchens, with a particular focus on how emerging immersive and spatial*

computing technologies can support communication, collaboration, and shared presence in these environments. Rather than addressing XR in general or domestic computing in the abstract, the work concentrates on a specific socio-technical setting: the home kitchen as an active, materially dense, and socially meaningful space.

At a conceptual level, the thesis is concerned with how communication and coordination unfold when people cook, collaborate, and connect across distance, and how these practices might be reconfigured through XR-supported interactions. The focus lies on presence, awareness, and togetherness as experienced in everyday kitchen life, and on how these qualities can be mediated through ambient, synchronous, and asynchronous forms of connection. While the work necessarily engages with broader smart home discourses and ubiquitous computing visions, it does so through the concrete lens of kitchen practices rather than through a generalized model of domestic technology.

Methodologically, the thesis is scoped as a design-oriented, programmatic research effort that advances knowledge through a sequence of interrelated design experiments. It does not aim to produce universal design rules or predictive models, but instead offers situated, intermediate-level insights grounded in specific artifacts, scenarios, and studies. XR technologies are treated as epistemic and exploratory instruments that allow designers and researchers to stage, experience, and interrogate possible futures of domestic interaction. Implementation choices, hardware configurations, and software architectures are engaged only to the extent that they support the investigation of mediated spatial interaction, not as ends in themselves.

The thesis also occupies a particular position with respect to adjacent fields. It touches on family studies, domestic sociology, and Computer Supported Cooperative Work (CSCW) insofar as these domains illuminate how households communicate, negotiate roles, and build routines together. It intersects with research on smart homes, ambient computing, and telepresence by drawing on their conceptualizations of connected environments and mediated co-presence. However, it does not claim to offer comprehensive sociological theory, nor to provide a full account of smart home infrastructures or organizational collaboration. Its primary contributions are directed toward HCI and Interaction Design, especially in relation to domestic communication, hybrid presence, and designerly ways of engaging with XR.

Several areas that may appear adjacent at first glance are deliberately left outside the scope. The thesis does not engage in low-level engineering of XR systems, optimization of tracking or rendering pipelines, or hardware innovation. It does not address nutrition or food studies

as primary topics, even though cooking provides the backdrop for many interactions. Energy consumption, sustainability of appliances, and long-term market adoption of smart kitchen products are not examined in depth. The work also does not focus on professional, industrial, or commercial kitchens, where organizational structures, safety regulations, and productivity metrics dominate design considerations.

Within these boundaries, the scope of the thesis is intentionally narrow in setting yet broad in implication. By concentrating on the domestic kitchen and on mediated spatial interaction through XR, the work seeks to offer nuanced, design-relevant understandings that can inform future research and practice in HCI, Interaction Design, and XR-mediated domestic communication, while acknowledging that many technical, sociological, and commercial questions lie beyond what this thesis is designed to answer.

1.3 Aims and Research Questions

This thesis aims to explore how Extended Reality (XR) technologies can support mediated spatial interaction within domestic kitchens by addressing both experiential and design-oriented concerns. First, it seeks to identify and characterize the distinctive interaction challenges that emerge in kitchen environments (A1), including spatial constraints, sensory limitations, safety concerns, and communication breakdowns that become especially visible in hybrid or remote engagements. Building on these insights, the study aims to examine how XR-enabled interventions can mediate presence, awareness, and coordination in ways that support togetherness, safety, and everyday collaboration without displacing existing domestic practices (A2). Parallel to this experiential focus, the thesis also investigates XR as a methodological and creative resource for designers (A3), examining how immersive and speculative prototyping can support reasoning, envisioning, and evaluation of meaningful domestic experiences under real-world spatial and social constraints. Together, these aims position the kitchen not simply as a site of task execution but as a lived social space, and they frame XR not as a technological novelty but as a lens for rethinking communication, care, and co-presence in everyday domestic life.

To structure and focus this investigation, a set of interrelated research questions was formulated. These questions reflect both the experiential dimensions of mediated spatial interaction in domestic kitchens and the role of XR within design practice.

RQ0. How can Extended Reality (XR) technologies support mediated spatial

interaction in domestic kitchens, addressing both everyday communication challenges and the design of meaningful domestic experiences?

RQ1. In what ways can XR technologies address the interaction challenges that emerge in domestic kitchen environments, particularly during hybrid or remote engagements?

RQ1 foregrounds the experiential dimension of mediated spatial interaction, focusing on how everyday communication and coordination unfold within the lived complexity of a domestic kitchen. Rather than treating the kitchen as a neutral technology deployment site, this question approaches it as a dynamic social environment shaped by movement, multitasking, emotion, care, and safety considerations. By examining where communicative, spatial, and sensory breakdowns occur and how people navigate them, the thesis investigates how XR might meaningfully intervene, not by replacing established practices, but by supporting more coordinated, legible, and socially attuned forms of interaction.

- **RQ1a.** What communicative, spatial, and sensory breakdowns do users encounter in kitchen contexts?
- **RQ1b.** How can XR mediate presence, awareness, and coordination in the kitchen environment, and with what implications for collaboration, safety, and togetherness?

RQ2. How can designers meaningfully employ XR as a tool for envisioning, prototyping, and evaluating domestic experiences in kitchen contexts?

RQ2 shifts the focus from interaction outcomes to design practice, examining XR not only as an end-user medium but also as a designer-facing instrument. It investigates how XR can function as a methodological tool that enables designers to reason about kitchens as spatial, embodied, and constraint-rich environments, and how immersive and speculative prototyping reshapes design thinking, experimentation, and evaluation. Rather than treating XR as a representational aid, this question explores how designers think with XR; using spatial simulation, embodied perspective, and speculative scenarios to surface interactional dynamics that are difficult to access through abstract or screen-based methods alone.

- **RQ2a.** How do designers use XR as a methodological tool to reason about spatial and interactional dynamics in kitchens?

- **RQ2b.** What unique design insights arise from immersive and speculative XR prototyping?

Taken together, these research questions foreground both the lived experience of domestic interaction and the designer’s role in shaping its future. They allow the thesis to investigate kitchens not only as sites of technological deployment but as complex social environments in which XR has the potential to mediate presence, communication, and shared experience in new ways.

This thesis is intentionally scoped to the domestic kitchen, a site of rich, embodied interaction that combines practical tasks with social connection. Unlike commercial or industrial kitchen environments, the home kitchen represents an intimate, variable, and often multitasking space where people cook, talk, collaborate, and care for one another. The focus here is not on automation, efficiency, or control systems, but on enhancing human connection, coordination, and everyday togetherness through XR-mediated spatial interaction.

1.4 Expected Contributions

This thesis is expected to contribute to the literature in four areas:

- **Conceptual.** The thesis will advance two conceptual contributions. First, a reframing of the domestic kitchen as a *spatial connection hub* for future homes, positioning social interaction as a primary design concern. Second, a reframing of the kitchen’s spatial structure as a *cluster of active workzones* rather than a homogeneous room.
- **Design knowledge.** The thesis will contribute intermediate-level design knowledge for XR-mediated kitchen interaction, articulated as four design suggestions (RQ1) for using spatial interaction and technologies to support communication and collaboration in the domestic kitchen. These suggestions are grounded in the interaction frictions identified with kitchen inhabitants in the contextual study.
- **Methodological.** The thesis will demonstrate how XR can serve as a design-research tool for domestic interaction, supporting envisioning, prototyping, and in-situ evaluation as an intermediate “living” testbed. This contribution is articulated as four further design

suggestions for design practice (RQ2) and is instantiated in *KitchXR*, the custom-built environment used across several studies in this thesis.

- **Artifacts.** The thesis will contribute a set of XR artifacts: *KitchXR* as a methodological tool that operationalizes the living testbed, together with the speculative probes *Portal*, *MagMemo*, and *Ambient Cooking* as provocative design exemplars rather than deployable systems.

1.5 Methodological Framework

This thesis approaches the domestic kitchen as a complex sociotechnical environment shaped by embodied activity, spatial organization, material arrangements, and everyday social relations. Investigating such environments requires methods that can live with ambiguity, accommodate multiplicity, and engage practices that unfold over time. Conventional Human–Computer Interaction (HCI) approaches, that often oriented toward task efficiency, usability optimization, and system performance, struggle to account for emerging notions of mediated presence, hybrid communication, and future-oriented domestic practices. In response, this research adopts a methodological stance that privileges exploration over validation and articulation over optimization.

To examine the evolving design space of mediated spatial interaction in kitchens, the thesis is grounded in a **Design-oriented Research** paradigm [22] and shaped by a **Programmatic Design Research** framework [23, 24]. Here, design is not merely a means of producing solutions but a way of thinking and knowing. Rather than resolving predefined problems, the research uses design to articulate and probe a complex design space through a series of interconnected design interventions. The work unfolds as a coherent *research program* composed of multiple design *experiments*, each contributing partial and situated insights. This approach is particularly suited to the domestic kitchen, where embodied labor, social negotiation, and technological mediation intersect in what Rittel and Webber describe as “wicked problems” [25]. Instead of isolating variables or testing narrowly framed hypotheses, knowledge is built through iteration, comparison, and reflection across studies.

Within this framework, design artifacts are treated as **epistemic instruments**. Each study contributes an *exemplar*, such as the *KitchXR* design tool or speculative *Portal* concepts, that operates as a material argument about how mediated spatial interaction in kitchens might be

configured. These artifacts do not seek universal rules, but generate intermediate-level knowledge by demonstrating how hybrid domestic interactions can be constructed, experienced, and reasoned about in practice. The programmatic structure also enables movement across multiple design modes, as articulated in Fallman’s Interaction Design Research Triangle [26]: contextual inquiry grounds the work in lived domestic practice; design tools intervene in professional design processes; and speculative explorations materialize alternative futures. Rather than functioning as separate methodological silos, these modes inform and reshape one another across the research trajectory.

By structuring the thesis as a sequence of connected inquiries within a shared methodological framework, the work supports cumulative knowledge production. Early empirical insights inform later speculative and tool-building efforts, which in turn deepen engagement with mediated spatial interaction. Extended Reality technologies are therefore positioned not only as topics of study but also as methodological resources for investigating how future domestic communication and togetherness might take shape. Ultimately, the contributions of this thesis emerge not from any single artifact or study, but from the relationships and reflections that develop across them.

1.6 Thesis Structure

This thesis is structured as a paper-based thesis that brings together a set of interconnected studies conducted over the course of the doctoral research. Each study contributes a distinct perspective on mediated spatial interaction in domestic kitchens, while together they build a coherent research trajectory that connects empirical inquiry, speculative exploration, and design-oriented knowledge production. Rather than a linear problem–solution narrative, the thesis unfolds as a progressively developing research program, in which each study both responds to earlier insights and opens new conceptual ground for the next. Readers may therefore approach the thesis either sequentially, to follow the evolution of ideas and methods, or focus on individual studies of interest, using this introductory chapter as a guide to their conceptual and methodological relationships.

1.6.1 Project Context and Contributors

This doctoral research was initiated within a university–industry collaboration between Koç University and Arçelik A.Ş. R&D, carried out under the coordination of the Koç University Arçelik Research Center for Creative Industries (KUAR) as part of a TÜBİTAK-supported BİDEB 2244 project. This BİDEB project, numbered as 118C095, is designed to explore future domestic experiences through immersive and spatial computing technologies, with a particular emphasis on kitchens as complex and socially meaningful environments. The project aimed to answer following research questions: (1)How can the physical device experience of users be enriched through AR/VR/MR? (2)How can efficiency be increased by integrating AR/VR/MR technologies into the Product Development / Production / Service stages?

Within this collaborative framework, the thesis developed as an academic investigation that both benefited from and contributed to industry knowledge, while maintaining an independent research agenda grounded in design research and Human-Computer Interaction. The work has been shaped through close collaboration with academic supervisors, industry advisors, R&D, UX departments, designers, and study participants who contributed their expertise, experiences, and critical reflections across different stages of the research. Additionally, the outcomes of these studies are shared within R&D and UX department Specific authorship, collaboration roles, and publication contexts of each study are explicitly detailed in their corresponding chapters.

1.6.2 Linking the Studies

Across the four studies conducted in this thesis, a coherent research trajectory emerges, one that progressively builds an understanding of how communication, collaboration, and hybrid presence unfold in domestic kitchens through XR technologies. Each study addresses a distinct layer of the problem space, and together they develop the contributions consolidated in Chapter 6: a friction lens on kitchen interaction, two reframings of the kitchen itself, the construct of tunable presence, and XR as a methodological living testbed.

The first study, Kitchen Talks (Paper I, section 4.1), provides the thesis’s inhabitant-grounded empirical foundation. Through observational fieldwork and ideation workshops, it surfaces the recurring breakdowns (visibility, audibility, spatial awareness, multi-sensory communication, spatial adaptability, and control over privacy and identity) that the later chapters consolidate into the friction lens. It also establishes the two reframings the thesis carries for-

ward: the kitchen as a socially central connection hub rather than a private cooking facility, and as an active, materially dense space of distributed workzones between which inhabitants continually move.

The second study, the Portal investigation (Paper II, section 4.2), expands the inquiry in a speculative direction by examining phygital thresholds that bridge physical and virtual realms. Through immersive workshops, it conceptualizes portals as hybrid boundary objects shaped by identity flexibility, real-time responsiveness, and strategic spatial placement. These qualities seed the identity and space axes of tunable presence; the temporal axis is substantiated by the asynchronous channel in the fourth study.

The third study, KitchXR (Paper III, section 4.3), shifts perspective from inhabitants to professional designers and contributes the thesis’s methodological core. It demonstrates how a custom-built XR environment can serve a design research instrument: a context-rich setting in which designers reconstruct the kitchen’s spatial complexity, enact interaction in situ, gather behavioral and bio-sensory data, and revisit early decisions under realistic constraints. KitchXR is both the central artifact the thesis contributes and the instrument that makes the subsequent evaluation possible.

The fourth study, Hybrid Connected Kitchens (Paper IV, section 4.4), integrates the preceding insights by developing and evaluating three speculative phygital artifacts that operationalize an ambient, synchronous, and asynchronous communication model. Evaluated with designers inside KitchXR, these artifacts consolidate the thesis’s design knowledge through the asynchronous channel in particular, substantiate the temporal axis of tunable presence.

Taken together, these four studies form a continuous chain of inquiry, from grounding the problem in inhabitants’ experience, to conceptualizing a speculative threshold, to building a methodological environment, to evaluating phygital artifacts within it. It is this cumulative trajectory, rather than any single study, that the concluding chapter abstracts into the thesis’s contributions.

1.6.3 AI Use Disclosure

Several Artificial intelligence (AI) tools, including platform-embedded AI features, were used in this thesis as limited support for writing and organization, rather than to generate research contributions. AI-assisted tools were employed selectively for proofreading, improving clarity

and narrative flow, and occasionally supporting the preliminary structuring of author-produced qualitative insights. All conceptual reasoning, methodological decisions, analytic judgments, interpretation of findings, and design conclusions were developed by the researcher. Where AI-assisted outputs influenced the manuscript in a substantive way, they were critically reviewed, revised, and integrated through the researcher’s scholarly judgment.

AI tools were also used in a limited manner to support literature discovery (e.g., identifying potentially relevant keywords, venues, and candidate publications) to broaden the scope of the review. All AI-suggested literature and related claims were manually verified against the original sources. No section of this thesis was automatically generated and accepted without the researcher’s manual revision and full authorship responsibility.

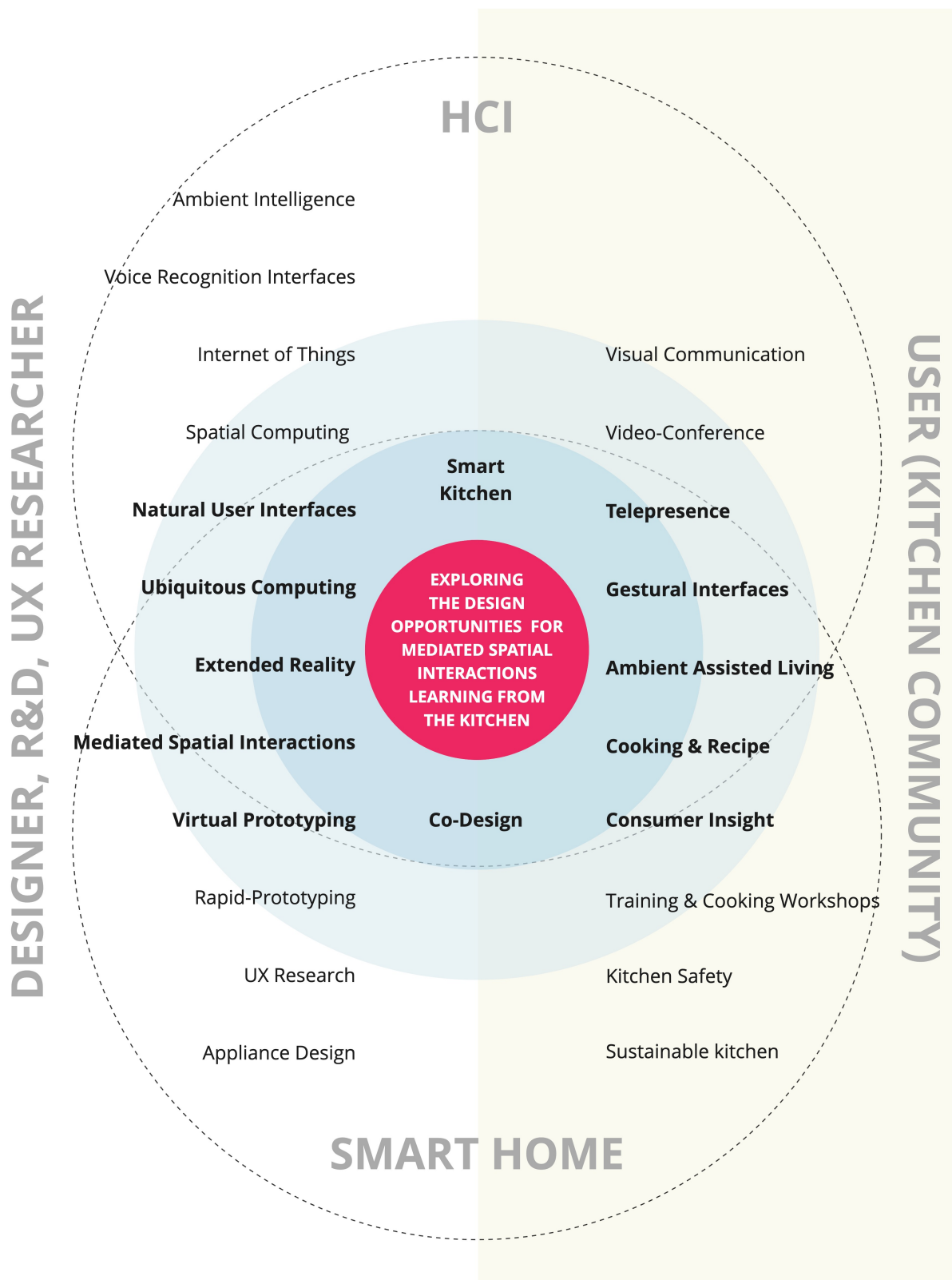


Figure 1.1: The Literature Scope of the Thesis.

Chapter 2

Literature Review

This chapter situates the thesis within scholarship on domestic life, ubiquitous computing, smart homes, and spatial computing, with particular attention to how interaction unfolds in the domestic kitchen. The kitchen has long been understood as both a productive workspace and a deeply social environment. Contemporary research mirrors this duality: some traditions pursue efficiency, automation, and task performance, while others foreground memory, care, learning, and relational experience. Against this backdrop, the chapter traces how ubiquitous and pervasive computing evolved from automation-driven visions toward concerns with appropriation, negotiation, control, and lived practice in connected homes and smart kitchens. In parallel, research on spatial computing and XR provides concepts such as embodied action, spatial awareness, anchoring, and collaboration for understanding how interaction is reorganized when digital content is registered to physical space. Taken together, these strands frame mediated spatial interaction in kitchens as a socially embedded design problem, not merely a technical one.

Structurally, the chapter begins by establishing the kitchen’s cultural and historical significance, then reviews key trajectories in ubiquitous computing, smart homes, and smart kitchens that shape domestic computing imaginaries and deployments. It then turns to spatial computing, first outlining its definitions and core tensions, and then focusing on the bodies of work most relevant to this thesis, spatial and smart kitchens, and spatial collaboration. The chapter concludes by reviewing prototyping and user-insight methodologies that inform how such systems can be designed, staged, and evaluated in practice.

2.1 Kitchen, Home, and Domestic Computing

This section reviews literature that frames the kitchen as a central site of domestic computing, where everyday practices, social relations, and technological systems intersect. It traces how research in HCI and related fields has conceptualized kitchens and homes as sociotechnical environments, moving from cultural and ethnographic accounts toward ubiquitous computing, smart home, and smart kitchen paradigms. Together, these perspectives establish the domestic kitchen as a complex interactional setting in which computing technologies are embedded, negotiated, and appropriated through everyday life rather than simply adopted as tools.

2.1.1 Domestic Kitchens

Historically, the kitchen has functioned as a symbol of economic development, technological progress, and socio-cultural values. It has served as a marker of status, hospitality, culinary skill, and material sufficiency across cultures [7]. A canonical example is the 1959 American National Exhibition in Moscow, which hosted the famous “kitchen debate” between Nixon and Khrushchev during the presentation of the RCA Whirlpool Miracle Kitchen [27]. Beyond its role as a media spectacle, the exhibition positioned the kitchen as a proxy for modern living, where domestic technology, comfort, and efficiency symbolized broader socio-economic progress. The Miracle Kitchen framed household work as something that could be optimized through integrated appliances and automation imaginaries, foreshadowing many themes later revisited in smart kitchen research.

Ethnographic studies have been central to understanding domestic life, particularly the dynamics of kitchen spaces. Through long-term observation and interviews, this body of work captures the routines, cultural practices, and social interactions that structure everyday kitchen activity. Warde [16], for example, shows that cooking practices are not merely functional acts but are deeply embedded in cultural identity, social relationships, and shared meaning. By examining the kitchen as a site of both labor and leisure, ethnographic research reveals how food preparation and consumption rituals reflect broader social norms and values.

Within this tradition, researchers such as Bell and Valentine have highlighted how technological change reshapes domestic kitchens and family practices. They argue that modern appliances and convenience technologies have gradually shifted the kitchen from a shared, communal space toward more individualized patterns of use, raising concerns about the erosion of

intergenerational knowledge transfer and shared culinary identity [28]. This tension is captured in their observation:

“Passing on recipes and particular cooking techniques from one generation to another (usually from mother to daughter) is one way in which some households have traditionally reproduced their ‘identities’ over time, although some commentators have claimed that new food technologies (e.g. the microwave) and convenience foods are destroying this tradition.”

At the same time, other studies complicate this narrative by showing how kitchens continue to function as dense social hubs. Miller et al. [29], using a programmatic design research approach, demonstrate that many concurrent family activities, such as cooking, cleaning, homework, and conversation, coexist in kitchens, even when households claim to value other rooms as primary social spaces. Their findings suggest that the kitchen remains a key site of informal coordination and social cohesion, underscoring the importance of designing technologies that enhance, rather than disrupt, these shared practices.

Taken together, this literature establishes the kitchen as an interactional ecology in which productive work, care, identity, and everyday sociability are tightly entangled rather than separable into “task” and “social” layers. Breakdowns and negotiations routinely emerge around routines, roles, interruptions, and shared use of tools and space, making the kitchen a particularly demanding site for technology-mediated interaction. For this thesis, the key implication is that the kitchen cannot be reduced to efficiency or automation alone: its social functions and relational dynamics are as central as task-related activities. This perspective directly informs RQ0 and RQ1 by prioritizing the study of communication and interaction within the domestic kitchen.

2.1.2 Ubiquitous Computing for Home

This subsection reviews foundational HCI visions and evaluations of computing technologies as they relate to domestic life. While these visions span several decades—from early ubiquitous computing proposals to more recent smart home systems, they increasingly converge around the challenge of integrating computation into everyday environments without disrupting existing practices. In this thesis, I use spatial computing as an integrative lens for examining a subset of domestic computing in which interaction is organized around spatial models, embodied action, and spatially embedded information, allowing analysis across smart home infrastructures and XR-mediated experiences as kitchens become hybrid interaction spaces.

Ubiquitous computing (UbiComp), articulated by Weiser in 1991 [30], proposed a future in which computation is embedded throughout everyday environments and recedes into the background of attention. Rather than focusing on explicit interaction with screens, UbiComp aimed to create smart environments that provide context-aware services in an unobtrusive and intuitive manner [31,32]. This vision strongly influenced HCI research on domestic technologies, framing the home as a primary site for studying how computation becomes part of daily routines. A persistent challenge identified in this literature concerns understanding what people actually do in everyday contexts. Rogers [33] argues that designing UbiComp systems requires grappling with people’s motivations, timing, routines, and improvisations—factors that are difficult to model computationally. Related work on implicit interaction emphasizes that many interactions are not intentional commands but actions interpreted by systems within a specific context [34,35]. This places significant demands on context-aware systems, which must infer meaning from partial, noisy, and situated data.

To address this, Dey [32] formalized context-aware computing around three core capabilities: presenting relevant information and services, automatically executing actions, and tagging contextual data for later use. Subsequent research explored how sensing and AI techniques—such as activity recognition and human–object interaction detection—could support these goals in domestic settings, including kitchens [36–39]. While technically promising, such systems often remain focused on recognition accuracy and task assistance rather than on broader interactional and social consequences.

Parallel developments in HCI emphasized materiality and embodied interaction. Norman’s vision of “invisible” computing [40] inspired Tangible User Interfaces, introduced by Ishii and Ullmer [41], which sought to bridge physical and digital space by coupling bits and atoms through interactive surfaces and ambient media. These approaches foregrounded the role of physical space and material interaction, prefiguring later spatial computing paradigms. Ethnographic work further complicated purely technical visions of domestic UbiComp. Frissen [42] noted persistent frictions in adopting communication technologies in the home, even when they promise functional benefits. Crabtree et al. [43] elaborated this by identifying domestic “habitats”, “activity centers”, and “coordinate displays” as recurring organizational structures through which communication is managed. Their analysis shows how technologies must align with existing routines and spatial arrangements, such as kitchen tables acting as coordination hubs, if they are to be meaningfully integrated into domestic life.

Across these developments runs a common conceptual foundation, which Dourish [44] names *embodied interaction*: the creation, manipulation, and sharing of meaning through engaged, practical involvement with the world. For Dourish, “embodiment” is not a matter of being merely *physical* or *bodily* but of being *present and participative* in a world of people, artifacts, and settings. So, a conversation is as embodied as the handling of a tool. Seen this way, the material strand traced above, from Norman’s invisible computing to Ishii and Ullmer’s tangible interfaces, and the situated, routine-bound practices documented by Frissen and Crabtree et al. are not separate concerns but two faces of one principle: interaction becomes meaningful through our coupling to the world, and through the ways people appropriate technologies into the ongoing, intersubjective negotiation of shared activity. This framing is directly pertinent to the kitchen, where cooking, coordinating, and caring for others unfold as continuous, materially situated practice rather than as discrete operations on an interface.

2.1.3 Smart Home

As UbiComp ideas moved from speculative vision to deployable systems, the smart home emerged as a concrete instantiation of embedded, context-aware domestic computing. Advances in sensing, networking, and IoT infrastructures enabled researchers to operationalize UbiComp concepts within real or simulated homes, prompting a shift toward empirical evaluation in situated settings. Early smart home research often relied on living laboratories, such as the Georgia Tech Aware Home, to observe user behavior and test system concepts in instrumented domestic environments [45–47]. Coskun et al. [48] categorize this work into three strands: living-lab studies of interaction, studies of existing smart home users, and investigations of potential users’ expectations. While these approaches produced valuable insights into usability and functionality, they also revealed limitations in scaling such systems beyond controlled settings.

Crucially, HCI researchers increasingly recognized that homes are not neutral deployment sites but socially organized spaces where control, privacy, care, and coordination are continuously negotiated. Foundational work by Lee et al. [49] demonstrates that questions of “who controls what” in smart homes are relational and political rather than purely technical. More recent studies further emphasize that smart homes should be understood as inhabited ecologies rather than collections of devices, foregrounding how residents adapt, resist, and reinterpret domestic technologies over time [50].

This shift has brought renewed attention to spatiality and architecture as first-class design concerns. Socio-technical IoT perspectives stress that value emerges from the coupling of technological capabilities with social practices and place-based contexts [51]. Complementary spatial design work explicitly treats domestic space as an interaction substrate, arguing that device ecologies should align with mobility, room-to-room routines, and embodied activity [52]. These concerns are particularly salient for kitchens, where interaction unfolds through movement, divided attention, and multi-actor coordination, and where early exemplars such as dual-perspective interactive counters illustrate how interfaces can be embedded into work surfaces to support situated action [53].

2.1.4 Smart Kitchen

Early smart kitchen research primarily focused on assisting cooking tasks through recipe presentation and instructional support. Systems such as CounterActive [54] projected digital recipes onto kitchen counters, replacing conventional input devices with touch-sensitive surfaces. This study is a part of a vision named *Active Workspaces* [55]. It is a definition of context-specific environments in which people work with physical tools and materials to produce tangible artifacts, augmented with displays, sensors, and computation that inform, guide, and respond to the user’s actions rather than performing the task on their behalf. In this thesis, Ju contributes with design principles (user control, adaptability, feedback, appropriately calibrated translucency, and coherency) together with process guidelines for augmenting everyday physical work with embedded computation. Similarly, EyeCook [56] explored hands-free interaction using gaze and speech to reduce attentional demands during cooking. Lee and Bonanni’s work on spatial user interfaces extended this line of inquiry by treating the kitchen itself as an interactive surface, using projected AR to reveal contextual information such as appliance contents or procedural guidance [57–59].

While technically innovative, these systems also highlighted the importance of attention management and safety in cooking contexts. Lee et al., drawing on context-aware design principles [60], explicitly aimed to minimize distraction during risky actions, for example by using ambient cues rather than explicit instructions.

Parallel research examined everyday coordination and communication in domestic life, including studies of household communication patterns [43], refrigerator magnets as coordination artifacts [61], and long-term technology adoption [62]. Taylor and Swan’s notion of “artful orga-

nizing systems” describes how families assemble and maintain flexible combinations of artifacts to manage routines and sustain social order [61]. These systems do not merely organize tasks; they actively shape family roles and relationships. Consistent with this view, Lee et al. [49] argue that smart home technologies should enrich family activities and emotional bonds rather than impose rigid structures.

2.1.5 Relevance for the Thesis: Kitchen, Home, and Domestic Computing

This body of literature establishes the kitchen as a socially and materially complex interactional ecology in which technologies are continuously negotiated rather than simply adopted. For this thesis, the central gap is that much domestic computing research still treats the kitchen either as a site of task optimization or as a site of social meaning, without offering interaction design approaches that can hold both simultaneously under real kitchen conditions such as divided attention, mess, and multi-actor routines. My studies take this tension as a starting point and investigate how mediated spatial interaction can support communication and co-presence without reducing the kitchen to a productivity interface or a purely symbolic social space.

2.2 Spatial Computing and Extended Reality

This section situates XR and spatial interaction research within the HCI literature that informs this thesis. The thesis focuses on the design and evaluation of interactive XR systems, how they are conceptualized, prototyped, and studied in relation to spatial interaction and user experience, rather than on physiological or perceptual foundations of immersive media. Accordingly, the section is organized around three strands: taxonomies and classification frameworks for positioning XR experiences, technical research that enables spatial interaction capabilities, and empirical studies examining how these capabilities shape use and experience.

2.2.1 MR Taxonomy: Reality-Virtuality Continuum

In 1994, Paul Milgram and Fumio Kishino introduced the Reality–Virtuality (RV) continuum and the term mixed reality (MR) as a taxonomy for emerging display systems [21]. Because the paper was written to classify display technologies, the original framework focuses primarily on visual display configurations. Despite this scope, the RV continuum has remained a widely

2.2 SPATIAL COMPUTING AND EXTENDED REALITY

cited lens for reasoning about the design space of immersive experiences, and later work has extended the discussion toward interaction and experience.

The RV continuum (Figure 2.1) represents a spectrum between a real environment, defined as “consisting solely of real objects”, and a virtual environment, defined as “consisting solely of virtual objects.” Any configuration between these endpoints is considered mixed reality [21,63]. Within this space, experiences that overlay virtual content onto the real world are typically described as augmented reality (AR), while experiences that are predominantly virtual but retain some inclusion or awareness of real-world elements are described as augmented virtuality (AV) [21,63].

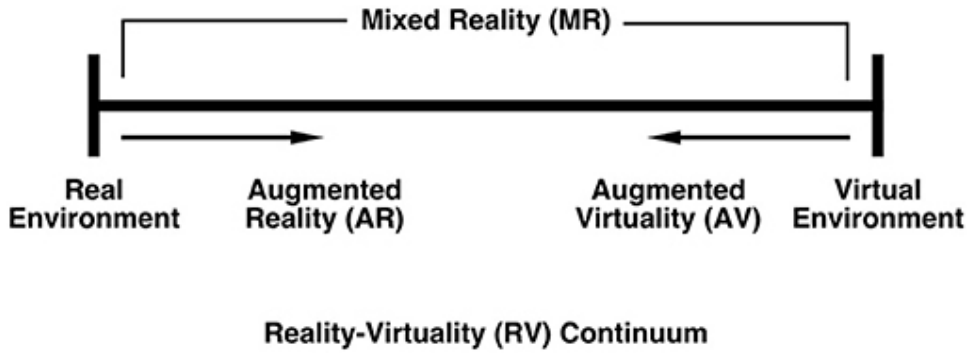


Figure 2.1: Original illustration of Milgram’s Real-Virtual Environment taxonomy [21].

Alongside the RV continuum, Milgram and Kishino proposed additional dimensions for analyzing MR systems: (1) Extent of World Knowledge (EWK) (“how much do we know about the world being displayed?”), (2) Reproduction Fidelity (RF) (“how realistically are we able to display it?”), and (3) Extent of Presence Metaphor (EPM) (“what is the extent of the illusion that the observer is present within that world?”) [21].

Skarbez et al. revisit these ideas in light of twenty-five years of technological development and critique the RV continuum as effectively discontinuous, arguing that a “Matrix-like” perfectly virtual endpoint is not achievable in practice; as a result, many experiences commonly labeled as VR can be interpreted as forms of MR [63]. They further contend that RF and EPM are closely related and propose merging them into a single dimension.

In the same work, Skarbez et al. propose an updated three-dimensional taxonomy (Figure 2.2.1) consisting of: (1) *Extent of World Knowledge* (EWK), expanded beyond “what/where” sensing to include “how” (how things work) and “when” (when things happen); (2) *Immersion* (IM), combining RF and EPM; and (3) *Coherence* (CO), describing the degree to which the

experience is internally and externally consistent, and aligning with discussions often framed in terms of fidelity or authenticity [63].

Critique A: Limited operationalization for consistent classification. Although Milgram and Kishino’s RV continuum and associated dimensions provide a valuable vocabulary for positioning MR display systems [21], both the original framework and later revisions remain difficult to operationalize as consistent and repeatable classification schemes across studies and contexts. In practice, placing an experience along axes such as EWK and RF/EPM or IM and CO in Skarbez et al.’s revision, requires interpretive decisions about what counts as “immersion”, “coherence”, and relevant capability under specific usage conditions [63]. This ambiguity resonates with broader critiques in immersive media research: Nilsson et al. show that “immersion” is used inconsistently across communities and is often conflated with related constructs such as presence and engagement [64]. Norton et al. likewise motivate a quantifiable framework for immersion because existing definitions often support only relative comparisons rather than an operational description [65]. Taken together, these issues suggest that RV-continuum-based taxonomies are most dependable as analytic lenses for comparative reasoning, rather than as standardized rubrics that yield stable classifications across contexts.

Critique B: Under-specification of interaction and contemporary system capabilities. A second limitation concerns scope. Milgram and Kishino’s taxonomy was developed to classify MR *visual display* systems [21], and while Skarbez et al. update the framework, interaction remains largely treated indirectly through system properties supporting “valid actions” and coherence [63]. This is increasingly restrictive given contemporary XR systems in which interaction modalities and sensing configurations are major differentiators. For example, Kim’s chronological review documents a shift from controller-dominant input toward hand- and gaze-based interaction and, more recently, multimodal interaction shaped by sensing stacks and SDK support [66]. Complementarily, Shakil et al. demonstrate how interaction-focused taxonomies can classify gaze techniques by input type and practical constraints (e.g., accuracy and interface requirements), providing structured distinctions that RV-continuum-oriented frameworks do not make first-class [67].

In this thesis, I therefore use RV-continuum-based taxonomies primarily for conceptual positioning rather than as classification tools. I use *Extended Reality* (XR) as an umbrella term encompassing VR, AR, and MR, and I deliberately adopt XR in several places instead of label-

2.2 SPATIAL COMPUTING AND EXTENDED REALITY

ing the work strictly as MR, to keep attention on continuity between physical-world situations and digitally mediated spatial interaction rather than categorical boundaries. The increasing importance of interaction modalities and sensing configurations is addressed later when discussing virtual prototyping and UX evaluation.

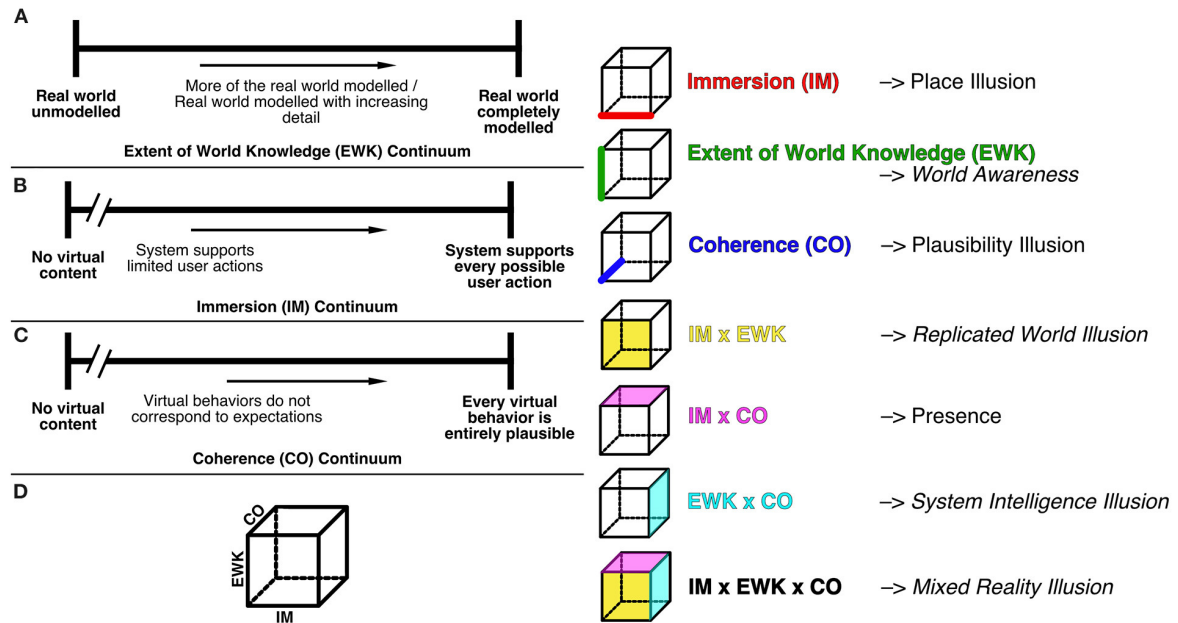


Figure 2.2: Original illustrations from Skarbez et al.'s taxonomy [63].

2.2.2 Spatial Computing Paradigm

Building on the taxonomic lenses above, this subsection shifts from classification toward a contemporary framing often described as spatial computing. In HCI and immersive systems research, spatial computing refers to interactive computing that is situated in, and responsive to, the three-dimensional physical world: systems sense environments, track bodies and objects, and align digital content to spatial structure so interaction can be organized around embodied action rather than screen-centric control. Instead of treating immersive technologies primarily as display categories (VR/AR/MR), the spatial computing perspective foregrounds enabling capabilities (world sensing, spatial mapping, anchoring, and multimodal input) that make space itself an interface substrate and a coordination resource.

”But the concept of spatial’ in the context of digital design is not new. It evolved from explorations of text-based virtual places like multiuser dungeons, to the initial hype surrounding VR in the late 1980s, to the emergence of 2D graphical virtual

environments, to immersive CAVEs, to collaborative virtual environments (CVEs) and inhabited information spaces’ in the 1990s and early 2000s.” [68]

In parallel to smart home trajectories, spatial computing has emerged as a complementary paradigm that foregrounds computation *in and about space*: systems model physical environments, track bodies and objects, and register digital content to locations so interaction can be structured through spatial relationships and embodied reference [69, 70]. From an HCI perspective, this paradigm builds on decades of AR/VR interface research showing how immersion, tracking fidelity, visual encumbrance, and interaction technique shape usability and evaluation practices [71]. At the collaboration layer, mixed reality has long been positioned as a promising medium for CSCW because it can preserve real-world referents and face-to-face cues while introducing shared virtual objects [72]. Collaboration is developed further in the following subsections.

Recent accounts frame spatial computing not only as technical convergence (computer vision, sensor fusion, AI), but as a capability that can reshape how people work, learn, and communicate as interaction moves beyond screen-bound interfaces [70]. This framing becomes particularly relevant for domestic settings such as kitchens, where coordination depends on ongoing activity, partial attention, and shared understanding of objects, workflows, and constraints in the immediate environment.

From an infrastructure perspective, this literature highlights cloud-based spatial computing as enabling shared access to spatial content across devices and locations, including real-time collaboration on shared spatial data, models, and applications [70]. In parallel, edge computing is presented as supporting low-latency responsiveness and contextual processing by keeping computation close to sensors and users, while also offering privacy advantages by reducing the need to transmit sensitive data and enabling offline operation when connectivity is limited.

At the interaction layer, spatial computing research emphasizes AI/ML capabilities that expand what systems can interpret and support in real time: object recognition and scene understanding for spatial mapping, gesture and pose recognition for natural input, and mapping/reconstruction approaches such as SLAM for localization and environmental alignment. These capabilities are often tied to opportunities for adaptive interfaces and predictive assistance that tailor interaction to user and environmental cues.

Zhang’s “A Tale of Two Interfaces” [68] synthesizes these discussions into a set of persistent tensions, arguing that spatial computing is not a straightforward translation of familiar 2D UI

patterns into 3D scenes (Figure 2.3). He articulates five paired conflicts (objectives, immersion, mediation, attention, and affordance) contrasting mainstream digital interface conventions with architectural and spatial logics. A conflict of objectives distinguishes task-oriented interfaces optimized for discrete command–response loops from activity-oriented spatial arrangements that support unfolding routines across time and scale. This framing connects to a conflict of immersion: conventional interfaces are operated as external objects, whereas architectural experience is enveloping, situated, and viewpoint-dependent.

Zhang extends these contrasts into implementation-relevant concerns. A conflict of mediation emphasizes that interaction remains structured by sensing and translation layers that convert intent into computable input and output; even when spatial interaction feels direct, the mediation chain persists, creating design challenges around legibility, accountability, and user understanding of system behavior. A conflict of attention contrasts engagement-seeking digital incentives with architectural environments that often support activity as an ambient background; if attention-capture norms carry into spatial computing, domestic environments risk becoming saturated with anchored content, producing overstimulation and eroding the calm periphery everyday activity relies on. Finally, a conflict of affordance contrasts digital minimalism with the situated, material affordances of physical environments, implying that overly abstract spatial UIs may strip away contextual cues that make action discoverable and learnable in place [68].

Taken together, these tensions motivate a domestic spatial computing agenda that prioritizes activity-centered organization, embodied coordination, legible mediation, careful attention management, and context-sensitive affordances, rather than simply re-hosting screen-era interaction patterns in volumetric scenes.

A further implication of treating space as the interaction substrate is the possibility of *cross-realm continuity*: interfaces can support transitions and coordination across physical and virtual contexts without forcing users into a single isolated “world.” Speculative work on phygital bridging interfaces (e.g., portal-like thresholds) highlights needs such as maintaining persistent context across realms, reducing workflow fragmentation through unified control surfaces, and placing spatial interfaces strategically within everyday environments to support co-creation and social presence.

Bringing these threads together, spatial computing can be read as an interaction-level complement to connected-home research. Where smart home work clarifies the socio-technical con-

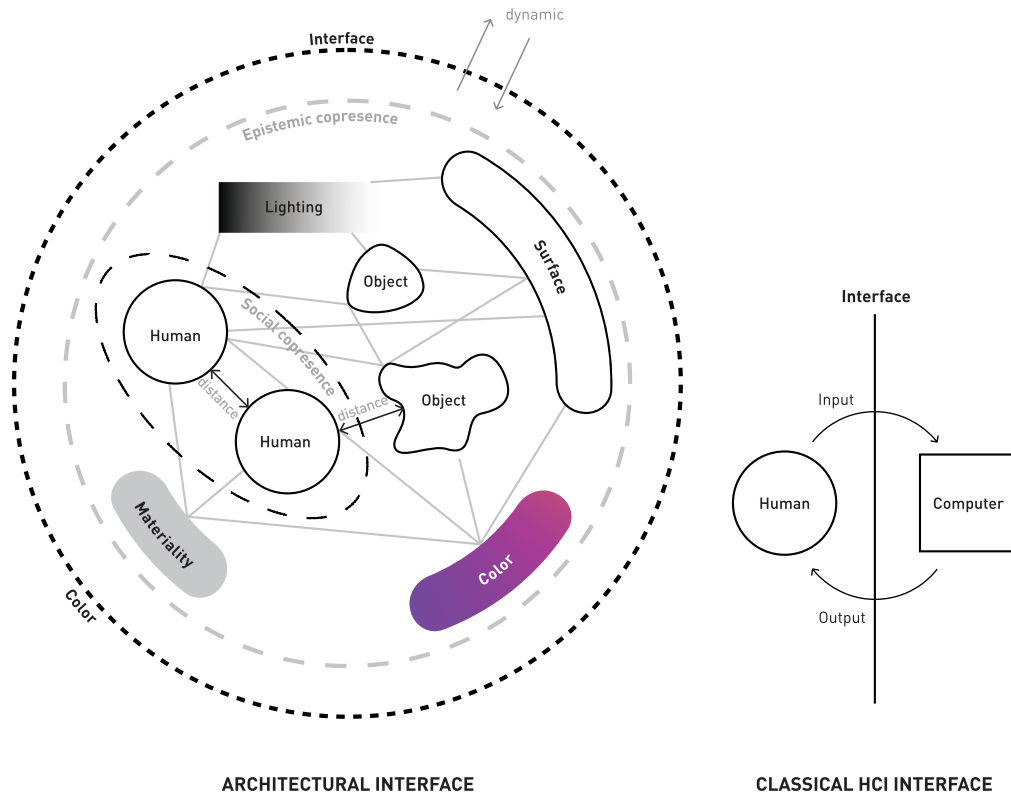


Figure 2.3: Comparison of HCI and architectural interface logics, adapted from [68].

ditions of adoption and the politics of domestic control, spatial computing offers mechanisms to make device ecologies legible and actionable *in situ*. Rather than concentrating interaction in dashboards or phone apps, spatially embedded interfaces can align information, guidance, and communication with everyday activity zones, supporting coordination through embodied reference, shared spatial cues, and context-sensitive presentation [69,73]. For kitchens in particular, combining connected appliances with spatially anchored guidance and collaborative XR interaction suggests a shift from automation-centric “smartness” toward interaction designs that support movement, multi-actor teamwork, and moment-to-moment decision making at the work surface [53,74].

2.2.3 Spatial & Smart Kitchen

HCI research on smart kitchens has increasingly framed cooking not only as a task to optimize, but as a site of communication and coordination, and the literature offers a range of interaction techniques and system configurations that are designed, prototyped, and evaluated. This work has also shifted toward the broader notion of the *connected home*, where the transformation is not only autonomous appliance control but the expansion of communication, media, and networked social relations across domestic space [75]. In parallel, systems-oriented surveys describe the smart home as an assemblage of devices, sensing, computation, and service layers, while highlighting recurring challenges around interoperability, security, and reliability as capabilities scale [76]. Together, these perspectives position the home as a socio-technical ecology shaped as much by routines and meaning-making as by technical infrastructure [75, 76].

Terrenghi et al. introduce *Computer-Supported Cooperative Cooking* (CSCC), adapting CSCW concerns (coordination, awareness, and articulation work) to the time-critical, materially grounded practice of cooking. CSCC foregrounds how multiple actors (co-located or remote) collaborate through shared artifacts (e.g., recipes, timers, shopping lists), distributed roles, and continuous negotiation of attention between food, tools, and social interaction. From a remote-communication perspective, CSCC systems also emphasize lightweight presence cues, turn-taking and handoff across stages of a meal, and the sharing of tacit know-how that is difficult to convey through text alone. Related cooperation and collaboration work using spatial interaction is discussed further in Section 2.2.4.

PersonalChef [53] addresses co-cooking by introducing a kitchen equipped with an interactive counter and a dual-display, dual-perspective setup. The system supports novices through a combination of text, visuals, and audio distributed across a counter display and a second display behind the stove. This setup allows beginners to observe expert technique through a face-to-face perspective while also interacting with procedural guidance on the work surface. Scheible et al. similarly conceptualize a smart kitchen that integrates multiple assistive systems to support interdisciplinary research, using in-situ projection and AR approaches to explore different interaction strategies [77].

In *Living Cookbook* [78], Terrenghi explores how cooking experiences can be recorded and shared through an interactive digital cookbook, treating documentation and sharing as part of domestic sociability and learning. More recently, Zhou et al. evaluate multi-user tablet interfaces in home cooking scenarios, comparing collaborative performance in triadic family

tasks [79].

Changing conditions of domestic life also create new situated practices for HCI to study. Weber et al. explore the design space of virtual cook-along sessions mediated through videoconferencing platforms, an activity that gained prominence during the COVID-19 pandemic [80]. Their study proposes six design implementations addressing tensions such as automation versus user control, hands-free interaction, privacy, preserving experiential qualities of cooking, reducing stress, and supporting conjoint activities. This framing and prototyping approach aligns closely with the aims and design strategy of our *Kitchen Talks* study (Section 4.1).

2.2.4 Spatial Collaboration

Extended Reality (XR) technologies have expanded how people collaborate across physical and digital spaces. Compared to conventional CSCW systems, collaborative XR enables embodied presence, spatial referencing, and shared virtual objects anchored to environments, opening collaboration possibilities that are difficult to achieve through screen-based tools. This subsection reviews theoretical foundations, technological developments, and emerging research directions in collaborative XR.

Theoretical Foundations and Historical Context The theoretical foundations of collaborative XR build on CSCW frameworks for understanding collaborative activity. Grudin’s time–space matrix distinguished collaborative systems by temporal (synchronous/asynchronous) and spatial (co-located/remote) dimensions [81]. As MR collaboration matured, however, researchers argued that these dimensions were insufficient to capture cross-reality collaboration. Ens et al. [82] provide a comprehensive review of three decades of collaborative MR research and reconcile it with CSCW theory. They document how the field evolved from early engineering constraints (tracking, registration, display) toward human factors and interactional dynamics. They extend CSCW framing with dimensions specific to MR collaboration: **symmetry/asymmetry** (differences in roles and capabilities), **artificiality** (the blending of physical and virtual elements), and positioning along the reality–virtuality continuum. They also identify four future research directions: complex collaboration structures, transitional interfaces, empathic collaboration, and collaboration beyond physical limits. These themes structure much of the subsequent literature.

Asymmetric Collaboration: Bridging Heterogeneous Realities A prominent line of work examines **asymmetric collaboration**, where collaborators use different devices, perspectives, or interaction modalities. This reflects real deployment conditions in which XR ecosystems are heterogeneous. Recent work addresses both technical and design challenges of bridging these differences. De Bäuw et al. [83] propose conformal mapping to preserve spatial relationships across workspaces of different physical dimensions. Kumaravel et al. introduce **TransceiVR**, enabling desktop collaborators to interact with VR users at the platform level without modifying application source code [84]. Empirical studies show that asymmetry can be both a constraint and a resource. Schröder et al. [85] document how dyads transition between modalities based on task demands, and how complementary perspectives can improve problem solving. At the same time, asymmetry creates challenges around spatial awareness and deixis, since pointing and reference become harder when collaborators perceive the shared space through different representational and interaction regimes.

Remote Expert Guidance and Adaptive Instruction Remote expert guidance is one of the most mature application areas for collaborative XR, particularly in maintenance, assembly, training, and technical support. Recent systems move beyond video calls toward context-aware instruction and responsive spatial annotation. Zhang et al. [86] demonstrate instruction generation adapted to user skill and task progress, reporting lower error rates compared to static manuals. Baumgartner et al. [87] integrate responsive digital twins with spatial annotations that appear registered in a local AR view. Across this literature, adaptive guidance is repeatedly shown to reduce cognitive load, completion time, and errors compared to manuals or conventional video communication. Marques et al. [88] emphasize that AR guidance must carefully balance information richness and cognitive simplicity, motivating interfaces that dynamically adjust the level of detail based on expertise and task complexity. Related evaluation studies also show that design details such as view stabilization, spatial awareness cues, and reliable anchoring strongly shape coordination and effectiveness [73, 89]. These findings reinforce that spatial computing is not simply “content in space”, but the design of spatial *coordination* across bodies, viewpoints, and constraints.

Communication Cues and Immersive Co-presence A core challenge in remote collaboration is establishing co-presence. Collaborative XR systems address this by spatializing collaborators and enabling embodied cues such as gaze, gesture, and movement. Gaze visualization,

in particular, has been shown to support coordination and reduce miscommunication. Jing et al. [90] demonstrate that bi-directional gaze-ray visualization improves spatial awareness and task performance. Kim et al. [91] show that combinations of cues (gaze, pointing, hand visualization, sketching) are more effective than any single cue, suggesting the value of multimodal communication channels. Work on embodiment and avatar representation further indicates that representation asymmetries can affect social presence and collaboration dynamics. Reski et al. [92] show that mismatched embodiment (e.g., one collaborator as a full avatar, another as a cursor) can create inequities in interaction and co-presence, motivating representation strategies that sustain social balance across heterogeneous setups.

Transitional Interfaces and Cross-Reality Workflows As collaborative XR systems mature, an emerging theme is that effective collaboration often involves moving across modalities rather than remaining inside one “reality.” Transitional interfaces aim to preserve continuity of work state as users shift between desktop, AR, and VR based on task demands. Jetter et al. [93] describe transitional blended realities connecting video conferencing and MR to support these handoffs, with users gravitating to different modalities across phases (e.g., brainstorming in VR, refinement on desktop, validation in AR). Tahmid et al. [94] extend this logic to academic work through CoLT, showing how hybrid synchronous-asynchronous workflows can support distributed collaboration across time zones.

Asynchronous Collaboration and Temporal Flexibility Most collaborative XR research has focused on synchronous interaction, but there is growing recognition that asynchronous modes are essential for realistic deployment. Mayer et al. [95] show that MR “ghost” recordings of expert actions outperform conventional video instructions because they preserve spatial context and allow replay from multiple perspectives. Cho et al.’s *Time Travellers* system supports hybrid synchronous and asynchronous collaboration through recording and temporal navigation of actions and annotations [96]. At the same time, asynchronous spatial collaboration introduces unresolved challenges around versioning and conflict resolution. Pintani et al. [97] address this in CIDER through layered editing and branching strategies, but the problem remains open for complex spatial scenes and larger groups.

2.2.5 Relevance for the Thesis: Spatial Computing and Extended Reality

Spatial computing and XR research provide the conceptual and technical vocabulary for thinking with space (anchoring, embodied reference, multimodal input, and cross-reality collaboration) which directly informs the interaction metaphors explored in this thesis. The state-of-the-art gap is that much XR work is still shaped by general taxonomies, lab-driven interaction technique studies, or collaboration in professional and industrial contexts, with limited attention to domestic routines where attention is fragmented and where presence must be lightweight, safe, and negotiable. This thesis brings spatial computing into the kitchen not as a “3D UI upgrade”, but as a way to examine how communication, coordination, and co-presence can be shaped through spatially embedded mediation under everyday domestic conditions.

2.3 Digital Prototyping and XR for UX

This section surveys user-insight and prototyping approaches that are commonly used when designing and studying emerging interactive systems: Wizard-of-Oz, video prototyping, living labs, and virtual prototyping. These methods differ in how they trade off implementation effort, experiential fidelity, and ecological validity. We review all four to establish methodological context, but the empirical studies in this thesis rely primarily on Wizard-of-Oz-style staging and virtual prototyping, rather than on long-term living lab deployments or non-interactive video demonstrations. This positioning reflects a practical constraint in domestic XR research: many of the most interesting interaction concepts depend on sensing, connectivity, and spatial registration that are difficult to deploy robustly in real homes, yet they still need to be experienced in context to be evaluated meaningfully.

2.3.1 Wizard of Oz

Wizard-of-Oz (WoZ) prototyping is a widely used method for early-stage user studies in HCI. In a WoZ setup, participants interact with what appears to be a functioning system, while key system behaviors are enacted behind the scenes by one or more human operators. Rather than treating this as deception, HCI work typically uses WoZ as a controlled technique to prototype interactive behavior before reliable automation exists. This makes it possible to evaluate interaction concepts, task flows, and user expectations with relatively low implementation cost and within short iteration cycles [98]. WoZ is particularly useful when the core research question

2.3 DIGITAL PROTOTYPING AND XR FOR UX

concerns experience, communication, or coordination—and when the technical implementation (e.g., robust sensing, intent recognition, or adaptive assistance) is either unavailable or would dominate the research effort.

At the same time, WoZ introduces methodological responsibilities. The operator needs scripts and response rules that keep behavior consistent across participants, and the study design must acknowledge that the “system” is being performed rather than executed. In domestic settings, this matters even more: participants quickly notice timing, attention demands, and interruptions, so the credibility of the prototype depends on careful orchestration. When designed well, WoZ allows researchers to test near-future interaction ideas while still observing how people make sense of system behavior, negotiate control, and adapt their practices.

2.3.2 Video Prototyping

Video prototyping uses pre-recorded footage to depict an interaction concept, either by showing users interacting with a prototype or by illustrating how a system might behave in a plausible scenario. Because it can combine live action with drawing, animation, photography, stop-motion, and post-production effects, video prototyping supports rapid exploration of design space and can communicate speculative capabilities that are not yet technically feasible [99]. It has a long tradition in HCI for envisioning novel interfaces and eliciting early feedback, especially when working with stakeholders who need to grasp an experience quickly [100].

However, video prototypes are fundamentally non-interactive, which limits what they can reveal about embodied action, timing, breakdowns, and user appropriation. They tend to foreground a designed storyline and can therefore under-represent uncertainty and negotiation—precisely the dynamics that often shape domestic technology use. For this reason, video prototyping is often most effective as a complementary method: it can be used to set a shared frame in participatory design, to compare alternative concepts, or to enrich WoZ studies by illustrating system behaviors that would otherwise be difficult to stage [101].

2.3.3 Living Labs & Smart Kitchen Prototypes

Living lab research offers a contrasting approach: rather than staging interaction in short sessions, living labs instrument domestic-like environments with sensors and logging infrastructure to support longitudinal observation and experimentation. Building real-life smart home or

2.3 DIGITAL PROTOTYPING AND XR FOR UX

smart kitchen prototypes is resource-intensive and typically requires interdisciplinary expertise in sensing, networking, software systems, and environmental setup. Deploying such infrastructures in actual homes can be difficult due to installation constraints, privacy concerns, maintenance, and the fragility of sensing in everyday conditions. In addition, many ubiquitous computing applications depend on training data and stable configurations to recognize activities reliably, which is challenging to achieve in naturalistic settings [46]. These factors help explain why purpose-built living laboratories have been attractive to ubicomp researchers.

A living lab can be understood as a naturalistic environment augmented with sensing and observational technologies, enabling researchers to study behavior in context while retaining experimental control [45, 102]. PlaceLab is a prominent example, instrumented across rooms to collect multi-modal data for studying everyday activity [46, 103]. Alongside these, several projects built high-fidelity kitchen prototypes in laboratory settings to evaluate interaction concepts. Siio et al. developed a “Kitchen of the Future” with embedded cameras, microphones, LCD screens, and foot switches [104]. La Cantina at the MIT Media Lab served as a research environment for exploring kitchen augmentation and interaction design [15, 57, 58, 105–108]. Chi et al. built a lab-based smart kitchen with displays, appliance and cabinet sensors, RFID readers, and sensing equipment to recognize cooking activities [109]. Ambient Kitchen at Newcastle University represents one of the most densely instrumented replicas, combining floor pressure sensing, projectors, cameras, RFID tags and readers, accelerometers, and networked computing infrastructure [110]. Importantly, Oliver et al. describe this environment not only as a test bed, but also as a design tool for designers and users, and as an observatory for collecting sensor data [110].

Living labs and instrumented kitchen replicas provide high ecological richness compared to conventional lab studies, but they also come with trade-offs: high cost, limited portability, and difficulty generalizing beyond the specific instrumented environment. For emerging XR and spatial interaction concepts, living labs can be methodologically powerful, yet they are not always feasible—especially when the research aims to iterate rapidly across multiple design alternatives and study experience-level questions rather than long-term infrastructure performance.

2.3.4 Virtual Prototyping

Virtual prototyping presents a design concept inside a virtual environment, typically using head-mounted VR, AR/MR devices, or CAVE-like systems. As immersive technologies have

2.3 DIGITAL PROTOTYPING AND XR FOR UX

become more accessible, virtual prototyping has gained traction as a way to evaluate spatial interaction, embodied navigation, and experience qualities that are difficult to represent with sketches or screen-based prototypes. It also allows controlled experimentation: researchers can standardize layouts, tasks, and system responses while still giving participants a sense of being “in” the space. When paired with logging and sensing, virtual prototypes can capture behavioral traces and interaction breakdowns in ways that are difficult to observe in physical mock-ups.

At the same time, virtual prototyping comes with practical and epistemic limitations. It can require substantial effort in modeling, scripting, rendering, and interaction implementation, and many prototypes depend on simulation rather than physical causality. The lack of physicality may reduce how participants handle objects, and simulations may omit important real-world effects such as inertia, collision, or material resistance [111]. Comparative work has therefore examined how well immersive environments approximate physical equivalents. Heydarian et al. report differences in performance across some test activities, while also noting strong feelings of presence in VR [112]. Bordegoni et al. demonstrate that combining modalities (visual, haptic, audio) can make virtual prototypes viable substitutes for physical artifacts in certain product contexts [113]. Bruno et al. similarly use VR-based setups for user interface design in a product domain, suggesting that virtual prototyping can support design decision-making when physical prototyping is costly or slow [114].

Industry adoption further illustrates why virtual prototyping matters for UX. Automotive companies were early adopters of VR in product development, with use tracing back to the mid-1990s, reflecting the complexity of vehicles and the need to coordinate multidisciplinary design workflows [115]. Zimmermann describes VR-aided design as supporting multiple milestones in product development, including virtual prototypes, digital mock-ups, assembly simulation, and data-control models [115]. In UX research, VR has been used for staged and simulated scenarios that would be difficult, expensive, or unsafe to study otherwise. Paredes et al. report WoZ-supported VR studies for speculative passenger experiences [116], and other work explores driving-related UX through simulation [117–119]. Volvo’s ARCAR tool similarly exemplifies how VR can support scenario exploration and investigation of concerns such as distraction, visibility, safety, reachability, and perceived quality [120]. Comparative evaluation by Pettersson et al. shows that UX questionnaire results can differ between VR simulation and field studies, highlighting both the promise and the limits of immersive prototyping as a proxy for real-world use [121].

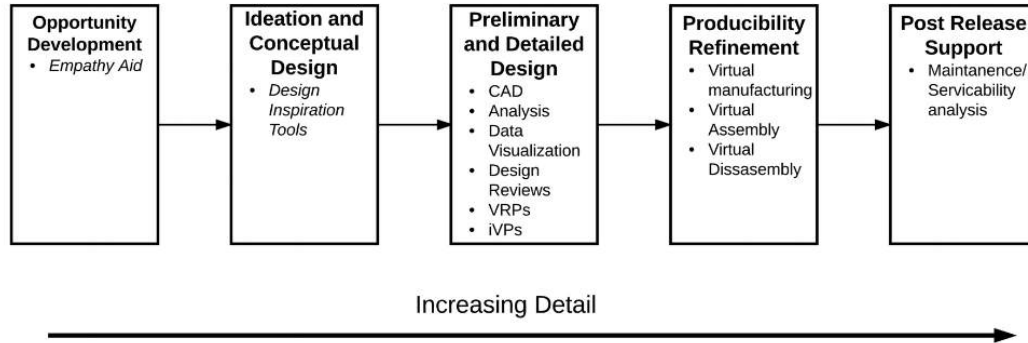


Figure 2.4: Overview of the design process with applications of VR previously explored. [122]

Because immersive systems require experiential testing, rapid prototyping tools have also become part of the methodological landscape. ExProtoVAR supports non-programmers in producing interactive AR prototypes [123]. Unity MARS introduced simulation and proxy-based workflows that bring real-world environment logic and sensor data into authoring, enabling designers to test AR interactions without repeated deployment steps [124,125]. Unity Hubble is positioned as a collaborative design tool with voice chat, avatars, annotations, and shared review of spatial ideas [126]. Tools such as ShapeXR (and earlier Tivori) enable designing and prototyping directly in VR, supporting storyboarding, collaboration, and passthrough-based work that can connect virtual ideation with real-world context [127,128]. Methodological reflections in the HCI literature also caution that VR integration into design processes is not automatic: Müller et al. discuss limitations and propose prototyping guidelines that emphasize interaction and visualization strategies [129]. A recurring theme is that, while point solutions exist, holistic toolchains that span from early needfinding and ideation through prototyping and user insight collection remain underdeveloped.

2.3.5 Relevance for the Thesis: Digital Prototyping and XR for UX

This literature clarifies why studying near-future domestic XR experiences cannot rely only on deployable smart-home infrastructures or fully functional implementations. In kitchens, interaction concepts often depend on sensing, context awareness, and spatial anchoring that are fragile in real homes, yet they still need to be experienced to evaluate how they shape attention, coordination, and social negotiation. The methodological gap is that many XR prototyping tools and evaluation practices remain tool-centric or scenario-demonstrative, prioritizing what can

be built and measured in controlled conditions over domestic ecological validity: the friction of routine breakdowns, the messiness of shared space, and the subtle negotiations around control and presence that emerge between household members.

This thesis responds by treating immersive prototyping as a research instrument rather than only a design convenience. I build a virtual kitchen environment (KitchXR) as a reusable context for enacting domestic routines and staging prospective interactions, allowing concepts to be explored as situated experiences rather than abstract features. Across the studies, Wizard-of-Oz-style staging supports speculative system behaviors that would otherwise require brittle automation, while virtual prototyping enables iterative refinement of spatial interaction metaphors and the collection of user and designer insight under consistent conditions. Taken together, these methods make it possible to evaluate mediated spatial interaction in kitchens in a way that remains speculative in implementation, but grounded in experience: attentive to everyday constraints, to negotiation, and to what it actually feels like to coordinate and communicate while doing domestic work.

2.4 Reflection and Synthesis

This chapter reviewed literature across domestic studies, ubiquitous and smart home computing, spatial computing, collaborative XR, and digital prototyping, with a particular focus on how interaction unfolds in the kitchen as a socially and spatially complex environment. Rather than treating these bodies of work as separate trajectories, the review traced how they collectively frame the kitchen as a site where embodied action, shared routines, technological mediation, and social negotiation are tightly intertwined.

Across this literature, a recurring tension emerges. Domestic computing research richly documents the social, cultural, and relational character of kitchen life, yet often operationalizes technology through automation, efficiency, or sensing-oriented paradigms. Conversely, spatial computing and XR research offers powerful interaction mechanisms for embodied presence, anchoring, and collaboration, but is predominantly developed and evaluated in professional, industrial, or laboratory contexts, with limited attention to everyday domestic constraints such as divided attention, safety, mess, and informal coordination. Methodological work on prototyping further highlights that while immersive tools enable experiential exploration of future systems, they rarely foreground domestic ecological validity or sustained social negotiation.

The gap identified in this chapter lies in the absence of design-oriented research that brings spatial computing into the kitchen as a lived, negotiated, and socially embedded setting. Existing work rarely addresses how mediated spatial interaction might support communication and co-presence without turning the kitchen into either a productivity interface or a speculative showcase. This thesis responds to this gap by investigating how spatially mediated interaction can be designed, prototyped, and studied within domestic kitchens, and by articulating both the interaction concepts and methodological tools needed to explore such futures. These concerns directly motivate the research program and research questions developed in the following chapters.

Chapter 3

Methodology

This thesis investigates the emerging design space of **Mediated Spatial Interactions** within the domestic kitchen, a context characterized by complex embodied activity, sensory richness, and evolving social needs. Traditional Human–Computer Interaction (HCI) methodologies, which often prioritize efficiency, usability, and optimization, offer limited capacity to address the nuanced social dynamics and “wicked problems” [25] associated with future domesticity. Engaging meaningfully with this domain therefore requires a methodological shift from validating predefined solutions toward exploring possible futures.

To navigate this complexity, the research draws heavily on a **Programmatic Design Research** philosophy [23]. The individual studies presented in this thesis are not isolated user evaluations; rather, they are interrelated *experiments* conducted within a unified *research program*. This structure enables artifacts such as the *KitchXR* tool and speculative *Portals* to operate as **exemplars** [24]. Instead of striving for universal principles, these exemplars function as material arguments that open up the design space of hybrid domesticity, demonstrating through constructive engagement how mediated spatial interaction may be configured and experienced.

3.1 Methodological Approach

This chapter outlines the methodological stance adopted in this thesis and clarifies how design serves as both the subject and vehicle of inquiry. It introduces the overarching research philosophy, positions the work within established design research frameworks, and explains how

3.1 METHODOLOGICAL APPROACH

the studies collectively operate as experiments within a unified research program. Through this lens, the chapter establishes how the thesis systematically investigates mediated spatial interaction in domestic kitchens.

3.1.1 Methodological Paradigm: Design-Oriented Research

This thesis follows the paradigm of **Design-oriented Research**, as articulated by Fallman [22]. In this approach, designing and building artifacts serve as the primary mode of inquiry, enabling the exploration of emerging technologies and evolving social contexts through practice rather than solely through analysis. This perspective differs from *Research-oriented Design*, which aims to produce finalized, commercially deployable systems. Here, the goal is the generation of knowledge. The artifacts developed in this thesis, such as the *KitchXR* tool and the speculative *Portal* and *Magmemo* concepts, act as epistemic instruments that surface otherwise hidden complexities of hybrid domesticity. Through their design, deployment, and reflection, these artifacts “open up” the design space of mediated spatial interaction, revealing how future domestic communication and social practices might take shape.

3.1.2 Research Positioning: The Interaction Design Triangle

Following Fallman’s [26] model, the activities in this thesis are positioned within the *Interaction Design Research Triangle*, traversing the three activity areas of **Design Practice**, **Design Exploration**, and **Design Studies**:

- **Design Studies (Theory & Context):** Represented in the contextual inquiries of *Kitchen Talks* (Paper I), where the focus is on building an intellectual understanding of domestic social frictions and grounding the research program in existing social practices.
- **Design Practice (Methods & Tools):** Represented in *KitchXR* (Paper III), which intervenes in professional design workflows to provide constructive tools for user insight, treating the design process itself as a site of inquiry.
- **Design Exploration (Speculation):** Represented in *Anchoring Hybrid Presence* (Paper II) and *Hybrid Connected Kitchens* (Paper IV), which use speculative design to materialize thought experiments and provoke discussion on future sociality by envisioning alternative domestic realities [26].

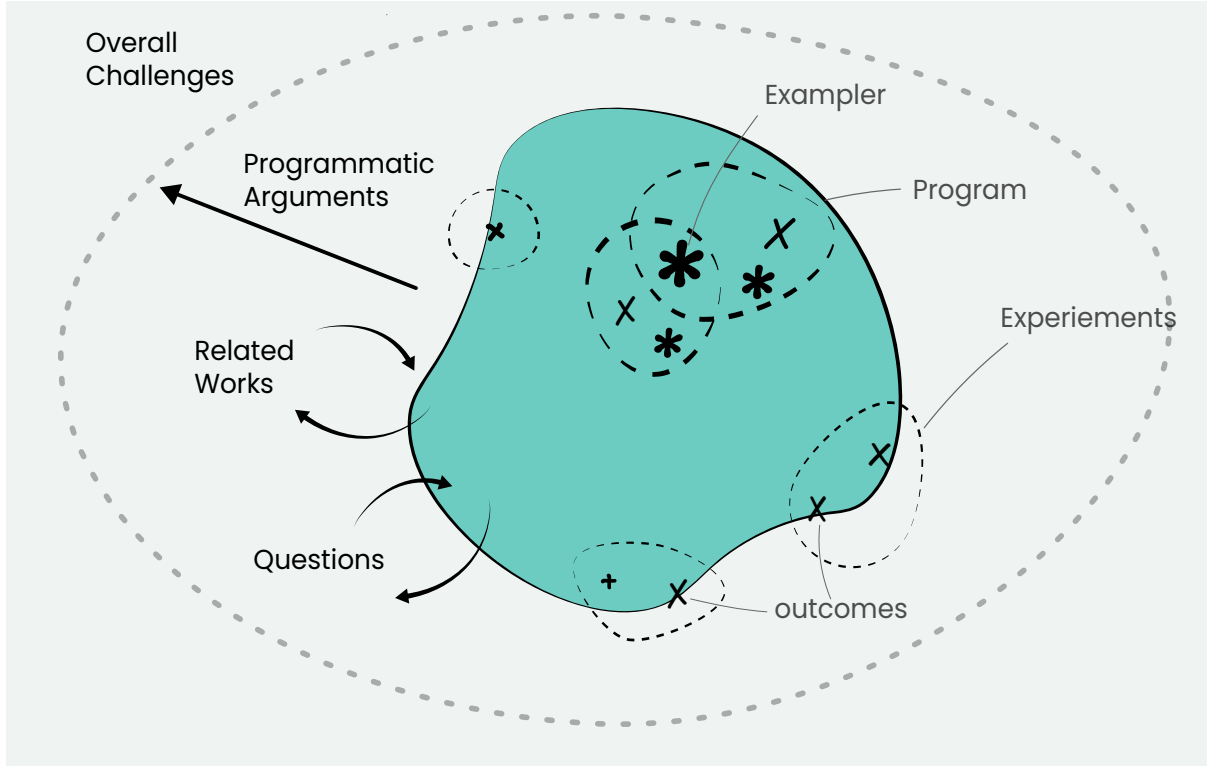


Figure 3.1: illustration of studies in programmatic design research framework scheme. Green area represents the program as morphing scope.

3.1.3 Research Framework: Programmatic Design Research

Given the complexity of the domestic domain and the inherent ambiguity of the design object, this thesis is heavily influenced by the framework of **Programmatic Design Research** [23,24], which it adapts to the demands of this inquiry rather than adhering to as a fixed template. It was chosen because the research questions demand generative, artifact-driven inquiry rather than descriptive or explanatory methods. The domestic kitchen, as I conceptualized such a site where embodied labour, social negotiation, and technological mediation intersect, constitutes what Rittel and Webber [25] describe as a wicked problem: one that resists decomposition into testable hypotheses and instead requires exploration through design.

This framework provides the methodological flexibility required to navigate an evolving research trajectory, enabling the deployment of diverse methods under a shared and coherent research agenda. Within this framework, each study is framed as a design **experiment** intended to materialize and interrogate the core research concerns articulated through a unifying **Research Program**. Rather than serving as isolated inquiries, the experiments collectively

3.2 METHODOLOGICAL STRATEGIES

contribute to the program, which is continuously refined through the insights and outcomes generated across studies. The following quote from “Designing everyday computational things”, illustrates the programmatic design mechanics.

“... we use experimental design as a way of finding a path into a problem that is poorly understood at the outset. A design program serves as a starting point and a framework for such investigation.” [130]

Accordingly, this research is structured not around a single, linear hypothesis, but as a *Research Program* entitled **Mediated Spatial Interactions in Domestic Kitchens**. By conducting multiple design *experiments* within a unified program, the research explores a “space of possibilities” [23] from complementary perspectives. In line with the principles of *Exemplary Design Research* [24], the artifacts developed through these experiments are treated as **exemplars**. These exemplars operate as material arguments, demonstrating how domestic communication and collaboration can be reconfigured through spatial computing technologies.

This programmatic structure allows insights from early contextual and exploratory studies (Paper I) to inform the development of design tools and infrastructures (Paper III), which subsequently enable the systematic evaluation of speculative future concepts (Papers II and IV). The research program is grounded in the premise that the future of domestic computing extends beyond two-dimensional screens toward “**Computational Composites**” [131], in which physical architecture, digital systems, and social practices are tightly interwoven. To operationalize this premise, the program is pursued through two parallel and complementary tracks of inquiry, enabling triangulation of the research questions:

- **The Constructive Track (Means):** The development of design tools, workflows, and technical infrastructures for prototyping complex spatial experiences (addressing RQ2).
- **The Speculative Track (Ends):** The envisioning and evaluation of future interaction metaphors aimed at supporting domestic communication and mitigating social isolation (addressing RQ1).

3.2 Methodological Strategies

Within this programmatic structure, two primary strategies are utilized to manifest the goals of the research:

3.2 METHODOLOGICAL STRATEGIES

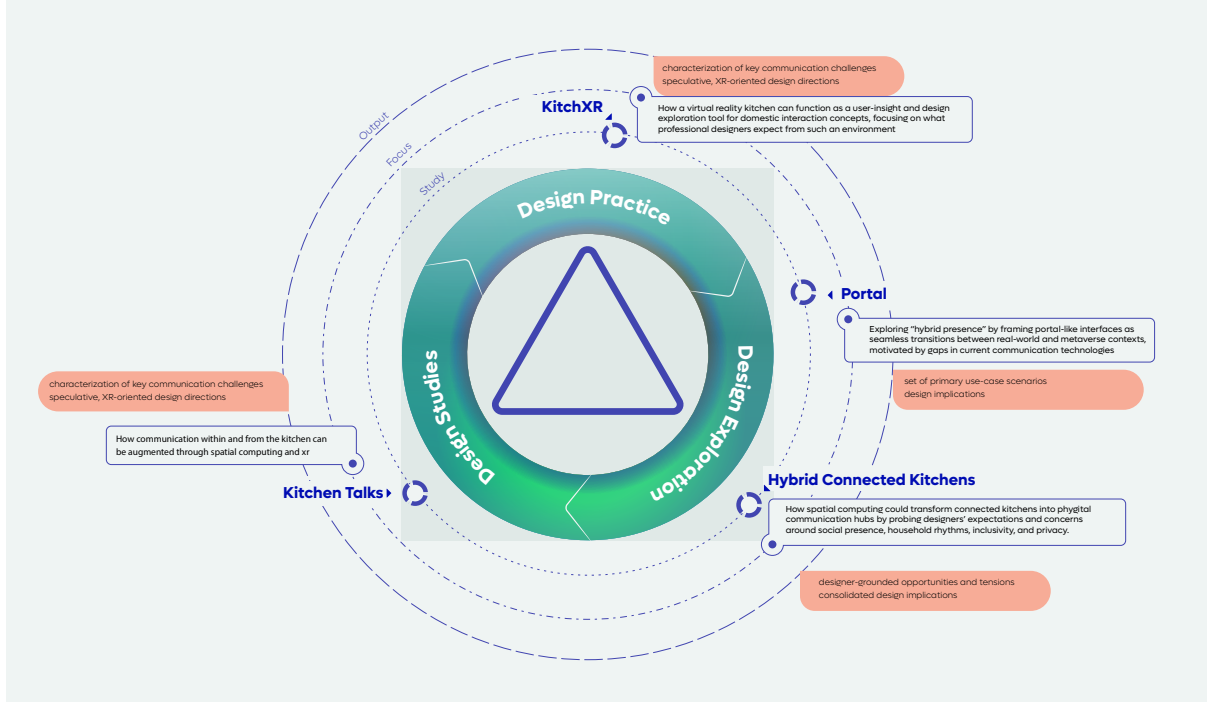


Figure 3.2: The diagram of this thesis structure in Fallman's Design-oriented Research perspective

3.2.1 Speculative Design (The "Ends")

To investigate **RQ1 (Future Forms)**, this research utilizes **Speculative Design** [132,133]. Since the technologies for seamless hybrid kitchens (e.g., lightweight mixed reality glasses, volumetric capture) are not yet ubiquitous, empirical user testing of commercial products is premature. Instead, I create "material speculations" [134] such as the *Anchored Portals* (Paper II) and *Phygital Companions* (Paper IV). These artifacts are not proposals for immediate production but are designed to provoke debate (Challenging the current paradigm of "screen-based" domesticity), to probe values (investigating how users trade off privacy, safety, and social presence in hybrid spaces), and to explore preferred states (Moving beyond "what is possible" to determine "what is desirable").

3.2.2 Constructive Design Research (The "Means")

To investigate **RQ2 (Design Tools)**, this research employs **Constructive Design Research** [135]. This strategy involves building working systems to understand the complexity of the design material. The development of **KitchXR** (Paper III) serves as the primary artifact in

this track. By constructing a custom Virtual Reality (VR) insight tool, this research reveals the specific methodological gaps in standard design software. The artifact itself is a research contribution, demonstrating that designing for spatial domesticity requires tools that can simulate **contextual friction** (mess, noise, and biometrics) rather than just visual aesthetics.

3.3 The Research Process

This thesis follows a programmatic, iterative research process structured across three inter-related phases. Rather than a linear progression, each phase builds upon insights, limitations, and design needs that emerged from the previous one. Together, these phases move from understanding everyday domestic practices, to constructing methodological means for design inquiry, and finally to exploring and evaluating speculative interaction concepts. While this section outlines the overall research logic and sequencing, the specific methodological details of each study are presented in depth in Chapter 4, within the corresponding paper sections.

3.3.1 Phase 1: Contextual Inquiry (The Problem Space)

The first phase focuses on understanding the lived realities of domestic kitchens as social, communicative, and activity-rich environments. Through qualitative inquiry, this phase investigates how people experience connection, coordination, and presence in everyday kitchen routines, and where existing communication technologies succeed or fail. Rather than starting from a technological solution, this phase establishes the problem space by foregrounding users' values, frictions, and tacit practices.

- **Associated Study:** *Kitchen Talks* (Paper I).
- **Method:** Participatory workshops and qualitative interviews.
- **Outcome:** Identification of a central friction: participants desire ambient, lightweight forms of connection but actively reject the attentional demands and intrusiveness of current 2D, screen-based communication tools. This phase establishes the need for alternative, spatial forms of mediation.

3.3.2 Phase 2: Tool Construction (The Methodological Bridge)

The second phase addresses a methodological gap revealed by Phase 1. Initial attempts to explore spatial alternatives exposed the limitations of existing design and prototyping tools, which lack the ability to represent domestic spatial context, embodied interaction, and situated use. This phase therefore focuses on constructing a research instrument that can support experiential, spatially grounded design inquiry.

- **Associated Study:** *KitchXR* (Paper III).
- **Method:** Iterative software development informed by expert review.
- **Rationale:** Existing prototyping tools were insufficient for exploring future kitchen interactions in context. The development of *KitchXR* provides a spatially situated, immersive environment that functions as a methodological bridge between empirical insight and speculative design.

3.3.3 Phase 3: Speculative Exploration (The Solution Space)

The third phase leverages both the empirical insights from Phase 1 and the methodological infrastructure developed in Phase 2 to explore possible futures of mediated interaction in domestic kitchens. This phase adopts speculative design as a means to articulate, test, and critique alternative interaction paradigms, focusing on how spatial mediation might support presence and communication without demanding focused attention.

- **Associated Studies:** *Anchored Hybrid Presence* (Paper II) and *Hybrid Connected Kitchens* (Paper IV).
- **Method:** Speculative prototyping and design critique.
- **Outcome:** Articulation of *Anchoring* and *Phygital Presence* as design concepts for ambient, negotiable communication in domestic space, later consolidated within the thesis's design suggestions

The Continuous Fuzzy Kano Model as a Shared Evaluation Method In this thesis, across three of the studies, reported as separate papers, the Continuous Fuzzy Kano Model

(CFKM) is used as the common method for evaluation to understand how the prototype features are perceived by users. This approach provides a consistent analytical instrument that allows the studies' evaluative findings to be expressed and compared in the same terms. This section introduces methodological grounds of that choice: the Kano model from which the approach descends, the limitations that motivate its fuzzy and continuous refinement, and the procedure by which the CFKM was applied within each study.

The approach descends from the Kano model of customer satisfaction [136,137], which classifies attributes by the asymmetric way each affects satisfaction using a paired functional/dysfunctional questionnaire and a standard evaluation table [138]. The classifications are named as must-be (expected; their absence dissatisfies, but their presence is taken for granted), one-dimensional (satisfaction rises with fulfillment), attractive (unexpected; they delight when present but are not missed when absent), alongside indifferent and reverse.

In its original form, however, the model is discrete and qualitative: it cannot express how strongly an attribute influences satisfaction, it forces each respondent into a single answer despite the imprecision of user requirements, and it suffers a discontinuity problem whereby near-identical responses can fall into different categories [139,140]

The fuzzy Kano model addresses the imprecision by allowing respondents to distribute membership degrees across the response options rather than choosing one [141]. The Continuous Fuzzy Kano Model (CFKM), introduced by [142], extends this with a continuous quantitative assessment that resolves the discontinuity problem and accommodates uncertainty at the same time. It is administered through a modified fuzzy Kano questionnaire and derives an Evaluation Index (EI) that locates each attribute on a continuum and supports prioritization by EI ranking or threshold [140].

Chapter 4

Papers - Understanding Kitchen Space and Users

This thesis is organized as a collection of papers. This chapter compiles the thesis's constituent outputs, including two published paper and additional manuscripts that are submitted or under review. Each paper is introduced with a brief positional summary, author contributions, and an explicit link to the thesis research questions, followed by a cross-paper synthesis.

4.1 Paper I - Understanding Kitchen Interaction Challenges

Preface

This paper contributes primarily to RQ1a by characterizing communicative, spatial, and sensory breakdowns in remote kitchen engagements, and it contributes to RQ1b by translating those breakdowns into design directions for mediating presence, awareness, and coordination across ambient, synchronous, and asynchronous modes of connection. Through an initial prototype evaluation it also offers early methodological input to RQ2b. As the thesis’s foundational empirical grounding, it provides an early account of domestic users’ practices and constraints in kitchen-based remote interaction, and it is the only manuscript in this thesis that involves direct participation of domestic kitchen users.

The paper examines remote kitchen communication through a three-stage mixed-methods study: (1) participant observation of online cooking workshops conducted during the COVID-19 pandemic (15 sessions, with tutors, organizers, and 5–12 attendees per session); (2) online ideation workshops with domestic kitchen users and professional cooks (7 sessions; 42 participants); and (3) an evaluation of three XR prototypes with nine professional cooks using a Continuous Fuzzy Kano analysis. The first two stages identify four recurring communication challenges and four design themes, including breakdowns in visibility, audio, and spatial awareness, and needs for hands-free, context-aware, and multisensory communication; they characterize constraints—limited camera viewpoints, ambient noise and hands-busy conditions, and deficits in shared spatial reference—that explain why conventional screen-based tools fall short. The third stage instantiates these directions as three speculative XR artifacts spanning ambient, synchronous, and asynchronous communication, and finds that privacy controls register as near-essential user requirements. These artifacts are introduced here as an initial, empirically grounded proof of concept; each is developed and examined in depth in the subsequent papers.

My role as first author involved leading the study design, conducting and analyzing the observations and workshops, building and evaluating the prototypes, and writing the manuscript. Co-authors contributed to refining the conceptual framing, supporting analysis, and reviewing the paper.

A version of the following section was published as: O. E. Öğretmen, S. Cılızoğlu, and A. E. Yantaç, “Augmented Kitchens: XR-Enabled Communication in Active Domestic Workspaces”, in Extended Reality: International Conference, XR Salento 2026, Otranto, Italy, June 16–20,

2026, *Lecture Notes in Computer Science*, Springer.

4.1.1 Manuscript 1: Kitchen Talks

Augmented Kitchens: XR-Enabled Communication in Active Domestic Workspaces

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Abstract. Human communication in active domestic spaces, such as kitchens, is inherently embodied, spatial, and multisensory. Yet current communication tools remain static, designed for seated and screen-bound interaction. This paper investigates how extended reality (XR) and ubiquitous computing can augment perception, presence, and communication in such dynamic contexts. Through a three-stage mixed-methods study, comprising observations of fifteen online cooking workshops, seven co-design ideation sessions with 42 participants, and a VR prototype evaluation with professional cooks, we identify key spatial, sensory, and social challenges of communicating in motion and translate them into three XR artifacts: a real-time kitchen-to-kitchen portal, a 3D asynchronous messaging system, and an immersive ambient cooking environment. A Continuous Fuzzy Kano analysis reveals that privacy controls emerge as near-essential requirements. By reframing the kitchen as a testbed for embodied, XR-enabled communication, this work extends human augmentation toward social, perceptual, and communicative dimensions.

Keywords: Extended Reality (XR); Virtual Reality; Spatial Computing; Embodied Communication; Human Augmentation; Domestic Environments; Co-presence; User-Centered Design

1 Introduction

Human communication extends beyond words; it is embodied, spatial, and richly multisensory. Yet most communication systems remain designed for stationary, screen-bound contexts, poorly suited for the dynamics of movement, material interaction, and sensory engagement. Emerging paradigms in *ubiquitous computing* and *extended reality (XR)* present new opportunities to augment human perception, expression, and social awareness in these complex settings. The ambition to amplify human capability through technology traces back to Engelbart’s vision of *Augmenting Human Intellect* [4]. In the decades since, research in human augmentation has evolved to encompass physical, cognitive, and perceptual enhancement through digital means. This paper situates itself within that lineage, focusing

on how ubiquitous and XR-based systems can extend human communication capabilities, supporting awareness, empathy, and co-presence across spatial and sensory boundaries.

We examine this question in what we term *Active Workspaces* [11]: environments such as kitchens, workshops, or studios where people engage in overlapping physical and social tasks. These spaces challenge existing communication paradigms because they require users to interact while multitasking, moving, or manipulating tangible objects, yet conventional touch-based or screen-focused systems often disrupt rather than enable these embodied exchanges. The kitchen, in particular, serves as a compelling context; it is globally familiar, culturally meaningful, and inherently multisensory, representing a microcosm of human coordination where communication unfolds through gesture, proximity, sound, and shared attention. Remote communication in such contexts is increasingly common through virtual workshops, live-streamed tutorials, and distributed family activities, making the kitchen an ideal environment for examining how embodied, hands-free, and multisensory communication can be technologically augmented.

Prior work has explored how domestic environments can be augmented through mixed and augmented reality. Knierim et al. [12] investigated how users envision AR augmentations in everyday home life, identifying themes of assistance, enhanced perception, and social activity, primarily focused on human–environment interaction. In contrast, our study turns toward human–human communication within these same everyday environments, reframing augmentation as a social, perceptual, and embodied phenomenon. By situating kitchens as live, sensorily intense workspaces, we extend domestic AR research from object- and task-based augmentation to communicative augmentation and multisensory co-presence.

This study explores how spatial computing and XR technologies can *augment communication* in active domestic spaces. Unlike prior research on smart kitchens, which has largely focused on automation, monitoring, and assistive functionality, we approach technology as a means of extending human presence, perception, and social connection. Through a three-stage mixed-methods study, we investigate real-world communication breakdowns, user-driven design visions, and XR prototype evaluations in kitchen environments. The first stage observes 15 online cooking workshop sessions, Figure 1), which served as natural experiments in remote co-presence, revealing breakdowns and adaptations in embodied communication. The second stage comprises seven ideation workshops where 42 participants envisioned how ubiquitous and XR technologies could enhance awareness, coordination, and expression in the kitchen. The third stage evaluates three VR prototypes with professional cooks using a Continuous Fuzzy Kano analysis to assess user requirements. Our research addresses three central questions:

1. What challenges do users face when communicating in active, embodied domestic environments?
2. How can ubiquitous computing and XR augment perception, awareness, and presence in such contexts?

3. What design opportunities emerge for multisensory, context-aware, and socially meaningful augmentation of communication?

We present four observation-based challenges and four ideation-based design themes (Figure 2) that collectively articulate a framework for *communication augmentation*, grounded in empirical observation and validated through XR prototyping. By treating the kitchen as a testbed for ubiquitous and XR-enabled communication, we contribute to the broader discourse on human augmentation, demonstrating how technology can enrich human experience, empathy, and connection in the fabric of everyday life through emerging ubiquitous and spatial computing.



Fig. 1: Key challenges observed during the cooking workshops. Visibility issues appear in multiple sessions due to limited camera angles and phone orientation. In some sessions, tutors used extra cameras or assistants to demonstrate techniques, while some participants cooked outside the kitchen or chose only to observe.

2 Related Work

This paper situates itself at the intersection of three overlapping domains: Smart Homes and Internet-of-Things (IoT), Extended Reality (XR) and spatial computing for embodied interaction, and design research on communication in domestic environments. Within these domains, we focus specifically on how ubiquitous, spatial, XR technologies can *augment human communication and perception* in active workspaces—environments where users are physically and cognitively engaged with tasks, tools, and materials. While prior studies have explored automation and task support within smart kitchens, comparatively little research has examined how such environments might serve as testbeds for augmenting

human-to-human communication through spatial and multisensory augmentation. In this paper, we use the term spatial computing to denote systems that sense, interpret, and respond to human activity in physical space—through sensors, projection, or XR interfaces—aligning with contemporary definitions of ubiquitous and embodied computing in HCI.

Researchers have approached the Smart Kitchen from diverse perspectives. Eom et al.[5] introduced a system using smart Tupperware as an IoT solution to automate shopping lists and provide recipe suggestions, while Balaji et al.[2] proposed integrating image recognition into kitchen wardrobes for similar purposes. Oliver et al.[18] developed the *Ambient Kitchen*, a lab-based replica incorporating RFID sensors, projectors, and cameras to support ubiquitous computing research in cooking contexts. Patterson et al.[23] and Pham et al.[24] leveraged RFID for motion recognition, while Hilliges et al.[10] explored mid-air interaction through depth-sensor cameras. Similarly, Panger[21] and Hashimoto et al.[9] used vision-based systems for recognizing gestures and cooking actions. These works illustrate the growing ambition of ubiquitous computing to embed digital intelligence into domestic activity, though most focus on efficiency and automation rather than on augmenting perception, communication, or co-presence.

Ju [11], in her doctoral thesis, synthesized work from concept kitchens such as *CounterActive* and *Origami Desk* to propose “active workspaces” as sites where tangible and digital interaction converge. Building on this foundation, Bonanni’s *CounterIntelligence* project [14] employed projection-based guidance systems to assist cooking processes, while Kosch et al.[13] designed a Kinect- and projector-based kitchen for cognitively impaired users, showing how spatial augmentation can enhance coordination and independence. Recent work in the Augmented Humans community has expanded this perspective through the concept of *Assistive Augmentation*—interfaces that integrate seamlessly with body and behavior to extend perception and action [17]. Nanayakkara et al. proposed a two-dimensional design space (bodily integration $\times$ ability), framing augmentation as a continuum from assistive to superhuman. This framing helps position kitchen communication systems as *assistive augmentations*—supporting not only efficiency but also sensory and social expansion within everyday routines.

Knierim et al.[12] examined AR interfaces in home environments, extending domestic interaction into augmented visual layers. The *Telelife* project [19] speculated on future domestic XR applications, envisioning connected kitchens with enhanced remote communication capabilities. Similarly, the *CoDine* studies [32,31] proposed multisensory co-dining systems that support remote family meals, illustrating how XR can augment both sensory perception and social connectedness. Beyond cooking assistance, Mennicken et al.[16] embedded interactive displays into kitchen counters for *First Person Cooking*, and Terrenghi et al.[27] explored social recipe sharing via the *Living Cookbook*. Patibanda et al.[22] further introduced the notion of *intra- and inter-corporeality* in augmented human systems, emphasizing shared bodily control and mutual coordination across multiple users. This perspective aligns with our focus on co-present kitchen interactions, where communication involves not only cognitive and sensory exchange but also

bodily synchrony and spatial negotiation. Together, these projects demonstrate how XR and spatial technologies can be leveraged to augment not only task performance but also embodied social experience.

Further research has investigated learning, assistive, and conversational augmentation in kitchen contexts. Preston et al.[25] transformed the kitchen into a smart learning environment for language acquisition, combining sensor-based feedback with everyday activity. Conversational agents such as *Foodie Fooderson* [1] and *SmolKat* [7] aimed to assist with nutrition, meal preparation, and task sequencing through speech-based guidance, while Hamada et al.[8] developed *Cooking Navi* to manage multi-recipe coordination. These studies highlight the growing role of multimodal, context-aware systems in augmenting cognitive and behavioral capabilities during cooking. However, most remain focused on instrumental or assistive augmentation, leaving the social and communicative dimensions underexplored. The perception and acceptance of augmentation itself have also become a growing concern in HCI. Villa et al.[28] conducted a mixed-method study on how people perceive augmented humans across sensory, motor, and cognitive dimensions, showing that social attitudes toward augmentation depend strongly on perceived fairness, safety, and authenticity. Their findings underline that augmentation is not merely technical but inherently social—a view our work extends by exploring *communicative and relational augmentation* in shared domestic spaces.

Designing technologies for domestic environments also requires sensitivity to their spatial and cultural dynamics. Kitchens have long been central to design research, from Lillian Moller Gilbreth’s early “kitchen work triangle” optimizing efficiency through spatial layout [26], to contemporary inquiries emphasizing collaboration, emotion, and well-being. For example, Paay et al.[20], in *Cooking Together*, used proxemics and F-formation theory to analyze co-located and remote coordination in kitchen activities. Qi et al.[26] synthesized quantitative and qualitative findings to evaluate how user-centered kitchen design contributes to well-being under the PERMA model. Wang et al.[30] applied space syntax analysis to study aging-in-place, identifying how layout influences accessibility and daily performance. Mathis et al.[15] recently examined how AR systems intersect with *everyday life challenges*, arguing for contextual, situated, and socially grounded augmentation rather than purely functional enhancement. Their insights resonate with the domestic, improvisational settings of our study, where augmentation operates within the rhythms and constraints of everyday practices. These works broaden the understanding of domestic design from efficiency toward inclusivity, sociality, and emotional resonance—key factors for human-centered augmentation.

Despite this substantial body of work, few studies examine synchronous, embodied communication where users must coordinate, speak, and move simultaneously in sensory-rich domestic contexts. Our three-stage mixed-methods inquiry addresses this gap by (1) characterizing communication challenges in motion, (2) identifying user needs and design opportunities for XR-enabled augmentation,

and (3) evaluating XR prototypes that extend human perceptual, social, and communicative capacities beyond efficiency and automation.

3 Methodology

This study employs a three-stage mixed-methods methodology to investigate how ubiquitous computing and extended reality (XR) technologies can augment communication practices and embodied interaction within kitchen environments as active workspaces. As illustrated in Figure 2, the research was structured around three interconnected stages: (1) observational fieldwork conducted during 15 online cooking workshops, (2) a series of 7 ideation workshops designed to elicit user needs and speculative design opportunities, and (3) a VR prototype evaluation with professional cooks using a Continuous Fuzzy Kano analysis to assess user requirements. The first stage identifies the essential communication challenges users face in dynamic domestic contexts. These findings informed the second stage, which explores how emerging modalities such as ubiquitous, spatial computing and XR might augment embodied communication. The third stage translates these insights into three XR prototypes, evaluating how they meet user needs and expectations. Through this iterative process, we ground speculative design in empirical observation, generating a framework for embodied, multisensory, and context-aware communication in domestic environments.

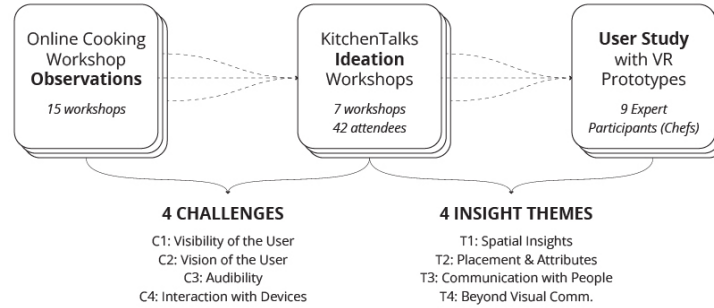


Fig. 2: Overview of the three-stage KitchenTalks study: 15 cooking workshop observations yielding four challenges (C1–C4), seven ideation workshops with 42 participants yielding four design themes (T1–T4), and a VR prototype evaluation with 9 expert participants.

3.1 Stage 1: Cooking Workshop Observations

To explore communication challenges in the kitchen, we observed a series of online cooking workshops organized by a non-profit initiative gathering people to cook, share recipes, and discuss cultural themes. One researcher joined each session as

a silent observer with prior approval from the organizers, documenting verbal and non-verbal exchanges through detailed field notes. Each session involved three roles: *the tutor*, who conducted cooking demonstrations; *the organizer*, who managed audio-visual coordination and participant moderation; and *the attendees*, who joined from their own kitchens (5–12 per session), sometimes cooking along, sometimes observing. After each workshop, a 30–45 minute debrief consolidated findings and identified recurring issues. Using an inductive thematic coding approach, we categorized communication challenges such as device placement, visibility, and multitasking limitations, which directly informed the ideation phase and provided empirical grounding for subsequent XR augmentation opportunities.

3.2 Stage 2: Ideation Workshops

Seven online ideation workshops were conducted with 42 participants, recruited through an open call including everyday kitchen users and four professional cooks. Workshops were held via Zoom, using Miro for collaborative tasks, and structured around three phases. *Warm-up*: Participants sketched their kitchen layouts using provided templates, identifying areas for activity, communication, and ideal technology placement. This activated spatial memory and revealed physical constraints affecting communication flow. *Reflection on daily habits*: Participants described their cooking practices, task rhythms, and communication behaviors, such as whom they communicate with, which devices they use, and under what conditions, deepening understanding of multitasking demands and informing design considerations for context-aware and hands-free augmentation. *Group ideation*: Participants conceptualized speculative tools for kitchen communication, exploring how spatial computing and XR could enhance perception, co-presence, and awareness, revealing both functional needs and imaginative visions for future communication augmentation.

3.3 Stage 3: User Study through Virtual Prototypes

In the final stage, we designed and built three VR prototypes based on findings from the previous stages. Artifacts were modeled in *Blender* and integrated as interactable objects into a virtual kitchen scene in *Unity*. The *Record3D* iOS app was used to capture high-resolution 3D video messages via the device depth camera, and a Unity package *Portals for VR* [6] was used to implement portal interactions.

To evaluate the design insights emerging from the KitchenTalks ideation workshops, each prototype addressed a distinct communication dimension identified across the four themes: immersive ambient experience, real-time remote presence, and asynchronous spatial messaging. Virtual reality was chosen as the prototyping medium because it allowed us to simulate situated, kitchen-embedded interactions that would be technically complex and costly to deploy in real environments at this stage, enabling participants to experience and evaluate speculative communication artifacts in a realistic spatial context. In the user study, participants were first briefed on the study purpose, the experience flow, and potential risks. They

were then introduced to the basic interactions in VR and asked to freely explore the virtual kitchen for five minutes to familiarize themselves with the environment. After this familiarization period, participants experienced each prototype individually in the following order: (1) Immersive Kitchen, (2) Kitchen Portal, and (3) Kitchen Annotation. Following the experience, participants completed a Continuous Fuzzy Kano Model questionnaire to evaluate the prototypes and their features [33,34]. We employed this approach and analysis method by considering the characteristics and conditions of the prototypes, guided by the assessment framework proposed by Violante and Vezzetti [29].

(1) *Immersive Kitchen*: Derived from ideation workshop suggestions (e.g., tools that boost mood, play context-aware music, or enhance cooking with augmented sounds), this prototype places the user in a pizza-making scenario where the environment responds dynamically: themed music plays, ambient spotlights activate, and a projected backdrop of an Italian coastal town appears behind the counter, creating a multi-sensory immersive cooking atmosphere.

(2) *Kitchen Portal*: Inspired by the portal metaphor, this prototype enables real-time kitchen-to-kitchen communication through a fixed in-situ device. It supports parallax view for enhanced depth and presence, gesture and voice control for hands-free operation, and transmission of spatial cues such as pointing and highlighting between remote spaces. In the multi-user VR experience, participants appeared as Meta Avatars, could see each other’s gestures and the remote kitchen through the portal, and were able to reposition and reshape the portal within their kitchen space.

(3) *Kitchen Annotation*: Inspired by fridge magnets, this prototype supports asynchronous communication through 3D-recorded messages that appear pinned to the fridge door. Users can view messages in 3D at their convenience and manage them by hiding, moving, or saving them. For the study, six short 3D video messages were pre-recorded, covering everyday scenarios such as requesting a meeting, asking for a recipe, and noting a delivery.

3.4 Analysis

Two complementary analyses addressed the spatial and narrative dimensions of the data. Kitchen layout drawings were examined using a custom clustering method to identify common spatial relationships, while textual data from Miro boards and discussions underwent reflexive thematic analysis [3], capturing both physical configurations and conceptual patterns relevant to communication augmentation.

Kitchen Plan Analysis Kitchen plans varied widely in size, shape, and structure. We developed a clustering approach emphasizing relational proximity among activity zones, devices, and communication points (Figure 3), allowing comparative interpretation of how spatial arrangements influence communicative

behavior. The synthesized relationships were then visualized as 4D diagrams (Figure 4), representing how work areas, device positioning, and communication flows intersect across time and activity.

Thematic Analysis Following Braun and Clarke’s reflexive thematic analysis framework [3], two authors independently coded the qualitative data through six iterative phases, comparing and refining codes in collaboration with HCI researchers. Key themes included embodied communication barriers, adaptive practices, and speculative augmentation concepts, forming the basis for a design framework of *ubiquitous and XR-enabled communication augmentation* in domestic environments.

User Study Survey Analysis For the survey, we adopted a hybrid quantitative approach grounded in the Continuous Fuzzy Kano Model (CFKM) [33]. Category assignment was inspired by the Analytical Kano model [34] and operationalized through Euclidean distance-based classification. Specifically, we computed the average functional and dysfunctional values for each feature and measured their respective distances to the four ideal Kano categories: Attractive (A), Performance/One-Dimensional (O), Must-Be (M), and Indifferent (I). The questionnaire included assessment items for each overall prototype (Table 2), as well as their specific features (Table 3 & Table 4). Ultimately, this analytical framework classifies user requirements based on the extent to which each feature impacts overall user satisfaction.

4 Findings

This section presents communication challenges and design opportunities emerging across the first two stages of our study. The cooking workshop observations revealed everyday challenges in active domestic environments, including misaligned attention, occluded visibility, and fragmented coordination, while the ideation workshops built upon these insights to explore how XR technologies could address them through speculative design.

The synthesis of both stages yielded four overarching thematic areas: **(1) Spatial Insights**, showing how kitchen layouts and movement patterns shape perception and interaction; **(2) Placement & Attributes**, analyzing where and how devices are positioned and perceived; **(3) Communication with People**, examining the social and embodied dimensions of coordination; and **(4) Beyond Verbal Communication**, exploring speculative, multisensory, and ambient modes of interaction. Together, these themes frame the kitchen as a testbed for *communication augmentation*: technologies that extend perception, awareness, and presence beyond efficiency, toward expressive, socially attuned interaction (see Table 1).

4.1 Cooking Workshop Observations and Challenges

Across workshops, four recurring challenges—*visibility*, *vision*, *audibility*, and *interaction with devices*—revealed how domestic cooking spaces constrain commu-

Table 1: Summary of findings from cooking and ideation workshops

Study Phase	Key Challenges and Themes
Cooking Workshop Observations and Challenges	Challenge 1: Visibility of the user — Limited camera angles and device placement hindered users’ ability to be seen clearly. Challenge 2: Vision of the user — Small screens and distant placement of devices made it hard to follow others. Challenge 3: Audibility — Background noise from kitchen activities disrupted verbal communication. Challenge 4: Interaction with devices — Wet or busy hands and unfamiliar interfaces caused usability issues.
Ideation Workshop Findings	Theme 1: Spatial Insights — Diverse kitchen layouts showed mobile device use and spatial separation of work and communication zones. Theme 2: Placement & Attributes — Participants requested mobile, wall-mounted, or durable communication tools for kitchen environments. Theme 3: Communication with People — Tools should support real-time interaction, ease of expression, and group coordination. Theme 4: Beyond Verbal Communication — Envisioned expanding kitchen communication through multisensory and ambient experiences. Theme 4.1: Communication with Taste — Speculative use of edible 3D-printed messages and cultural food rituals for emotional connection. Theme 4.2: Augmented Cooking Experience — Use of ambient sound, video, or light to enhance immersion and communication. Theme 4.3: Immersive and Ambient Experience — Smart kitchens could amplify cooking sounds for engagement and awareness.

nication. These breakdowns are not merely usability deficits but invitations for *augmentation*: ways to extend perceptual, auditory, and bodily capacities under realistic working conditions. Camera framing that dynamically tracks bodies and tasks (*perceptual augmentation*), audio systems that separate appliance noise from speech (*auditory augmentation*), and input modalities that remain usable with wet or occupied hands (*multimodal augmentation*) all point to communication as a co-adaptive process between humans and tools. Participants’ improvised solutions—using dual devices, mounting phones on jars or tripods, asking others to act as mediators—already hint at the need for *distributed* augmentation: multiple cooperating sensors and displays that sustain mutual awareness as people move through space.

Challenge 1: Visibility of the user Participants frequently vanished from view when turning toward the sink, leaning over a pot, or stepping aside to retrieve ingredients. These moments highlighted the fragility of single, static viewpoints

in active spaces. Instructors often asked, “Can we see what you’re doing?” while participants awkwardly adjusted devices mid-task. Such breakdowns indicate a need for *multi-perspective capture*—wearable cameras, steerable tripods, or depth-aware framing that automatically centers relevant gestures or artefacts. Beyond solving occlusion, visibility becomes a lens for *embodied legibility*: designing systems that maintain the readability of human action as a key communicative channel.

Challenge 2: Vision of the user While visibility concerns what others see, vision concerns what participants themselves can perceive. Many positioned devices away from the main work area for safety or space, leaving remote collaborators literally out of sight. Participants would lift dough-covered hands toward screens to check cues, then resume work with delayed feedback. This points to a demand for *proximal, glanceable, and projective* visual channels that keep partners “in the corner of the eye.” Under an augmentation lens, this includes *ambient or situated displays*—for instance, low-resolution projections on cabinets or head-mounted overlays—that preserve social presence without fragmenting attention between screen and workspace.

Challenge 3: Audibility Sound emerged as a constant negotiation. The roar of blenders, boiling pots, and echoing kitchens routinely masked speech, while remote participants struggled to locate who was speaking. Some muted appliances or raised voices; others relied on improvised hand signals. These coping strategies underscore the need for *context-aware audio augmentation*: directional microphones that track speakers, adaptive noise suppression that distinguishes between utensil clatter and voice, or lightweight push-to-talk gestures triggered by motion or gaze. Augmentation here is less about perfect fidelity than maintaining conversational continuity amidst acoustic turbulence.

Challenge 4: Interaction with devices Finally, interaction breakdowns exposed the embodied friction of managing tools mid-task. Wet or flour-covered hands made touchscreens unreliable, while interface icons were too small to manipulate at a distance. Participants resorted to elbows, knuckles, or requests for help from family members—revealing how bodily constraints shape communication. Under an augmentation frame, this invites *hands-free interaction grammars*: voice or foot input for simple controls, proxemic gestures near work zones, and micro-automations such as auto-muting when appliances start or spotlighting the cook during key steps. Such features would transform device handling from an interruption into an integrated rhythm of motion and attention.

Taken together, these four challenges delineate the lived complexity of “communicating in motion.” They make visible the latent design space where ubiquitous sensing and XR technologies could enhance not just task efficiency but the perceptual, auditory, and expressive synchrony that underpins domestic collaboration.

4.2 Ideation Workshop Findings

Building on the empirical insights from the cooking workshop observations, the second phase of the study invited participants to envision how emerging technologies could address those communication challenges. Through seven ideation workshops, we explored user-driven proposals that reframed kitchen interaction as a space for perceptual, spatial, and social augmentation—transforming everyday coordination into opportunities for expressive and embodied communication.

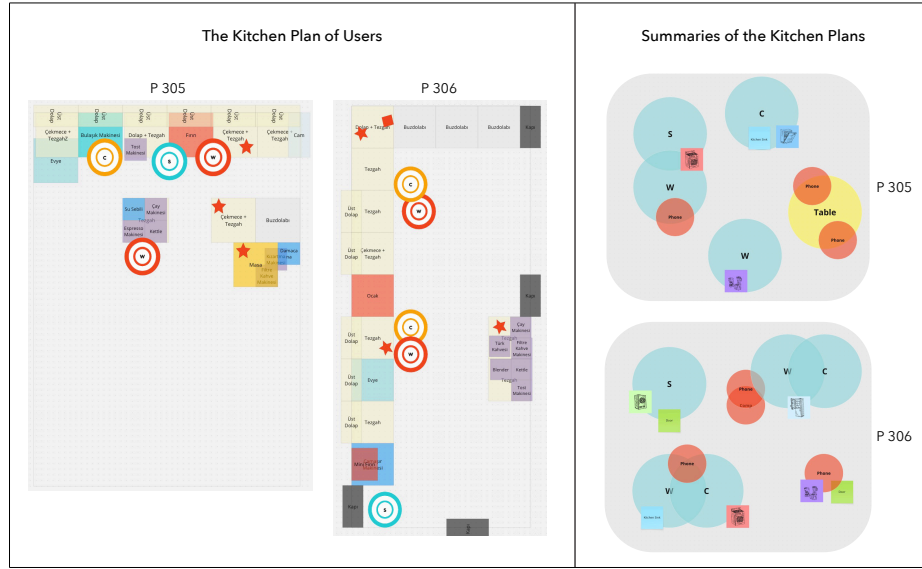


Fig. 3: Two participants’ kitchen plans (P305 and P306) and their abstracted summaries. Original layouts (left) show working zones and device placements, while the summaries (right) cluster key functional areas, Work (W), Storage (S), and Communication (C), to visualize spatial relationships across diverse kitchen configurations.

Spatial Insights Participants’ spatial diagrams and discussions revealed that the kitchen’s geography fundamentally shapes communication—where it happens, how it flows, and where it breaks. Conversations naturally gravitated toward *tables and shared surfaces*, which served as informal command centers for social interaction, while the most intensive physical work clustered around counters, stoves, and sinks. This polarity between social and functional zones points toward distinct augmentation opportunities: *situated, low-footprint display zones* near tables to facilitate shared awareness, and *lightweight, peripheral overlays* near work anchors to sustain coordination without disrupting motion. The

recurring insistence on keeping workspaces “clean” or “uncluttered” underscores the preference for *ambient*, non-dominant XR visuals or sounds that hover at the perceptual edge, surfacing only when context demands it.

Diagram 1: Table-centric clusters emphasize that communication thrives around social anchors. Participants imagined “co-present social augmentation”—shared glanceable displays or subtle lighting cues indicating who is speaking or seeking attention—to preserve conviviality and turn-taking without the cognitive burden of device management.

Diagram 2: Overlaps between current phone placements and proposed solution areas show how augmentation can *piggyback on existing ecologies*. Rather than introducing alien devices, participants preferred enhancements; mounts, docking spots, or wearables; that extend their habitual rhythms. This reflects a user-driven logic of continuity: augmentation that grows from lived practice rather than overwriting it.

Diagram 3: Avoidance of placing devices directly in high-intensity work areas reflects a concern for safety and fluidity. Participants proposed technologies that *reach into* those zones—steerable cameras, projected pointers, head-worn micro-overlays—so that guidance and feedback remain accessible while hands stay free. This frames augmentation as a distributed presence that orbits rather than occupies the workspace.

Diagram 4: The omnipresence of phones in all diagrams positions them as *mobile augmentation hubs*. Participants envisioned the phone not as a fixed screen but as a “roaming coordinator” orchestrating other peripherals—clip-on microphones, auto-tracking cameras, ambient projectors—creating adaptive *situational assemblages* that reconfigure as the activity unfolds.

Placement & Attributes Participants translated their daily irritations—tiny screens, slippery devices, muffled speakers—into concrete design demands: *hands-free control*, *glanceable feedback*, *directional sound*, and *ergonomic camera mobility*. These are not mere wishlist items but expressions of lived bodily negotiation with tools. The “laptop + phone” workaround observed among tutors, where one device handled visuals and another managed voice, prefigures a *distributed augmentation paradigm*: a multi-node system that mirrors human division of attention. During ideation, participants refined this concept through proposals for wall-mounted displays, context-sensitive tracking views, and flexible camera mounts. Collectively, these envision the kitchen as an orchestrated communication ecosystem rather than a single smart appliance—where each node contributes to seamless continuity as users circulate through different zones.

Solution Approach: Always with Me / Using Additional Cameras Participants repeatedly voiced the desire for technology that “follows me” and maintains

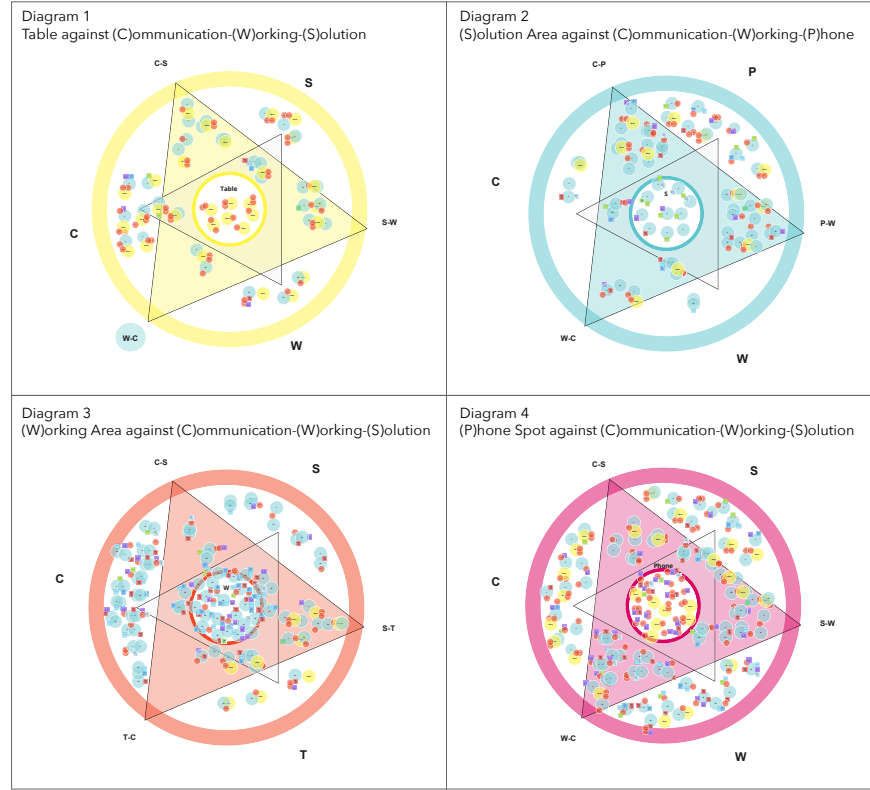


Fig. 4: Each diagram represents clustered spatial relationships between Tables, Phones, Communication Zones, Working Areas, and proposed Solution Points (S) for a new communication device.

“continuous visual connection.” Their concepts evoke *viewpoint augmentation*: cameras capable of auto-tracking, gaze- or voice-steered framing, and adaptive switching between close-up and contextual views. Instead of demanding attention, these systems would quietly keep the communicative thread intact through task transitions—moving from chopping to stirring to plating without a single touch. Such solutions transform visual continuity from an effortful act of camera management into a form of embodied awareness.

Communication with People Across ideation sessions, participants reimagined kitchen communication as a negotiation between *social rhythm* and *task rhythm*. They emphasized the frustration of having to choose between maintaining conversation and staying focused on the meal. In response, their solutions prioritized *social fluency*: lightweight cues that visualize who has the floor, soft notifications

for turn-taking, or adaptive modes that differentiate between addressing the instructor and speaking privately to peers. This fine-grained control of presence reflects a shift from automation toward *communicative orchestration*—systems that understand tempo, intent, and context rather than just transmitting sound.

Solution Approach: Mute & Unmute Default muting, though practical, broke the natural flow of talk. Participants proposed “smart muting”: voice-toggled microphones, context-based noise filtering, and gesture-triggered unmute actions to restore spontaneity. Some imagined context-aware mixing that isolates voices from background appliance noise or channels speech selectively (“talk to instructor,” “talk to group”). These ideas represent a form of *auditory augmentation for social coordination*, where systems listen with the user’s body and task, dynamically routing audio to sustain fluid, embodied dialogue.

Beyond Verbal Communication The most speculative yet emotionally charged ideas reached beyond words, suggesting that communication in the kitchen is inherently multisensory and affective. Participants envisioned XR and ubiquitous systems that could translate mood, taste, and atmosphere into new communicative textures—where cooking becomes both conversation and performance. Read through an augmentation lens, these concepts propose expanding communicative bandwidth through sensory richness.

Solution Approach 1: Communication with Taste Participants imagined *edible signals*—flavors, textures, or plating gestures—that could mark celebration, gratitude, or learning milestones. These “messages through food” express *material communication*, linking sensory pleasure to social meaning. As design speculation, such ideas frame augmentation not as substitution but as amplification of cultural ritual and embodied expression.

Solution Approach 2: Augmented Cooking Experience Participants extended the idea of sensory augmentation into atmosphere. They described responsive environments where ambient light, projection, or music adjusts to the recipe, region, or collective pace of the group. This *situated ambience augmentation* synchronizes digital cues with bodily tempo—deepening the sense of shared presence while preserving concentration.

Solution Approach 3: Immersive and Ambient Cooking Experience Sound was frequently reimaged as a communicative substrate. Participants proposed enhancing or spatializing natural kitchen sounds—the hiss of oil, the rhythm of chopping—as *auditory beacons* for coordination, safety, and empathy. Hearing another’s cooking progress became a subtle yet powerful way to stay connected. Such approaches exemplify *awareness through augmentation*: systems that expand perception without demanding attention.

Table 2: Kano Analysis for each artifact. Values represent the Euclidean distance to the ideal centers defined by the Kano model. Smaller numbers indicates aligns stronger to the category. Bold cells indicates the smallest number of the feature and close scores are both emphasized which indicates hybrid profile; Attractive and Performance.

	A	O	M	I	Result
Kitchen Portal	0.199	0.304	0.563	0.514	Attractive
Kitchen Portal on Cooking	0.224	0.277	0.559	0.535	Attractive
Kitchen Annotations	0.070	0.435	0.644	0.480	Attractive
Immersive Kitchen	0.163	0.535	0.612	0.338	Attractive

4.3 User Requirements from VR Prototype Evaluation

In the final phase of the study, nine professional cooks experienced three virtual prototypes in VR and evaluated them through a survey.

Table 2 shows the Kano analysis results for each artifact. While all artifacts are classified as Attractive, Kitchen Annotations exhibits the shortest distance, indicating the strongest alignment with this category. The Attractive category indicates that features are exciting for users but their absence does not cause dissatisfaction. This is understandable since these artifacts do not exist in today’s kitchen and participants only used them briefly in a virtual environment, so they did not develop a strong connection with them.

Conversely, Kitchen Portal on Cooking is technically classified as Attractive, but its high proximity to the Performance coordinate indicates a hybrid profile. Among all evaluated artifacts, it sits closest to the Performance threshold, indicating that users are excited by it but its absence would cause dissatisfaction especially during cooking. Some features were not experienced directly during the session; instead, they were presented as hypothetical features. Table 3 reports each feature’s Euclidean distance to the ideal points and highlights hybrid classifications in red.

The results show that privacy features on the Kitchen Portal, such as turning the camera on/off and the recording status indicator, as well as the privacy feature on Kitchen Annotations, cluster closer to Performance, indicating that their absence is likely to cause dissatisfaction. For the Kitchen Portal, Portability is also classified as a Performance feature, and Gesture Control emerges as a hybrid case. In contrast, Parallax Interaction appears as a hybrid between Indifferent and Attractive, suggesting it is comparatively low priority: participants do not see it as essential, and its absence would not be particularly upsetting. For Kitchen Annotations, features are predominantly Attractive, especially those supporting richer communication and message management such as 3D avatar messages, video messages, and archiving.

In summary, the Kano analysis suggests that privacy controls are near-essential requirements, and participants are most excited by communicating through 3D

avatar and video in both synchronous and asynchronous modes, which they perceive as particularly valuable during cooking.

Table 3: Kano analysis of Portal’s features. Values indicate Euclidean distance to the ideal Kano category centers: Attractive (A), One-Dimensional/Performance (O), Must-Be (M), and Indifferent (I). Features with similarly close distances to two categories are treated as hybrids.

Categories	Feature	A	O	M	I
Interactions and Indicators	Spatial Cues: Pointing to objects	0.170	0.382	0.538	0.416
	Parallax Interaction	0.308	0.410	0.407	0.304
	Privacy: Turning off the camera/video	0.398	0.113	0.468	0.604
	Gesture Control: Hand/eye tracking	0.292	0.302	0.425	0.419
	Voice Command: Voice control	0.224	0.386	0.483	0.367
	Recording Status: Status indicator	0.259	0.288	0.471	0.454
Mobility and Placement	Portable	0.305	0.195	0.536	0.585
	Placement behind the countertop	0.059	0.492	0.659	0.442
	Wall-mounted placement	0.186	0.393	0.521	0.389
	Placement on top of the cabinet	0.276	0.573	0.550	0.224
	Placement on the refrigerator	0.322	0.683	0.644	0.228
Form	Circular Portal	0.428	0.796	0.710	0.234
	Square Portal	0.399	0.596	0.458	0.118

5 Discussion

This study investigated how communication in active domestic workspaces, particularly kitchens, can be augmented through user-centered design and emerging XR technologies. Through three sequential stages, we identified communication challenges in remote kitchen interactions, explored user-driven design visions, and evaluated XR prototypes to assess user requirements. Our findings, structured across four observation-based challenges and four ideation-based thematic categories, extend existing literature and open speculative design pathways for spatial computing and XR applications in everyday life.

From Challenges to Design Insights The observational study revealed four key challenges: *Visibility*, *Vision*, *Audibility*, and *Device Interaction*, reflecting the embodied, multisensory, and often chaotic nature of the kitchen as an active workspace. Participants struggled with camera placement, screen limitations, appliance noise, and interaction breakdowns due to messy or occupied hands, challenges not merely technical but deeply entangled with spatial and bodily constraints, highlighting the inadequacy of conventional screen-based, touch-reliant

Table 4: Kano analysis of Kitchen Annotations’ and Immersive Kitchen features. Values indicate Euclidean distance to the ideal Kano category centers: Attractive (A), One-Dimensional/Performance (O), Must-Be (M), and Indifferent (I).

Feature	A	O	M	I
Kitchen Annotations				
3D Avatar / Object	0.097	0.424	0.612	0.453
Video message from the kitchen	0.094	0.453	0.615	0.427
3D avatar from the kitchen	0.101	0.420	0.608	0.451
Message from a mobile phone	0.222	0.323	0.498	0.439
Saving / Storing messages	0.113	0.393	0.609	0.479
Hiding messages / Privacy	0.338	0.203	0.437	0.514
Moving / Carrying messages	0.243	0.330	0.470	0.413
Immersive Kitchen				
Ambient Light	0.150	0.403	0.557	0.413
Theme Music	0.140	0.391	0.571	0.439
Ambient Images	0.155	0.448	0.561	0.371
Ambient Chef	0.158	0.342	0.605	0.523
Haptic Feedback	0.175	0.347	0.547	0.457
Audio Alerts / Cues	0.188	0.322	0.554	0.488

interfaces. The ideation workshops responded with speculative proposals, including wearable or spatially embedded communication tools (Theme 2: *Placement & Attributes*) and hybrid input modalities like voice commands and continuous camera tracking, aligning with Ju’s notion of Active Workspaces [11]. *Visibility* and *Vision* map to perceptual augmentation (e.g., multi-perspective capture informed by [10,18]), *Audibility* to auditory augmentation (noise-aware, role-sensitive capture as in [18]), and *Device Interaction* to hands-free, ubiquitous control consistent with kitchen-focused spatial systems [14,13]. The Kano analysis further confirmed these directions, with privacy controls emerging as near-essential and 3D avatar communication as the most exciting feature. Workshop ideas such as the “Always With Me” device extend this trajectory by suggesting lightweight XR or distributed display/audio arrangements that sustain communication without interrupting action, an augmentation-first stance rather than a mere usability fix.

Communication as a Situated Social Practice Beyond technical constraints, our findings underscore that communication in kitchens is deeply social. Theme 3 (*Communication with People*) captures how users desired systems supporting not only task coordination but also spontaneous conversation, shared experiences, and emotional resonance. The role of the workshop organizer as an ad-hoc facilitator who maintained group flow and mediated muting/unmuting revealed a critical gap in current HCI literature: communication management interfaces for synchronous, social use in active settings, where social fluency requires surfaces for lightweight turn-taking and selective addressing, especially when bodies and tools are in

motion. Prior systems like *Living Cookbook* [27] and *First Person Cooking* [16] illustrate the value of visual augmentation during cooking, yet do not explicitly handle the social orchestration seen in our workshops. Our findings close this gap by exposing fine-grained interactional dynamics, such as who needs help, who is plating, and who is waiting to speak, that can guide XR overlays and ubiquitous audio-visual cues toward augmented co-presence rather than simple instruction delivery.

Spatial Reasoning and Embodied Device Use Our kitchen layout analysis (Theme 1: *Spatial Insights*) revealed how users think through space when imagining new technologies. Communication tools were commonly associated with central, shared areas like tables, while intensive work areas (e.g., stove, sink) were treated as off-limits for tech placement, offering valuable insights for spatial computing: not all surfaces or zones in a kitchen are equal in their communicative affordances. The spatial diagrams created from participant-drawn plans serve as a methodological contribution, mapping embodied routines in a way that informs spatially adaptive XR interfaces where AR systems dynamically adjust notification positioning based on proximity to work or social zones. This orchestration extends the logic of projection- and sensing-based kitchens [14,18] and complements assistive spatial setups [13] by emphasizing augmentation of awareness and social continuity. While Knierim et al. [12] primarily addressed human–environment interaction, our findings expand this by showing how spatial reasoning and proxemic awareness underpin human–human communication itself, where layout, body orientation, and visibility shape the rhythm and success of collaboration, positioning the kitchen as a living testbed for *situated augmentation*: ubiquitous sensing and XR overlays that adapt to proxemic dynamics, respect task intensity, and sustain legibility among people, artifacts, and actions.

Speculative Futures of Multisensory Communication Theme 4 (*Beyond Verbal Communication*) pushes the boundary of what counts as communication in kitchens. From edible 3D-printed messages to ambient audio augmentation, participants envisioned ways of connecting through food, sound, and atmosphere, expanding the role of the kitchen into a site for affective, symbolic interaction that resonates with multisensory dining and remote togetherness explored in *Telelife* [19] and *CoDine* [32,31]. Dynamic lighting, projected ambience, or culturally themed soundscapes act as communicative scaffolds that deepen co-experience and synchronize group tempo, foregrounding augmentation of presence, feeling with and feeling there, rather than automation.

Towards Hybrid Interaction Together, these findings point to a framework for hybrid interaction in active domestic spaces, one that: (1) embraces spatial and bodily context as a core design input, (2) prioritizes low-friction multimodal control (e.g., voice, gesture, spatial audio), (3) enables ambient social awareness (e.g., subtle cues for turn-taking and attention), (4) blends asynchronous and synchronous modalities (e.g., quick voice snippets or presence markers), and (5) incorporates playful and sensory layers for richer co-experience. As work on smart,

sensing-rich kitchens [18,10] and domestic AR [12] has shown, everyday spaces are fertile ground for understanding embodied interaction. Our results suggest that augmenting communication, rather than solely instructing or automating, provides a transferable blueprint for other *Active Workspaces* [11], where ubiquitous computing and XR converge to enhance human perceptual, social, and communicative capabilities in daily life.

6 Conclusion

This study explored the challenges and opportunities of communication in kitchen environments, an exemplar of active workspaces, through a three-stage mixed-methods inquiry combining cooking workshop observations, co-design ideation sessions, and VR prototype evaluation. Addressing three research questions, we identified limitations in device placement, usability, and social interaction; translated user needs into three XR communication prototypes; and evaluated them using a Continuous Fuzzy Kano analysis. Our contributions include an empirical grounding for future kitchen-to-kitchen communication tools, a speculative design framework for XR-enabled domestic communication, and evidence that privacy controls are near-essential requirements in such systems. These findings frame communication not simply as information exchange but as a form of perceptual and embodied augmentation, enhancing awareness, coordination, and presence in motion.

This research has several limitations. The observational findings were not fully shared with ideation participants, preserving autonomy but limiting cross-pollination. Participants were not directed toward specific technologies during ideation, resulting in broader rather than XR-focused outcomes. Additionally, the study was conducted within a single cultural context, and findings may not generalize across diverse cultural or spatial configurations of kitchen use.

Future work should examine cross-cultural comparisons of communication behavior in active domestic workspaces and develop functional XR prototypes enabling hands-free, spatially aware, and multisensory interactions. Hybrid communication scenarios integrating asynchronous and synchronous modalities also warrant further investigation. The originality of this work lies in reframing the kitchen from a site of automation and assistance into a living laboratory for human-to-human communication and embodied augmentation. By extending the discourse of human augmentation into the social and communicative realm, this paper contributes a perspective that emphasizes augmentation not only of function, but of relation, expression, and shared experience.

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4.2 Paper II - Conceptualizing Phygital through Portal

Preface

This paper explores communication through a speculative artifact and contributes primarily to RQ1b by examining how phygital interfaces may support hybrid presence—forms of mediated connection that combine physical and virtual engagement. Through its two-phase design, pairing speculative co-creation with a structured prototype evaluation, it also contributes to RQ2b, demonstrating how speculative probes and immersive prototyping can function as design research methods. Specifically, the work investigates whether a portal-like interface can enable the communication qualities and use cases associated with hybrid presence, including awareness, coordination, and socially meaningful connection across remote settings.

The Portal is a speculative XR concept, which is an anchored, bidirectional threshold embedded in everyday space, deployed first as a design probe and then as a functional prototype. While the artifact was informed by design considerations emerging from the previous paper, the speculative phase intentionally generalizes the concept beyond domestic settings to surface broad cross-realm qualities, before the evaluation phase re-grounds it in a kitchen-to-kitchen scenario that reconnects the concept to the thesis’s domestic focus. Within the programmatic design research trajectory, this paper functions as a conceptual and design-space articulation that moves from interaction imaginaries toward empirically grounded, evaluable design qualities.

The study follows a two-phase mixed methodology. In the first phase, immersive speculative workshops were conducted on the metaverse platform Spatial.io with 18 participants of diverse backgrounds (6 XR specialists, 6 product/architectural designers, and 6 non-expert users), analyzed through reflexive thematic analysis. This analysis foregrounds two use-case domains in which bridging becomes most meaningful (Co-creation & Prototyping and Social Presence) and distills two design qualities: enabling anonymous and identified interactions, and strategic placement of portals. In the second phase, building on the initial portal proof-of-concept introduced in Paper I, a dedicated, higher-fidelity functional portal prototype was built and evaluated with 17 daily kitchen users using the Continuous Fuzzy Kano Model. This evaluation positions real-time responsiveness, especially bidirectional visibility, as a foundational prerequisite; strategic spatial placement as the primary experiential differentiator; and identity management as a secondary, contextual concern. Collectively, these contributions offer

empirically grounded design directions for situated, bidirectional XR interaction that reduce discontinuities in hybrid engagement.

My role as first author involved leading the study framing and workshop design, conducting the sessions, building and evaluating the functional VR prototype, leading the analysis, and writing the manuscript. Co-authors contributed to workshop and study design, refining the conceptual framing, supporting analysis, and contributing to the writing.

4.2.1 Manuscript 2: Portal

Anchoring Hybrid Presence: Designing Speculative Portals for Cross-Realm Interaction in XR

ANONYMOUS AUTHOR(S)



Bridging physical and immersive digital environments remains a central challenge for embodied XR interaction. We present Portal, a speculative XR concept exploring how spatial interfaces can mediate hybrid presence across virtual and real spaces. Following a two-phase mixed methodology, we first conducted immersive speculative workshops in Spatial.io with 18 participants. Thematic analysis revealed two value domains — co-creation/prototyping and social presence — and two design qualities: enabling anonymous and identified interactions, and strategic portal placement. In the second phase, we evaluated a higher-fidelity prototype with 17 participants using the Continuous Fuzzy Kano Model, finding that real-time responsiveness functions as a foundational prerequisite, strategic spatial placement serves as the primary experiential differentiator, and identity management operates as a secondary contextual concern. By employing a speculative interactive prototype as a design probe, this study contributes empirically grounded interaction principles for situated, bidirectional XR interaction in future hybrid ecologies.

CCS Concepts: • **Human-centered computing** → **Mixed / augmented reality**; **Collaborative interaction**.

Additional Key Words and Phrases: phygital, XR interaction, cross-realm interactions, spatial interfaces, mixed reality, speculative design, hybrid presence, portal

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1 Introduction

Connectivity across physical and digital environments has become a foundational requirement in contemporary interactive systems. Where this connectivity succeeds, immersive systems become genuine extensions of everyday life; where it fails, they remain isolated, context-bound tools demanding constant, disruptive switching. Yet physical and virtual environments remain experientially discontinuous, what Guan et al. term the “connectivity gap” [24]: a structural limitation that persists despite rapid advances in immersive technologies. Mixed reality systems, pass-through VR, and projection-based environments each address parts of this gap, but none support the embodied, situated, and socially meaningful continuity that everyday cross-realm interaction demands [53].

The need for principled cross-realm interaction design is pressing for several reasons. First, immersive technologies are increasingly present in everyday contexts, from remote work and distributed collaboration to domestic life and social connection, yet the interaction paradigms governing transitions between physical and virtual spaces remain underdeveloped [24]. Second, the consequences of poor cross-realm continuity are not merely technical. When users lose situational awareness, social context, or embodied grounding during realm transitions, the experiential costs include cognitive overload, social fragmentation, and loss of creative momentum [53]. Third, the design space for cross-realm interaction is still open: foundational questions about how interfaces should behave at the boundary between physical and virtual environments have not yet been answered, and premature commitment to specific architectures risks foreclosing promising directions before they are understood. A speculative probe approach, exploring design principles through imaginative engagement before committing to architecture, is therefore the appropriate starting point.

In this paper, we investigate how *phygital interfaces*, interactive artifacts that anchor interaction across physical and virtual spaces, can support intentional, real-time, and socially meaningful cross-realm exchange. We introduce the “Portal”: a speculative XR concept that draws on the metaphor of a spatial threshold, an anchored, bidirectional window embedded within everyday environments, and deploy it as a *design probe* to elicit interaction imaginaries around hybrid presence. Rather than proposing a deployable system, the Portal operates as a boundary object: a shared artefact through which diverse stakeholders can reason about future XR interaction. To investigate this concept, we followed a two-phase mixed methodology. In the first phase, we conducted immersive speculative workshops in Spatial.io with 18 participants spanning XR specialists, designers, and non-experts, analysed through reflexive thematic analysis. In the second phase, we developed a higher-fidelity Portal prototype and conducted a structured evaluative study to assess how participants experienced and prioritised Portal features under realistic immersive conditions.

The study is guided by two research questions: **(RQ1)** What use cases and user needs establish the value of bridging immersive digital and physical environments? **(RQ2)** Which interaction qualities spanning identity flexibility, real-time responsiveness, and strategic spatial placement should phygital interfaces prioritise to support bi-directional, situated XR interaction? This paper makes two contributions. First, we identify *Co-creation and Prototyping* and *Social Presence* as the primary contexts in which cross-realm phygital interaction is most meaningful, grounded in thematic analysis of workshop outcomes with 18 participants. Secondly, we articulate two design qualities: *Enabling Anonymous and Identified Interactions*, and *Strategic Placement of Portals* as the core interaction qualities that characterise how phygital portals can support hybrid presence, substantiated through both speculative co-creation and structured prototype evaluation.

2 Background

To ground our exploration of hybrid presence, we draw from three strands of literature. First, we examine how the reality-virtuality continuum and the concept of hybrid presence have been theorised and pursued in HCI research. Second, we review spatial interfaces and cross-realm interaction systems that attempt to bridge physical and virtual environments. Third, we trace the portal’s evolution from a navigational metaphor to a spatial interactive artifact, situating our speculative phygital approach within this trajectory.

2.1 Hybrid Presence and the Reality-Virtuality Continuum

The reality-virtuality continuum (RVC), introduced by Milgram and Kishino [40], describes a spectrum from purely physical environments to fully virtual ones, with mixed reality occupying the space in between. This foundational model has structured decades of XR research by providing a shared vocabulary for understanding how digital and physical elements coexist across different display configurations. A recent systematic review of 38 papers on transitions between realities confirms that the RVC remains the dominant conceptual framework in the field, while identifying persistent research gaps: fragmented terminology, a predominant focus on games and entertainment to the neglect of other application domains, and a lack of comprehensive models that address transitions at multiple scales simultaneously [42].

Within this continuum, achieving meaningful hybrid presence, the sense of being simultaneously engaged across physical and virtual realms, requires more than display fidelity. It depends on how users see, move, and act across environments. Avatars function as central instruments of cross-realm representation, shaping how users perceive themselves and others [32, 36, 41]. Design factors such as facial expressivity, body completeness, and motion fidelity influence both embodiment and interpersonal trust [4, 35]. Technologies such as Microsoft’s Holoportation and HoloLens 2 demonstrate how point cloud avatars and motion synchronisation can sustain embodied cross-realm interaction at scale [1, 50].

Hybrid presence has also animated broader visions of physical and digital convergence. Researchers have proposed diverse applications that extend across the continuum: avatar-mediated healthcare [48], memory-preservation environments for Alzheimer’s patients [45], and AI-generated avatars of the deceased [29] all reflect a shared interest in continuity of identity and relationship across realms. Frameworks such as Burns’ model of proto-Metaverses [12] and Lee’s notion of a real-world Metaverse [9] articulate visions in which digital and physical interaction are not parallel tracks but layered extensions of a single experiential continuum. Despite these advances, Xu identifies a persistent absence of shared, open, and perpetual virtual worlds in current systems [54], a gap that motivates research into interfaces capable of anchoring continuous interaction rather than merely enabling transition.

2.2 Spatial Interfaces and Cross-Realm Interaction

Research on spatial interfaces for cross-realm interaction has developed along two interconnected trajectories: the design of hybrid collaboration systems, and the investigation of persistent disconnectivity within and across those systems. Early work by Piumsomboon et al. examined proxemics and gestural techniques for MR co-presence [43], while Wright et al. proposed augmented duality, using camera-phone matching to align physical and digital spaces [51]. Zhou extended this by introducing reflective surfaces as spatial anchors for AR experiences, emphasising user perspective and orientation as key design parameters [55]. These trajectories converge in Guan’s concept of the Extended Metaverse, which leverages ubiquitous computing and IoT to enable seamless transitions across domains [23, 24], and in the Shared Metaverse model, which introduces adaptive avatars and privacy controls for contextual identity representation [5].

Volumetric telepresence represents the most technically advanced realisation of cross-realm interaction to date. Irlitti et al. present systems that support real-time, symmetrical, multi-user, partially distributed interactions by combining augmented and virtual reality technologies with face-to-face communication [30, 31]. Their evaluations demonstrate the successful integration of remote spaces through interpersonal positioning and gestural referencing, sustaining spatial coordination and grounded co-presence in hybrid workspaces. Despite this progress, these systems remain technically demanding and context-specific, and persistent disconnectivity gaps continue to emerge wherever users lose experiential continuity when moving between realms [25].

These gaps are not exclusive to XR environments. In hybrid meeting contexts, Dybbøe et al. identified three core design trade-offs, namely wide field of view versus clear locus of attention, polished design versus expressing limitations, and autonomy versus assistance, through a Research-through-Design process centred on a tabletop telepresence robot [17]. These tensions directly parallel the challenges facing phygital portal interfaces, where remote agency, spatial awareness, and co-located participation must be continuously negotiated. Borowski et al. address related challenges through Spatialstrates, a platform that bridges 2D desktop and 3D immersive environments via spatial transclusion, embedding a slice of one space into another to enable collaboration across heterogeneous interfaces [8]. Collectively, these works establish that sustaining meaningful interaction across physical and virtual realms requires interfaces that are not merely technically interoperable but spatially, socially, and contextually aware.

2.3 The Portal as Transitional and Interactive Spatial Artifact

The design of transitions between realities has been studied as both a technical and an experiential problem. Husung and Langbehn conducted one of the earliest systematic comparisons of VR scene transition techniques, evaluating six approaches, namely Cut, Fade, Dissolve, Transformation, Orb, and Portal, across presence, continuity, usability, and preference [28]. Their results established that interactive transitions providing a preview of the destination, specifically Portal and Orb, consistently outperformed passive techniques across all measures. Feld et al. revisited these findings under active task conditions and found that task load reverses some preferences: users prioritised efficiency over interactivity, favouring short transitions such as Cut when engaged in cognitively demanding work [19]. Pointecker et al. extended this line of inquiry by introducing a replica-enhanced transition concept, in which a digital twin of the physical environment mediates the passage from reality to VR, a process participants found cognitively supportive and spatially orienting [44]. Cools et al. approached transitions from the object rather than the environment level, introducing blended space techniques that allow objects to move between distinct VR and AR environments without requiring the user to transition themselves [13]. Das et al. examined the interaction modality of transitions, demonstrating that finger-worn button configurations, particularly the 2×2 arrangement, outperform continuous input methods in speed, accuracy, and preference for multi-reality navigation [14]. Gottsacker et al. synthesised these threads into an interaction model for VR world switching, showing that world-in-miniature previews support rapid spatial search while portal previews support fast post-transition orientation, and that direct, body-based interaction techniques yield the strongest usability and preference outcomes [22]. Taken together, these studies establish that transition effectiveness is not fixed but context-dependent: task load, transition frequency, environmental similarity, and interaction modality all shape whether a technique supports or disrupts cross-realm continuity.

Portals themselves have evolved from navigational shortcuts into a distinct class of spatial artifact. Culturally embedded in architecture, mythology, and science fiction, and technically realised in projects from Google’s Starline to the real-time inter-city video portals linking Lithuania and Poland (<http://portals.org/>), portals in XR research are conceptualised as spatial devices for navigation, co-presence, and transition. Zuo et al. describe XR portals as

soft gateways to services and spaces [57], while Bajorunaitė et al. show that embedding real-world cues in virtual environments reduces disorientation and strengthens immersion [7]. Medeiros et al. approach portals through proxemic interaction design, using them to maintain user awareness of the physical world during immersion [39]. Atkins et al. analyse teleportation’s cognitive effects in portal-based navigation [6], and Schmalstieg’s SEAMS system models inter-world connections through portal metaphors [47]. Portals also serve collaborative functions: Kunert et al.’s photoportal supports spatial referencing and shared awareness across distributed users [34]. Ablett et al. demonstrate that portals can function as persistent interactive windows for action at a distance, enabling users to interact with remote objects without teleporting themselves, outperforming Point & Teleport in speed, pointer actions, and cognitive load when physical obstacles are present [3]. However, Tseng’s work on perceptual manipulation identifies ethical and cognitive risks in immersive portal transitions, particularly where the perceptual boundary between environments is ambiguous [38].

Building on this landscape, our study repositions the portal not as a navigational mechanism but as a *speculative phygital interface*, an artifact designed to anchor interaction across physical and virtual realms and to provoke reflection on how future XR systems might support sustained, situated, bidirectional cross-realm exchange. Drawing on established design principles such as real-size representation [54], real-time interaction [15, 49], spatial anchoring [57], spatial translucency [8], and parallax-enabled depth cues [27], we frame the Portal as a site of social, experiential, and embodied continuity. Unlike the portal literature, which largely emphasises transition and navigation, our workshops explore how such an interface might sustain ongoing situated interaction, balancing visibility, privacy, identity, and co-presence within hybrid environments.

3 Methodology

This study investigates how speculative interfaces can inform the design of seamless, anchored experiences between physical and digital environments, and how the resulting design qualities can be evaluated through high-fidelity prototyping in immersive VR. We adopt a mixed methods design research approach that combines speculative co-creation with structured empirical evaluation, treating the two phases not as a linear validation pipeline but as a dialogic process in which functional experience reshapes, refines, and occasionally challenges what speculation alone had proposed. This approach is grounded in the tradition of Research through Design [56], which positions design artifacts as instruments of inquiry rather than merely outcomes of research. Within this tradition, speculative design methods allow researchers to explore interaction paradigms that do not yet exist, surfacing user needs, values, and imaginaries that conventional usability studies cannot access [16]. By deploying the Portal first as a speculative probe and later as a functional prototype, we move iteratively between the generative and the evaluative, using each phase to inform and interrogate the other.

The research unfolds across two phases. In the first phase, we sought to identify critical scenarios where a bridging interface could enable meaningful cross-realm interaction, and to explore the interaction qualities such an interface might need to support hybrid presence. This phase prioritises breadth and imagination, drawing on diverse stakeholder perspectives to open up the design space. In the second phase, we translated the design qualities identified in the first phase into a functional VR prototype and conducted a structured evaluation to assess how participants experienced and prioritised Portal features under realistic immersive conditions. This phase prioritises depth and specificity, using quantitative preference modelling to ground the speculative findings in measurable user response. Together, the two phases construct a research trajectory from imagination to evidence, offering both conceptual framing and empirical grounding for the design of future cross-realm phygital interfaces.

Table 1. Two-phase research design overview.

Phase	Method	Participants	Procedure	Output
Phase 1 Speculative Workshops	Speculative co-creation in <i>Spatial.io</i>	N = 18 XR Specialists (6) Designers (6) Non-Experts (6)	5-phase workshop protocol: (1) Orientation (2) Platform exploration (3) Individual ideation (4) Group discussion (5) Reflection <i>Analysis:</i> Reflexive thematic analysis	2 use case domains 3 design qualities: · Identity flexibility · Real-time responsiveness · Strategic spatial placement 2 speculative scenarios
Phase 2 Prototype Evaluation	Functional VR prototype + CFKM questionnaire	N = 17 Daily kitchen users (median XR familiarity: 2.2/5)	(1) Free exploration (10 min) (2) Facilitator-guided portal experience (3) Post-experience questionnaire <i>Analysis:</i> Continuous Fuzzy Kano Model	EI rankings features Feature prioritization

Note. The three design qualities identified in Phase 1 serve as the structural thread connecting both phases. Phase 2 translates these qualities into evaluable prototype features, operating as a dialogic process of refinement rather than linear validation.

4 Study1: Speculative Workshops

Speculative workshops served as our primary method to provoke imaginative engagement and surface interaction-relevant insights. Three sessions with two-hour workshops were conducted virtually. Two authors acted as facilitators, guiding discussions and maintaining engagement. To accommodate dispersed participants, all sessions took place in *Spatial.io* [2], chosen for its accessibility and spatial affordances, including spatial audio, scene creation, and avatar expressivity. These features supported situated, socially engaging interaction while allowing participants to speculate on hybrid presence through lived experience. The platform’s familiarity from use in virtual exhibitions and co-working events also aided onboarding. Sessions were scheduled flexibly to ensure full participation.

4.1 Study 1: Participants

We recruited 18 participants across three groups: designers without XR experience, XR specialists, and non-experts in both XR and design. Purposive sampling [18] ensured relevance to the study’s aims, complemented by snowball sampling to diversify perspectives. **Non-Users:** Individuals with little XR experience but general interest in technology or design. They contributed outsider perspectives on barriers, opportunities, and usability of immersive systems. **XR Specialists:** Professionals with at least two years of XR experience in interaction design, UX, or immersive storytelling. They offered insights into current limitations and emerging practices in spatial computing. **Non-XR Designers:** Practitioners from product, fashion, industrial, or architectural design (minimum two years of experience), whose expertise helped examine how physical design principles translate into virtual and hybrid contexts.

ID	Gender	Experience & Age	ID	Gender	Experience & Age
P1	Female	4 year, 26	P2	Male	3 year, 29
P3	Male	6 year, 28	P4	Female	3 year, 29
P5	Female	4 year, 29	P6	Female	6 year, 29
X1	Female	5 year, 30	X2	Female	6 year, 30
X3	Female	7 year, 30	X4	Female	6 year, 31
X5	Male	7 year, 32	X6	Male	5 year, 27
N1	Male	6 year, 32	N2	Male	12 year, 40
N3	Female	3 year, 25	N4	Male	10 year, 38
N5	Male	3 year, 28	N6	Female	3 year, 26

Table 2. Participant Demographics: **P**1–6 are Non-XR Designers, **X**1–6 are XR Specialists, and **N**1–6 are Non-Users.

4.2 Study 1: Workshop Structure

Workshops unfolded entirely within the virtual environment to stimulate creative envisioning of anchored hybrid experiences. Each followed a five-phase structure: **Phase 1: Introduction and Orientation (10 min)** After consent and orientation, participants were primed with a speculative scenario: “*Imagine a world where seamless, real-time connection between physical and virtual realms is ubiquitous.*” This prompted imaginative thinking unconstrained by current technology. **Phase 2: Exploration of Spatial.io (30 min)** Participants explored multiple 3D environments to familiarize themselves with spatial interaction affordances and reflect on usability gaps. This served as experiential grounding before ideation. **Phase 3: Individual Ideation (30 min)** Participants addressed three prompts focused on future cross-realm use cases: (1) when to access the physical world from the Metaverse, (2) when to connect instantly from physical to virtual, and (3) what features a bridging interface should include. **Phase 4: Group Discussion (40 min)** Participants were introduced to a speculative “Portal” embedded within *Spatial.io*—a mid-air, bidirectional gateway linking the virtual space with a photorealistic remote environment (e.g., a kitchen). Presented at a 1:1 scale with parallax depth, the Portal acted as a conceptual probe and spatial reference for embodied discussion. The Portal was not presented as a prototype but as a speculative design probe and boundary object. Participants were asked: “*What interaction qualities must this Portal support to realize these use cases?*”, prompting collaborative elaboration on embodied, spatial, and social dimensions of hybrid interaction. The conversational structure encouraged fluid dialogue and collective construction of ideas. **Phase 5: Reflection (10 min)** Participants reflected on both the speculative process and their evolving understanding of hybrid presence. This final stage captured meta-level insights into speculative methods as tools for surfacing design imaginaries and experiential needs.

4.3 Study 1: Analysis

Following Braun and Clarke’s reflexive thematic analysis [10], we conducted a multi-stage analysis to identify patterns in participants’ reflections on cross-realm interaction. Two researchers independently reviewed recordings, ideation boards, and field notes. Triangulation across modalities captured both verbal insights and embodied cues such as avatar movement and spatial gestures. The first coding cycle used open, inductive coding to identify phenomena like “*persistent context*”, “*single control interface*”, “*adaptive workspace boundaries*”, and “*privacy mode*”. A second, axial cycle clustered these into broader themes—*seamless integration*, *user control presence*, and *adaptive configuration*. Themes were refined through memoing and peer debriefing among co-authors specializing in design research and XR. This

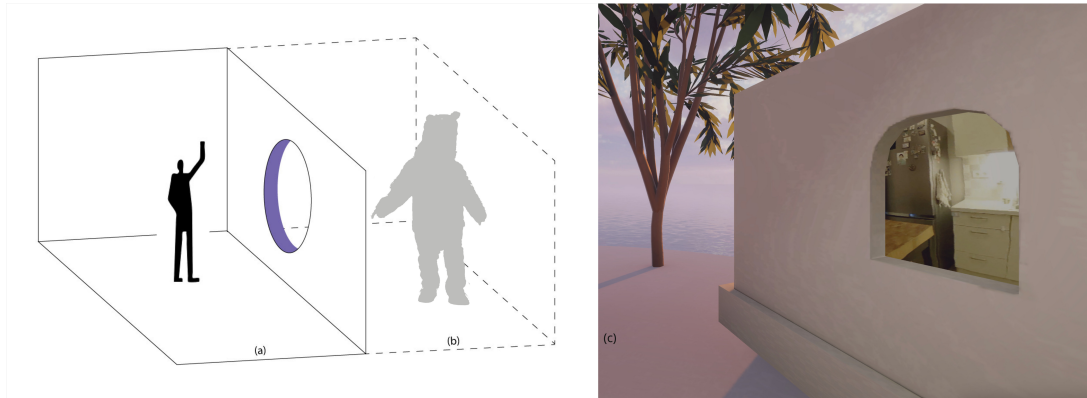


Fig. 1. **Speculative Bridging Interface:** (a) remote physical environment, (b) immersive Metaverse space, and (c) the speculative “Portal” within *Spatial.io*. The Portal appeared as a mid-air threshold linking users to a 180° panoramic representation of a physical space, provoking discussion on hybrid presence and cross-realm interaction.

iterative process enhanced analytic credibility grounding implications in participants’ situated imaginaries rather than abstract technological assumptions.

4.3.1 Speculative Scenarios. To translate analytic insights into design space, we developed two speculative scenarios derived from workshop themes. Using affinity mapping, authors organized ideas into shared motivations, emotional triggers, and contextual drivers. These highlighted recurring desires for spontaneous collaboration, embodied expressiveness, and context-aware responsiveness—capabilities not yet realized in current XR systems. The Portal functioned here as a *hybrid boundary object* embodying three identified design qualities: *real-time responsiveness*, *identity flexibility*, and *spatial anchoring*. Rather than prescriptive use cases, the scenarios act as critical provocations—plausible yet challenging futures that reveal how hybrid presence might be socially negotiated and materially grounded. They link participant imaginaries with the broader discourse on pervasive display ecologies, encouraging reflection on how XR interfaces can anchor human connection across realms.

4.4 Study 1: Findings

Our analysis foregrounds two emergent domains of use: **(1) Co-creation and Prototyping**, and **(2) Social Presence**. These domains capture overlapping needs—from seamless task execution across realms to nuanced forms of mediated togetherness. Drawing on thematic analysis of participant dialogue, ideation boards, and interactions with the speculative “Portal” interface, we identified patterns illustrating how hybrid presence might extend beyond current XR constraints. The analytic process leading to these findings is visualized in Figure 2. The findings are structured as narrative insights supported by direct quotes that expose user needs such as adaptive workspace boundaries, single control interface, and context-aware placement. For each domain, we articulate corresponding design implications and conclude with a speculative scenario. Grounded in participants’ perspectives, these scenarios serve as provocations—thought experiments that envision how hybrid experiences might feel, look, and operate in future systems. Together, they reveal the experiential gap between digital and physical spaces and propose conceptual directions for designing across it.

4.4.1 Use Case 1: Co-creation & Prototyping. A key theme to emerge was the desire for fluid, embodied creative engagement across physical and virtual environments. Participants emphasized the friction they experience when

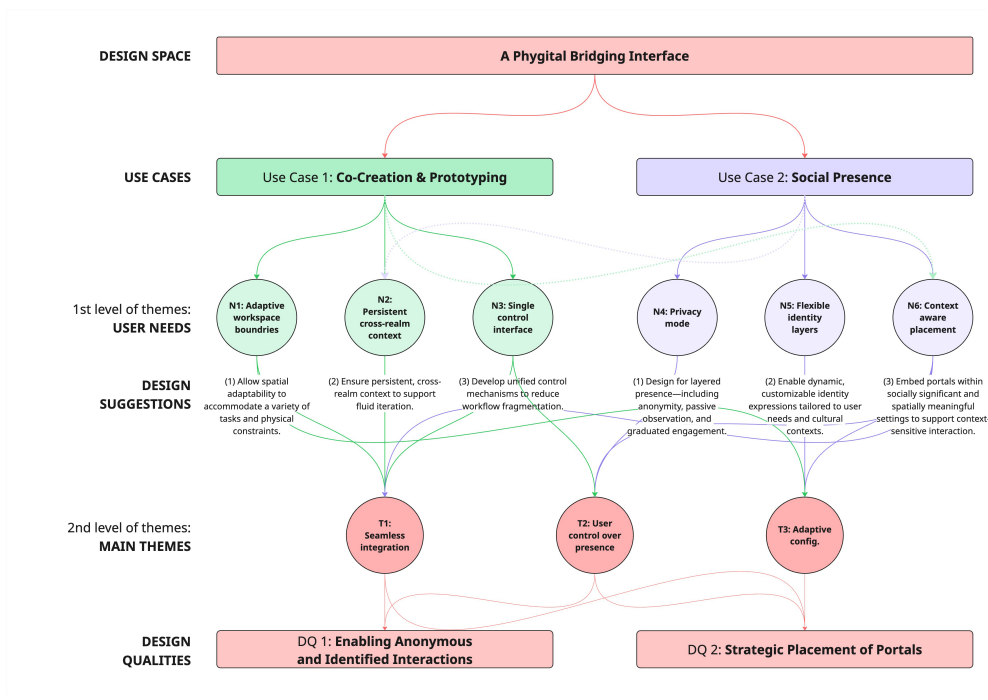


Fig. 2. Visual synthesis of thematic analysis process: from user needs (first-level codes) to main themes and design qualities. Based on speculative design workshops, we identified two primary use cases; each mapped through user needs, design suggestions, and cross-cutting themes. The diagram shows how layered insights were abstracted into three core interaction themes and two design qualities, forming a framework for future phygital bridging interfaces.

transitioning between immersive design and real-world prototyping. These discontinuities were especially evident in activities involving material interaction—such as cooking, handcrafting, product design, and choreography.

P1 noted, “I wish I could grab a virtual 3D model in VR and have it instantly fabricated on my workbench without exporting or importing files.” N5 described current workflows as “disjointed” and added, “A single interface that lets me stay in VR while controlling my CAD and CNC tools, so I don’t have to keep switching contexts.” N6 emphasized how such switching hinders creativity: “I would prefer to draw my designs in VR and see real-time updates on my physical prototype so my creative flow is not interrupted.” Similarly, N1 reflected on cognitive strain: “A phygital bridge that syncs my headset with my 3D printer and tools means I never lose my train of thought.” These comments highlight three recurring challenges in hybrid making: *multi-tasking* (P1, X1, N3), *fragmented creativity* (N3, N6), and *cognitive overload* (N1, X5)—underscoring a gap between current XR tools and the needs of material workflows. Participants imagined speculative systems that reframe co-creation itself. They desired hybrid platforms that preserve creative momentum, reduce mental switching costs, and enable tangible feedback loops between design and fabrication. While often grounded in professional contexts like engineering, these aspirations extended to everyday activities—cooking together remotely, performing dance routines, or crafting across distances. In all, the phygital interface was imagined as a connective threshold—anchoring a continuous, embodied design process across realities.

Implications for Future Phygital Bridging Artifacts. Participants outlined what such systems would require in terms of interface quality and experience (see Table 3). A major concern was the preservation of *persistent cross-realm*

context (N2)—synchronizing annotations, version histories, and spatial alignments across both digital and physical prototypes. This would prevent rework or confusion when switching contexts. A unified system could abstract control of both virtual toolsets and physical devices (e.g., CNC routers, 3D printers, or smart kitchen tools), reducing reliance on disjointed platforms. Participants also emphasized spatial adaptability. The notion of **adaptive workspace boundaries (N1)** recurred—allowing reconfiguration of interface size, scale, and position to accommodate different embodied practices. Equally valued was a **single control interface (N3)**, integrated within the immersive environment to ensure **seamless integration (T1)** and **user control over presence (T2)**. For instance, a dancer might collapse the interface to tabletop scale to fine-tune footwork or expand it for full-body rehearsals; a chef might project a remote assistant’s gestures onto a kitchen counter in real time. These reflections converge into three design suggestions: **(1) Adaptive Workspace Boundaries**: allow spatial adaptability across tasks; **(2) Persistent Cross-Realm Context**: ensure continuity between realms; and **(3) Single Control Interface**: unify control mechanisms to reduce fragmentation. Collectively, these suggestions outline how phygital portals might support embodied, uninterrupted, and distributed creativity.

User Need:	Qualities	Rationale
<i>N1: Adaptive Workspace Boundaries</i>	Dynamic scaling and configurable placement of the interface	Supports ergonomic interaction with varying prototype sizes and materials, extending beyond spatial limits of traditional MR.
<i>N2: Persistent Cross-Realm Context</i>	Bidirectional syncing of edits and annotations across digital and physical versions	Maintains design continuity, reduces context-switching, and prevents loss of intent between realms.
<i>N3: Single Control Interface</i>	Unified control panel abstracting virtual and physical tools	Simplifies workflow by reducing platform friction and cognitive overhead compared to siloed XR or desktop systems.

Table 3. Design considerations for phygital co-creation: key user needs, interface qualities, and rationales identified through speculative workshop scenarios. These insights inform the design of Portals as context-sensitive bridging interfaces between physical and virtual realms.

Speculative Scenario. Drawing on these insights, the following scenario illustrates how phygital bridging might enable new forms of hybrid co-creation:

Pawel, a drone engineer working remotely from Europe, spends hours daily in his company’s Spatial.io workspace—yet feels detached from his Silicon Valley team. To bridge this divide, the company installs a phygital interface: a life-sized, parallax-enabled window embedded in the lab, mirrored by a virtual counterpart in Pawel’s studio. Contextually placed (N6) beside fabrication tools, it remains accessible and purpose-aligned. Within VR, Pawel accesses his single control interface (N3) to synchronize inputs across realms—switching between live feeds, manipulating 3D schematics, and communicating via spatial audio. As he sketches a drone chassis, edits—through persistent cross-realm context (N2)—update instantly across both prototypes. To refine details, he invokes adaptive workspace boundaries (N1), zooming into a joint and resizing the interface to match his gestures. Around him, the portal adjusts to his movements, maintaining flow. In the lab, colleagues annotate in real time. The portal becomes not just a display, but an extension of their shared creative space.

4.4.2 Use Case 2: Social Presence. The second key domain centered on fostering rich, flexible social presence across physical and virtual contexts. Participants envisioned how a phygital bridging interface might support diverse modes of

being together—from ambient co-presence to anonymous observation, and from culturally sensitive participation to adaptive identity expression. Modulating presence was particularly valued. P5 noted, “I prefer to stay unobtrusive—observing my children’s plays in VR without anyone knowing I’m there.” N1 highlighted the therapeutic benefit of privacy: “I wish I could attend therapy in a private virtual booth with my virtual identity so I can speak more freely.” X4 underscored cultural contexts, noting that hybrid access could expand participation for groups often excluded from physical events. X6 envisioned inter-generational connection: “My parents could attend my virtual graduation without fully entering the Metaverse—they could appear through a portal at home and still feel present.” These reflections reveal recurring challenges in hybrid social interaction: enabling *genuine co-presence* (P5, P6, X1, X2, X4), allowing *anonymity and selective presence* (P2, P6, N1, N4, N5), and supporting *flexible identity presentation* (X3, X5, X6). Such affordances were seen as vital in contexts like performances, rituals, education, and therapy—where full immersion may be impractical or undesirable.

Implications for Future Phygital Bridging Artifacts. Participants proposed several characteristics for nuanced, accessible interaction across realms (see Table 4). They emphasized the importance of a *privacy mode* (N4)—including silhouette avatars, blurred presence indicators, and selective audio channels—to allow graded visibility and engagement, mitigating anxiety associated with constant exposure. Many also advocated for *flexible identity layers* (N5), enabling toggling among persona attributes such as age, gender, or cultural expression. These dynamic identities support contextually appropriate representation across hybrid settings. Finally, the placement of the interface itself mattered. Participants (P2, P3, P5, P6) emphasized *context-aware placement* (N6)—embedding the artifact within socially meaningful spaces such as living rooms, classrooms, or public squares. These features collectively sustain *user control over presence* (T2) and *adaptive configuration* (T3). From these reflections emerge three speculative design suggestions: (1) Privacy Mode: enable layered presence, anonymity, and graduated engagement; (2) Flexible Identity Layers: support customizable self-representation responsive to cultural and emotional contexts; (3) Context-Aware Placement: embed portals within socially and spatially significant settings to foster intuitive interaction. Together, these reframe phygital interfaces as relational infrastructure—enabling people to traverse social, emotional, and cultural boundaries with agency and care.

User Need:	Qualities	Rationale
N4: Privacy Mode	• Silhouette or “ghost” avatars • Selective audio channels	Supports passive observation and controlled engagement, reducing social anxiety compared to always-on presence.
N5: Flexible Identity Layers	• Quick toggles for gender, age, cultural markers • Contextual avatar presets	Enables users to adopt dynamic personas suited to varying social, cultural, or emotional contexts.
N6: Context-Aware Placement	• Embedded interface within relevant social spaces • Adaptive spatial audio	Aligns the phygital artifact’s location with intended social activity, promoting intuitive and meaningful engagement.

Table 4. Design insights for phygital social presence: user needs, associated interface qualities, and rationales addressing identity, privacy, and social interaction across physical and virtual realms—offering alternatives to conventional always-on or avatar-based systems.

Speculative Scenario. The following scenario illustrates how phygital social presence might unfold in practice:

Mariam, a retired teacher in Abu Dhabi, feels distant from her grandchildren who spend much of their time in virtual environments. To bridge this divide, her family installs a softly lit phygital portal—an elegant archway beside her kitchen

table. Equipped with depth cameras, directional audio, and a holographic display, it opens a window into her granddaughter Aisha’s virtual nursery. Mariam activates **privacy mode (N4)**: her presence becomes a translucent silhouette. She watches quietly, savoring the moment. When ready, she toggles to a contextual avatar resembling her younger self in traditional attire—enabled by **flexible identity layers (N5)**—appearing with warmth and cultural resonance. As Aisha draws with her stylus, Mariam’s soft voice echoes through, and their shared sketches persist across sessions via **persistent cross-realm context (N2)**. The portal’s **context-aware placement (N6)**—tuned to her domestic rhythm—anchors the exchange in everyday life. More than a screen, it becomes a cultural and emotional threshold—a bridge through which care and creativity flow between generations.

4.5 Design Qualities

This study explored how designers, XR specialists, and non-experts imagine bridging physical and immersive digital spaces through speculative use cases. By engaging participants with diverse backgrounds, we surfaced a layered understanding that combines technological foresight, interaction design, and lived experience. XR specialists emphasized emerging technical affordances; designers foregrounded embodied workflows and spatial reasoning; and non-users highlighted comfort, social dynamics, and relevance. Rather than treating the Metaverse as a destination or seamless immersion as the primary goal, we frame *phygital bridging interfaces* as tools for intentional, context-sensitive interaction across realms. Envisioned as “portals,” these interfaces operate as situated boundary objects, selectively activated, socially meaningful, and responsive to users’ environments and intentions.

Figure 3 visualizes how our analysis progressed from concrete user needs, such as adaptive boundaries and privacy control, toward overarching interaction themes and core design qualities. This framework articulates a design vocabulary for future phygital systems that support both task-oriented and socially situated interaction. We highlight two key qualities: (1) Enabling Anonymous and Identified Interactions—supporting flexible modulation from anonymous to symbolic or verified presence; and (2) Strategic Placement of Portals—embedding interfaces within meaningful physical or virtual locations to enhance legibility, comfort, and contextual fit. Although these qualities emerged within distinct use case domains (co-creation and social presence), they are mutually reinforcing and foundational to anchored hybrid interaction.

4.6 Design Quality 1: Enabling Anonymous and Identified Interactions

As users navigate hybrid environments, they encounter shifting dynamics of visibility, vulnerability, and self-expression. A core design quality identified in our workshops was the ability to fluidly transition between **anonymous and identified modes of interaction**. As shown in Fig. 3, this quality builds on user needs such as *privacy mode (N4)* and *flexible identity layers (N5)*, and is closely tied to the theme of *user control over presence (T2)*. We define this quality as a system’s capacity to support dynamic identity modulation, through verified, pseudonymous, or symbolic presence, with the ability to toggle not only at entry but also mid-interaction.

Avatars function as central vessels of embodiment [26]. While prior research has linked avatar fidelity to trust and presence [11, 20, 33, 35, 36, 46], our findings foreground *user control over visibility* as equally critical. Participants welcomed symbolic or abstract representations, alongside the option to transition to verified identities during collaboration or social engagement. Some (P1, P5, X3, X5, N3, N4, N6) described anonymity as enabling uninhibited expression, while others (P6, X5, N2) cautioned that ambiguous identity transitions could undermine trust—calling for intentional toggles and clear cues indicating current identity state. This ability to move between roles—observer, contributor, collaborator—reflects the layered interaction needs captured in our framework. Designing for fluid identity supports

both *privacy* and *accountability*, fostering inclusive participation in phygital contexts. Future research should examine how such transitions shape engagement, trust-building, and participation across situated XR interaction settings.

4.7 Design Quality 2: Strategic Placement of Portals

In hybrid environments where physical and virtual domains converge, the **placement** of a phygital portal plays a critical role in shaping how and when it is used. We define *strategic placement* as the intentional positioning of the interface to align with social, emotional, and practical context—whether for observation, co-presence, or active collaboration. Beyond logistical concerns, placement acts as a socio-spatial cue that informs expectations, regulates accessibility, and influences the emotional tone of interaction.

Participants conceptualized the portal not as fixed hardware, but as a configurable, responsive layer embedded within everyday routines and social contexts. P2 suggested positioning portals near informal gathering spaces such as kitchens or living rooms, framing them as soft, everyday presences—a “drop-in” window for casual cross-realm moments. P3 and P5 proposed embedding portals in public venues like festivals or community centers to invite shared experiences while preserving controlled participation. P6 emphasized agency over *where* and *how* the portal manifests: “If I can choose where it appears in my room, I might use it more often—it would feel less intrusive.” These reflections point to three concrete placement principles. First, *configurable placement*: portals for casual interaction belong near shared domestic spaces, while those for focused collaboration suit studios or offices. Second, *adaptive spatial behavior*: interfaces should flex to accommodate physical conditions and user activity, so that whether seated, standing, or moving, users experience the portal as integrated rather than intrusive. Third, *social legibility*: placement should communicate intent—semi-private configurations signal passive or observational use, whereas prominent, open-facing placements indicate readiness for engagement or co-creation.

These perspectives correspond to the user need of *context-aware placement* (N6) and connect to *adaptive configuration* (T3) in our framework (Fig. 3). They also intersect with co-creation scenarios—such as situating portals in flexible workspaces (N1) or supporting persistent context (N2)—illustrating that placement is socially as well as spatially adaptive. Together, these insights shift emphasis from technical immersion toward *situated usability*, underscoring that phygital interfaces must be *placed with purpose* to support diverse relational, emotional, and functional needs in hybrid life. Future work should explore how environmental affordances (e.g., lighting, soundscapes, furniture layout) and cultural practices shape placement preferences, and how these decisions influence interaction quality and affect in cross-realm experiences.

5 Study 2: Evaluation of Design Qualities

In the second phase, we evaluate the design qualities derived from the speculative workshops by means of a functional portal prototype deployed within a controlled VR environment. The context and environment is adopted from the speculative scenario of second use case, which was derived from the findings in the first phase. This phase evaluates the speculative outcomes by classifying user feedbacks based on satisfaction and dissatisfaction dimensions through presenting functional prototype.

5.1 Study 2: Development

To address the limitations identified in the first phase, we developed a VR experience featuring a fully functional portal prototype of higher fidelity. The implementation comprised two distinct kitchen environments, each populated

with fully interactive assets including appliances, a faucet, cabinets, drawers, and food items, enabling naturalistic, task-oriented interaction representative of everyday cross-realm scenarios.

The virtual environments and portal artifact were modeled in Blender and implemented in Unity. For user interaction, we integrated the Meta SDK and developed further custom interaction scripts. The portal artifact itself was built upon an adapted version of Tom Goethals' Portals for VR package [21], which provided custom shaders and core interaction algorithms that were further tailored to the requirements of this study. For multiplayer functionality, we integrated full-body Meta Avatars and Photon networking library and services. Although the visual fidelity of these avatars remains stylized, features such as estimated lower-body pose reconstruction and lip synchronization contributed to a stronger sense of co-presence compared to the first phase.

5.2 Study 2: Participants

We recruited 17 participants through a close-circle call, all of whom self-identified as daily kitchen users. The sample was predominantly female (13 women, 4 men), with ages distributed relatively evenly across three brackets: 18–24 ($n=4$), 25–34 ($n=7$), and 35–44 ($n=6$). Participants were purposively selected to reflect the domestic, task-oriented context of the portal prototype rather than XR expertise. Accordingly, while most reported only brief or incidental familiarity with XR technologies (median: 2.2/5), all identified as frequent kitchen users. Demographic profiling reinforced this orientation: participants spent a median of 1.57 hours per day in the kitchen and cooked on average 3.84 days per week, indicating meaningful lived experience of the domestic context under evaluation.

5.3 Study 2: Study Structure

Although the experience was designed to run as a standalone application on Meta Quest devices supporting simultaneous multi-user sessions, individual study sessions were conducted with one participant at a time to maintain methodological control. Prior to each session, participants were briefed on fundamental VR interaction mechanics and the structure of the experience. This onboarding also included disclosure of potential temporary side effects associated with VR use, in accordance with standard ethical practice. Each session was organized into two sequential steps. In the first step, lasting approximately ten minutes, participants were allowed to explore the virtual kitchen environment freely. This exploration period served a dual purpose: familiarizing participants with the core interaction mechanics and fostering a sufficient sense of immersion prior to the portal experience.

In the second phase, the facilitator joined the session, operating from the second kitchen environment and initiating the portal experience. The portal experience is structured in parallel to design qualities identified in Phase 1 — anonymous and identified modes of interaction^{4.5}, real-time responsiveness, and strategic spatial placement^{4.6} — ensuring that each quality was systematically foregrounded during the session. The facilitator and participant engaged in conversation through the portal, discussing kitchen-related tasks, communication, and plausible everyday use cases. To ensure participants experience each portal functionalities, the facilitator used two guidance strategies: direct guidance, in which specific features or functionalities were explicitly introduced; and indirect guidance, in which participants were prompted to locate objects outside their current field of view in the adjacent kitchen, organically motivating portal interaction without prescriptive instruction.

To evaluate the design quality of strategic spatial placement, participants were invited to reposition the portal across a predefined set of locations by selecting marked spots via point-and-click interaction. At each placement, the portal's shape, size, and position were systematically switched to expose participants to a range of spatial configurations. Additionally, replacing the portal instantiated unidirectional visibility conditions, in which participants could hear the

facilitator’s voice but could not see his avatar, which demonstrates evaluation of asymmetric presence and control over presence.

Following the interactive session, participants completed a post-experience questionnaire developed using the Continuous Fuzzy Kano Model (CFKM), as introduced by Wu and Wang [52]. The CFKM was selected over the classical Kano model and its discrete fuzzy variant (FKM) for its capacity to address the inherent imprecision and uncertainty in experiential preference reporting — particularly relevant in novel VR interaction contexts where responses may resist crisp categorical assignment. Unlike the standard Kano approach, the CFKM treats participant responses as continuous fuzzy values, enabling finer differentiation between features occupying ambiguous or transitional positions within the quality space. This approach also supports researchers by prioritizing the attributes based on the EI values, instead of traditional Kano classifications. The questionnaire comprised 32 items structured as 16 functional–dysfunctional pairs, where each functional item probed how participants felt about a feature’s presence and its dysfunctional counterpart probed its absence. Items were organized into four clusters. The first, *Baseline Portal Utility*, assessed general acceptance of the portal concept prior to feature-level evaluation. The remaining three clusters were aligned with the design qualities identified in Phase 1 — *Real-Time Responsiveness*, *Anonymous and Identified Interactions*, and *Strategic Spatial Placement* —the last of which additionally featured asymmetric presence as a distinct evaluative sub-dimension.

5.4 Study 2: Analysis

By following the analytical method of the CFKM [52], the Individual Satisfaction Score (S_n) for each participant was computed by first normalizing the raw functional (F) and dysfunctional (D) survey responses from the $[0, 100]$ scale to the interval $[-1, +1]$ using the transformation $x = (r/50) - 1$, where r denotes the raw score.

$$S_n = \frac{x + (1 - y)}{2} - 0.5 \quad (1)$$

To calculate the S_n for each participant, we employed bilinear interpolation rather than simply mapping normalized responses (F and D) onto a discrete 5×5 Influence Value matrix. Because participants provided responses on a continuous scale—rather than at the five discrete anchor points—bilinear interpolation allowed us to derive a precise Influence Value for each (f, d) pair 1. This yielded the individual S_n scores. The aggregate Evaluation Index (EI) for each attribute was then determined by calculating the arithmetic mean of all valid S_n scores across participants, as expressed in Equation 2:

$$EI_i = \frac{1}{N} \sum_{n=1}^N S_{ni} \quad (2)$$

where N denotes the number of valid responses for attribute i and S_{ni} is the Individual Satisfaction Score of participant n for attribute i , producing a continuous measure bounded within $[-1, +1]$. Following the classification thresholds established by [52], attributes with an $EI > +0.50$ were categorized as One-dimensional, $+0.50 \geq EI > +0.20$ as Attractive, $+0.20 \geq EI > -0.20$ as Indifferent, and $EI \leq -0.20$ as Must-be.

5.5 Study 2: Findings

The CFKM analysis yielded Evaluation Index (EI) scores for each of the 16 feature pairs, enabling ranked prioritization. Figure 3 presents the full results.

Baseline Portal Utility General acceptance of the portal concept yielded mixed results. Remote kitchen connection via the portal for general communication purposes (Q1) ($EI = 0.189$) was classified as Indifferent, whereas shared

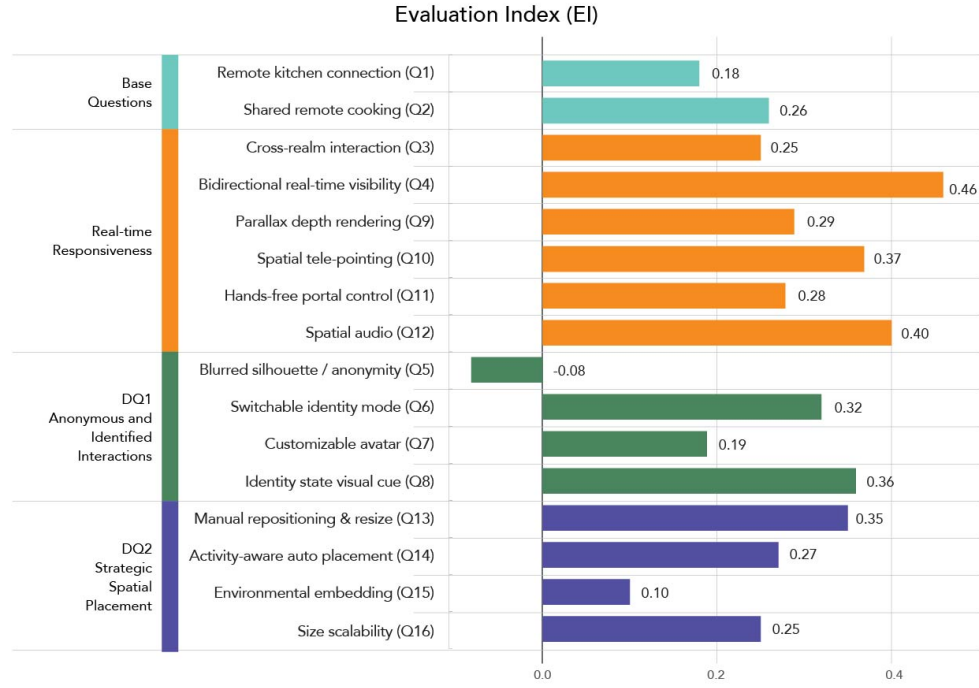


Fig. 3. Chart of the Evaluation Index (EI) of each question from CFKM methodology.

remote cooking with portal (Q2) classified as Attractive ($EI = 0.26$), suggesting that the portal's perceived value is contingent on meaningful use contexts — specifically, task-oriented collaboration rather than passive co-presence.

Real-time Responsiveness In this cluster, the results reveal that real-time responsiveness functions as a necessary technical substrate underlying. Bidirectional real-time visibility (Q4) has the highest score in the entire study with $EI = 0.46$, indicating that it is close to One-Dimensional classification in which satisfaction scales proportionally with its presence and dissatisfaction with its absence. The second and third features are also in this cluster; spatial audio (Q12, $EI = 0.40$) and spatial tele-pointing (Q10, $EI = 0.37$). These rankings indicate that multi-modal interaction methods are more desirable compare to Portal's fundamental features such as parallax effect and hands-free control.

Design Quality 1: Anonymous and Identified Interactions. Features within this cluster reflect participants' desire for fluid, controllable self-presentation across realms — consistent with the design quality's emphasis on dynamic identity modulation between anonymous and identified modes. Notably, the blurred silhouette anonymity mode (Q5) received the study's only negative EI score ($EI = -0.08$), indicating a net indifference tending toward mild aversion. This suggests that enforced or default anonymity is not a desirable configuration, and that participants value the *ability to choose* anonymity rather than anonymity as a default. This is corroborated by the comparatively higher scores for switchable identity mode (Q6, $EI = 0.320$) and identity state visual cue (Q8, $EI = 0.36$). The second, being the highest-priority feature within this cluster, indicates that *transparency* and *control* over one's identity state are the

core valued properties. Avatar customization (Q7, $EI = 0.19$) was classified as Indifferent, suggesting limited interest in avatar customization, possibly because participants preferred more realistic self-representation in the evaluated kitchen-to-kitchen scenario.

Design Quality 2: Strategic Spatial Placement. Features within this cluster consistently affirm spatial agency as a central valued property of the portal experience. Manual repositioning and resizing (Q13) emerged as the highest-priority feature in this cluster ($EI = 0.35$), underscoring that user control over the portal’s physical presence is a primary differentiator of experience quality. Size scalability (Q16, $EI = 0.25$) and activity-aware automatic placement (Q14, $EI = 0.27$) were similarly classified as Attractive, suggesting that both deliberate and context-responsive placement are valued affordances. Environmental embedding (Q15), however, was classified as Indifferent ($EI = 0.10$), suggesting that participants did not strongly differentiate between a spatially integrated portal and a floating interface. These results suggest that dynamic, spatially adjustable interfaces are preferred over those that are static and embedded in the environment.

Taken together, the EI rankings suggest a layered prioritization of portal design qualities. Features governing *Real-time Responsiveness* — particularly bidirectional visibility, spatial audio, and Spatial tele-pointing — establish the perceptual foundation of the portal experience, with bidirectional visibility standing close to the sole one-dimensional quality and thus the most critical threshold for user satisfaction. *Strategic Spatial Placement*, led by manual repositioning, emerges as the primary differentiator of experiential quality, indicating that users place significant value on agency and functionality over the portal’s physical presence in their environment. *Anonymous and Identified Interactions* features are broadly desired, yet the results nuance this: participants value transparency and dynamic control over their identity state more than anonymity per se, as evidenced by the negative EI of the blurred silhouette condition. These findings carry direct implications for the prioritization of design effort in future portal development, positioning responsiveness and spatial agency as foundational requirements and identity management as a contextually meaningful but secondary design concern.

6 Discussion

This study examined how phygital portal interfaces might bridge physical and immersive digital environments through a two-phase research design combining speculative co-creation with structured prototype evaluation. Phase 1 surfaced design qualities through participatory workshops with diverse stakeholders, while Phase 2 interrogated those qualities through direct interaction with a functional VR prototype evaluated using the Continuous Fuzzy Kano Model (CFKM). Together, the two phases construct a trajectory from speculative imagination to measured preference, not as a linear validation pipeline, but as a dialogic process in which functional experience reshapes, refines, and occasionally challenges what speculation alone had proposed. The findings collectively suggest that meaningful cross-realm interaction is not primarily a matter of technical immersion, but of *control* over identity, space, and the conditions of one’s own presence across realms. In the following, we reflect on the implications of these findings across three interconnected dimensions: the role of stakeholder perspectives and speculative methods in cross-realm design, the contribution of portal interfaces to adaptive and context-aware Cross-Reality systems, and the evaluative insights yielded by confronting speculative design qualities with functional prototype experience.

6.1 Reflections on Stakeholder Perspectives and the Role of Speculative Probes in Cross-Realm Design

To surface layered perspectives on cross-realm interaction, we brought together Designers (P1–P6), XR Specialists (X1–X6), and Non-Users (N1–N6) in speculative co-creation sessions within *Spatial.io*. This composition enabled

a multidimensional exploration of hybrid presence grounded in diverse forms of expertise, expectation, and lived experience. Distinct patterns emerged: XR Specialists emphasized real-time feedback and responsiveness; Designers foregrounded spatial arrangements, material metaphors, and symbolic affordances; and Non-Users highlighted cognitive load, unfamiliarity, and the importance of intuitive, opt-in interaction that respects autonomy and comfort. Speculative probes played a central role in bridging these perspectives. The shared immersive platform provided a neutral ground for dialogue, enabling participants to imagine beyond immediate constraints while remaining anchored in interactional concerns. Rather than enforcing consensus, the workshops surfaced productive tensions between immersion and control, continuity and separation, and abstraction and familiarity, all of which the Portal must negotiate. Designing *with*, rather than merely *for*, these perspectives yielded implications that accommodate complexity, enable configurability, and support inclusive, anchored XR interaction experiences.

6.2 Adaptive and Context-Aware Interfaces in Cross-Reality Systems

Beyond the specific use cases and design qualities identified in this study, our findings contribute to a broader discussion on adaptive and context-aware interfaces in Cross-Reality systems. As physical and immersive digital environments increasingly interweave, interaction design must move beyond static interface configurations toward systems that dynamically respond to users' spatial, social, and situational contexts. The Portal concept foregrounds this shift by treating the interface not as a fixed display or entry point, but as a responsive mediator whose behavior, visibility, and placement adapt to changing conditions across realms.

From a Cross-Reality perspective, the Portal operates as an interactional hinge between physical and virtual environments rather than a unidirectional gateway. This reframes cross-realm interaction as an ongoing negotiation of presence, identity, and attention, rather than a binary transition between worlds. The design qualities identified in our study, namely identity flexibility, real-time responsiveness, and strategic spatial placement, can be understood as core mechanisms through which adaptive behavior is enacted. Together, they point toward XR interfaces that sense and interpret contextual cues such as location, activity, social setting, and user intent, and adjust interaction modalities accordingly. Importantly, this adaptive framing emphasizes *intentionality* over automation. Rather than removing friction indiscriminately, participants envisioned systems that allow users to regulate how and when adaptation occurs, whether by modulating identity visibility, repositioning the interface, or shifting between observational and participatory modes. In this sense, context awareness is not merely a technical capability but an interactional resource that supports agency, legibility, and trust in hybrid environments.

6.3 Evaluation of Speculative Findings

The CFKM analysis enabled a structured evaluation of the speculative design qualities surfaced in Phase 1, translating participant imaginings into measurable preference data through direct interaction with a functional prototype. Rather than validating Phase 1 outcomes, this phase allowed us to interrogate them, revealing where speculative desirability holds under functional conditions, where it breaks down, and where new nuances emerge that speculation alone could not anticipate. The relatively low values in the data can be considered as indicators of insufficiency in the prototype implementation itself, or as signals prompting researchers to revisit and reinterpret the outcomes from the first phase. Additionally, the complexity of the interaction modalities and the novelty of such an interface may introduce a further evaluative barrier: participants lack established reference points in their daily experience against which to compare and assess these features, potentially compressing preference scores toward indifference rather than reflecting genuine disinterest.

The dominance of the Real-time Responsiveness cluster, anchored by bidirectional real-time visibility (EI = 0.46), spatial audio (EI = 0.40), and spatial tele-pointing (EI = 0.37), suggests that the perceptual immediacy of the connection functions as a foundational layer upon which higher-order experiential qualities depend. This aligns with prior work on social presence in mediated communication, where synchronous sensory feedback has been identified as a prerequisite for the sense of being there together [37]. Notably, bidirectional visibility approached One-Dimensional classification, meaning its absence would generate active dissatisfaction rather than mere indifference, a finding that positions it not as an enhancement but as an infrastructure-level requirement. The Strategic Spatial Placement cluster further extends this logic by demonstrating that once perceptual responsiveness is established, users prioritize agency over the portal's spatial configuration.

A particularly instructive divergence between the two phases concerns identity and anonymity. Workshop participants frequently expressed enthusiasm for privacy modes and silhouette-based anonymous presence, yet the CFKM evaluation revealed that enforced anonymity received the study's only negative EI score (EI = -0.08), indicating mild aversion rather than desire. This suggests that anonymity is valued as an option rather than a default condition, and that what participants imagined as desirable in the abstract did not translate uniformly into preference under functional conditions. This kind of divergence is precisely what the two-phase design was constructed to surface, and it illustrates the value of combining speculative and empirical methods rather than relying on either alone.

It should be acknowledged that the evaluation study was conducted within a deliberately constrained scenario in which two users connected through their respective kitchens. This scope limited the prototype's capacity to demonstrate interaction across heterogeneous realms, which is the cross-realm adaptivity that the broader portal concept envisions. Nevertheless, the results remain valuable as an empirical grounding for the design of novel interaction interfaces, offering transferable insights into user priorities that extend beyond the kitchen-to-kitchen configuration to wider contexts of spatially mediated communication.

Situating Portals within this broader Cross-Reality discourse positions them as speculative yet actionable explorations of how future XR systems might orchestrate adaptive behavior across heterogeneous environments. By grounding adaptation in spatial, social, and experiential considerations, this work contributes to an emerging understanding of XR interfaces as dynamic, context-sensitive assemblages, designed not only to connect realities, but to meaningfully align interaction with the rhythms and intentions of everyday life.

7 Conclusion

As physical and digital environments increasingly converge, the need to reconsider how immersive systems intersect with everyday life grows more pressing — not as parallel domains requiring switching, but as extendable, interleaved layers of human activity. This study examined how designers, XR specialists, and non-users imagine bridging interfaces that support intentional, context-sensitive interaction across physical and digital realms. Through speculative workshops conducted in *Spatial.io*, participants articulated use cases and interaction paradigms for a conceptual phygital “Portal.” Rather than treating seamless immersion as an unquestioned objective, they emphasised selective engagement, spatial sensitivity, and social legibility as central design concerns. The workshops surfaced not only technological imaginaries, but also emotional, cultural, and practical perspectives that shape hybrid presence in cross-realm settings.

Our analysis identified two domains in which phygital bridging is perceived as most meaningful: (1) *Co-creation and Prototyping* and (2) *Social Presence*. Across these domains, two design qualities characterize how phygital portal interfaces can support agency, contextual adaptability, and trust-based interaction: *Enabling Anonymous and Identified Interactions*, which foregrounds dynamic, user-controlled modulation between visible and anonymous modes of presence; and

Strategic Placement of Portals, which positions the interface as a socio-spatial artifact embedded within the rhythms and contexts of everyday life. Underlying both qualities, real-time responsiveness — particularly bidirectional visibility — emerged not as a design differentiator but as a foundational prerequisite: the perceptual infrastructure without which neither quality can be meaningfully enacted.

The second phase extended these speculative findings into empirical terrain by implementing a functional portal prototype and evaluating it through the Continuous Fuzzy Kano Model. This evaluation confirmed real-time responsiveness as an infrastructure-level requirement, established strategic spatial placement as the primary differentiator of experiential quality, and revealed that identity management operates as a contextually meaningful secondary concern. Critically, the evaluation also surfaced a meaningful divergence from workshop speculation: anonymity, which participants frequently expressed enthusiasm for in Phase 1, received the study’s only negative EI score in Phase 2, indicating that participants value the *option* of anonymity rather than anonymity as a default condition. This divergence is precisely what the two-phase dialogic design was constructed to surface, and it demonstrates the value of combining speculative and empirical methods rather than relying on either alone. The resulting EI-based hierarchy of portal features offers a practical prioritization framework for researchers and designers allocating design effort in this space.

Several limitations shaped the study. The speculative workshops were conducted entirely within *Spatial.io*, which offered a shared space for ideation but lacked affordances such as embodied sketching, real-time annotation, and shared manipulation that could further support co-creation. The platform could not fully reproduce the nuanced physicality and immediacy of hybrid or in-person sessions, and technical issues including intermittent lag and connectivity drops occasionally disrupted the flow — highlighting a central paradox of our inquiry: envisioning seamless hybrid presence while constrained by current XR infrastructures. The prototype evaluation was similarly bounded, conducted within a single kitchen-to-kitchen scenario that limited exposure to the cross-realm adaptivity the broader portal concept envisions. Future evaluations should test the Portal concept across more heterogeneous realm configurations and with populations holding greater XR familiarity, to assess whether currently Attractive features migrate toward One-dimensional classifications as baseline expectations mature.

Grounded in speculative scenarios, this work offers generative provocations rather than prescriptive solutions, using the Portal as a design probe to reason about future forms of anchored spatial interaction and hybrid embodiment. By foregrounding the experiential and relational conditions under which phygital connectivity gains value, this study contributes to HCI and XR research by offering empirically grounded design implications for situated, bi-directional XR interaction. Future work will extend this foundation through in-situ co-design with broader populations and longitudinal studies examining how adaptive cross-realm interfaces integrate into everyday practices. As physical and digital ecologies continue to converge, the design challenge ahead is not the pursuit of perfect immersion, but the careful orchestration of agency, ambiguity, and meaningful connection across hybrid thresholds.

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4.3 Paper III - Employing VR (KitchXR) as a User Experience Tool

Preface

In contrast to the earlier papers, which focus on domestic kitchen dynamics and remote communication practices, this work shifts the lens to professional designers and UX practitioners. It introduces KitchXR, a VR kitchen prototype conceived as a user-insight tool, and examines it through an expert evaluation with 17 design professionals. The study assesses the feasibility and perceived value of using VR to support user-insight generation and iterative evaluation for domestic-appliance experiences, with particular relevance to the industrial PhD context and the thesis’s applied research agenda.

Through hands-on sessions followed by interviews and survey measures, the paper investigates how designers envision using VR to access user insights and what workflows, data types, and evaluation scenarios they consider most valuable. The findings highlight constraints in current industry UX research practice—including restricted access to users, time pressure within development cycles, challenges in simulating complex domestic scenarios, and logistical barriers across sites. In response, the paper identifies three opportunity areas for VR-based insight tools: (1) context-rich testing of complex scenarios, (2) integration of biosensory and behavioral data, and (3) earlier and more frequent iterative revisitation of designs. It also distills design principles for VR user-insight tools, including seamless CAD-to-VR workflows, support for multi-user collaboration, realistic interaction fidelity, and efficient capture and analysis of evaluative data.

Within the thesis’s programmatic design research trajectory, this paper functions as a methodological and infrastructural exemplar: it reframes XR not only as an application technology for the home, but as an environment designers can appropriate to research and shape domestic experiences. The paper primarily addresses RQ2, while providing secondary input to RQ1b by articulating workflow-level requirements and constraints that condition how hybrid presence and mediated connection can be designed, evaluated, and iterated in practice. I led the conceptualization and implementation of KitchXR, the study design, data collection, analysis, and manuscript writing; co-authors supported recruitment, technical refinement, and critical feedback on the conceptual framing.

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4.3 PAPER III - EMPLOYING VR (KITCHXR) AS A USER EXPERIENCE TOOL

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4.3.1 Manuscript 3: KitchXR

Exploring Designers' Expectations From a Virtual Reality User Insight Tool: KitchXR

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ABSTRACT

This paper investigates how designers envision using virtual reality as a spatial interaction tool for collecting user insights during product development. We developed KitchXR, a virtual kitchen prototype, to elicit expectations from 17 professionals in the home appliance industry, including UX/UI designers, industrial designers, and consumer insight specialists via an expert evaluation study. Through thematic analysis of semi-structured interviews, we identify key challenges in current UX research workflows, particularly limited access to user data and constraints on the timing of insight collection. Our findings highlight 3 opportunity areas for VR-based user insight tools: (1) enabling complex and context-rich testing scenarios, (2) integrating bio-sensory and behavioral data, and (3) supporting the revisitation of early design decisions. By centering designers as users of spatial interaction tools, this study contributes to emerging discourse at the intersection of extended reality, user research, and collaborative design workflows.

KEYWORDS

Virtual reality, user insight, consumer insight, virtual prototyping, user experience, expert evaluation study

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1 INTRODUCTION

Product design and development is a complex, multi-stage process that demands collaboration across diverse disciplines. Teams—including decision-makers, UX researchers, product designers, engineers, marketing, and finance—must stay aligned throughout, each bringing distinct priorities and working styles. From the earliest stages, consumer or user insights serve as a shared language, bridging disciplinary divides and informing key decisions. User Experience (UX) research has become central to driving innovation and producing meaningful outcomes in both product and service design.

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Figure 1: The KitchXR virtual environment, designed as a modular, immersive testbed for UX research in home appliance design. Scene shows one of several customizable kitchen layouts, where participants interacted with smart appliances, UI elements, and spatial arrangements.

As Hassenzahl [21] emphasizes, UX goes beyond functionality and usability to encompass emotional and experiential qualities that shape users' relationships with products. By leveraging prototypes of varying fidelity, designers gather insights to guide decisions before production. Beyond feedback, UX research also links design stages and facilitates collaboration across teams.

Researchers use various techniques to understand users and envision user expectations in context. For example, Oliver et al. [35] replicated a real kitchen in a lab, outfitted with sensors and projectors, to study user interactions. The Living Labs methodology enables user-centered evaluations of technologies in real-life settings, particularly in smart homes, IoT, and ubiquitous computing. PlaceLab [23], for instance, allows researchers to explore ubiquitous technologies in authentic domestic environments. Similar methods have been applied in studies on assisted cooking [43], IoT infrastructures [19], and well-being research in kitchen labs [41]. While valuable for producing ecologically valid insights, Living Lab setups are often resource-intensive, time-consuming, and lack flexibility for rapid iteration.

Another approach might be using virtual platforms and tools to collect user insights. Although a wide array of methods (e.g., contextual inquiries, diary studies, and eye-tracking systems) and software platforms (e.g., Figma, Axure, ShapesXR, and NVIDIA Omniverse)

already support digital prototyping and user insight collection, there remains a strong industry demand for swift, user-friendly, and easily integrable tools that embed HCI research methodologies within Extended Reality (XR) environments. A Virtual Living Labs approach may offer new opportunities to investigate interaction patterns, environmental constraints, and behavioral routines in situ.

Several studies have explored the early adoption of XR technologies in industrial settings; Berg and Vance [4] illustrate how Virtual Reality (VR) facilitates early-stage design evaluations and spatial collaboration; Fillatreau et al. [15] demonstrate the use of immersive 3D checklists for training and procedural guidance; and Wang et al. [42] review AR/VR implementations in digital twin systems for product development. Alongside these developments, studies have been questioning whether physical prototypes can be meaningfully augmented or supplemented by immersive XR technologies in UX workflows [39, 44]. With early-adopter companies increasingly embedding XR across various stages of the design pipeline [4], and as social VR and Metaverse platforms become more accessible, new opportunities are emerging for immersive, distributed collaboration and prototyping. More recently, Marques et al. [33] examined how XR can support hybrid collaboration by enabling remote participants to engage in co-located design tasks. In VRception, Reipschläger et al. [20] investigate how VR can be used as a prototyping environment for cross-reality experiences, spanning the Virtuality Continuum states proposed by Milgram [38].

Despite this momentum, existing literature continues to prioritize engineering, simulation, or training applications, offering limited insight into the specific expectations, challenges, and workflows of UX researchers and designers. In this rapidly evolving landscape, understanding the designer's perspective is critical, not only to inform the development of XR-based insight tools but also to ensure their meaningful integration into real-world, cross-functional design environments, particularly during early conceptual and ideation stages. This paper addresses that gap by focusing on product designers and researchers as the primary users of VR-based insight tools. With an expert evaluation approach, it investigates their expectations, challenges, and envisioned roles for immersive environments such as KitchXR in enabling context-rich, iterative UX research within multidisciplinary design teams.

This paper explores the following research questions: (1) What challenges and limitations exist in current UX research workflows for home appliances? (2) How might immersive VR extend or enhance these workflows for iterative testing? To investigate these questions, we developed a VR kitchen prototype and conducted an expert evaluation study with 17 professionals from diverse design and research related roles, including UX/UI designers, industrial designers, and consumer insight specialists. Following participants' interactions with the prototype, we conducted semi-structured interviews and analyzed their feedback using thematic analysis. The VR application we developed (Fig. 1) served both as a probe for ideation and a testbed for evaluating designers' expectations for immersive UX research environments.

Our findings indicate that VR can address two major challenges in existing UX research workflows; limited access to user data (especially across distributed contexts) and throughout different stages of the design process. Participants identified several domains where

VR can enhance insight collection—particularly in evaluating practicality of use, material quality and aesthetics, and user-product interaction. From our analysis, three key design opportunities emerged for future VR-based UX tools: (1) enabling complex, scenario-based testing environments; (2) integrating bio-sensory data for deeper behavioral analysis; and (3) supporting the revisitation of early design decisions through iterative prototyping. These findings contribute to the growing body of HCI literature on the role of XR technologies in professional design practice, emphasizing a designer-centered approach to immersive user research.

2 RELATED WORK

While XR technologies have long been applied in visualization, rapid prototyping, and simulation, a growing body of work has begun to explore their value in UX research. These studies highlight XR's practical benefits for immersive design reviews, accelerating prototyping, and collecting feedback in context-rich environments. We review two key threads of prior work: (1) the use of XR across different design stages, and (2) the role of multimodal and context-rich immersive environments in UX research. Together, these threads provide the foundation for our investigation into designers' expectations, challenges, and practices when using XR-based user insight tools like *KitchXR*.

2.1 XR in Different Design Stages

This subsection explores how XR, particularly VR, is integrated into various phases of product development—from early ideation and collaboration to prototyping and testing—revealing opportunities and limitations in its application for user-centered design. Despite persistent limitations in generalizable hardware/software configurations and open-source availability, VR has been widely adopted in the industry to support decision making and collaborative workflows throughout the design process [4, 15, 42]. Numerous studies demonstrate how VR enables designers and engineers to explore, evaluate, and iterate on concepts in early stages, often reducing cost and development time [11, 13, 14, 22].

Providing a spatially realistic and context-rich experience can facilitate more accessible and grounded decision making [8]. Berg [3] presents a case study of industry professionals using VR for immersive design reviews, allowing stakeholders to evaluate 3D models in a virtual setting that mirrors real-world spatial arrangements. This supports clearer evaluation of design issues and more informed improvements. Similarly, Krauß et al. [26] document recurring bottlenecks in XR prototyping workflows, identifying challenges such as tool interoperability, time/resource intensity, and a lack of integration into broader UX research pipelines. VR's role in co-design has also received growing attention. Wang et al. [42] emphasize VR's ability to facilitate real-time remote collaboration among inter-professional teams and users. In a foundational survey, Berg et al. [4] identify widespread use of AR and VR by designers and engineers, primarily for visualization or collaborative design tasks [2, 15, 31]. These studies emphasize design professionals' practical needs during ideation and prototyping stages, including effective communication and spatial exploration [26]. However, while these tools are often evaluated through the lens of end-user experience,

few studies investigate designers' experiences using XR to collect user insights during the design process [37, 45].

Muller [34] demonstrates VR's role in conceptual-stage ideation, illustrating how immersive tools can address gaps not yet covered by AR-based methods. These findings suggest that XR tools hold potential not only for visual communication but also as user insight platforms throughout the design journey—offering behavioral, verbal, and even physiological data. Beyond VR, other XR modalities—particularly AR—have been used to support iterative design processes. Cascini [10] shows how AR enhances "co-creative" workflows, facilitating dialogue between designers and end-users. Knierim [25] uses mobile AR to explore user expectations in home environments, employing it as a probe for ideation and iteration. Marner et al. [32] develop a SAR prototype to support communication between clients and architects, suggesting that even in spatially constrained domains like kitchen design, XR can support collaborative envisioning. Grünefeld et al. [20] introduce the VRception toolkit, a system that enables developers to rapidly prototype and test cross-reality experiences entirely within VR. The toolkit supports multi-user collaboration, voice communication, and avatar representation. Based on findings from user studies, the authors suggest that such an approach facilitates the development of cross-reality concepts by enabling rapid iteration and testing, as well as both remote and co-located collaboration. In another study, Giunta's review of AR in industrial design maps prior work onto the six phases of the "generic design process model" by Pahl et al. [36]—from Task Definition to Product Documentation—identifying persistent gaps in AR's application for early and final stages [17].

Building on these insights, our study positions VR not simply as a tool for design presentation or co-creation, but as a tool for collecting user insights that inform iterative design decisions. In the *KitchXR* study, we examine how designers and UX professionals envision immersive environments for collecting data—be it behavioral feedback, task performance metrics, or reflections—to shape ongoing design. Unlike studies that evaluate user experience within VR products, our work asks: how can VR serve as a research environment for designers themselves? This shift repositions designers as critical users of immersive research tools.

2.2 Context-Rich and Multimodal Approaches to Immersive UX Research

This subsection reviews how immersive environments—particularly those using multimodal and spatial interaction—are used to simulate real-life usage scenarios, fostering empathy and providing richer data for design research. A growing body of research shows that VR supports empathy-building and perspective-taking, both essential to understanding user experience in context [12, 40]. These qualities are especially useful in domains where real-world testing is costly, dangerous, or logistically challenging—such as automotive design, kitchen appliances, or healthcare.

In the ARCAR project [16], users wear a mixed reality headset with eye-tracking and hand-tracking capabilities while participating in driving simulations. This setup allows researchers to capture real-time data on user attention, gestures, and perception under simulated environmental conditions. Similarly, VR-OOM [18] uses

Wizard-of-Oz techniques in VR to understand passenger experiences in autonomous vehicles—focusing on how users interact with interfaces in situ. Bordegoni et al. [5, 6] show how visual, haptic, and auditory modalities can be combined to simulate appliance usage, enabling realistic virtual prototyping. Kuliga et al. [27] compare walkthroughs of real and virtual buildings, finding that VR can approximate empirical research conditions by providing strong spatial presence and contextual realism. Frameworks such as the XR Safety Kit by Burova et al. [9] further advance this approach by combining gaze tracking, behavioral logging, and task progression metrics into a single virtual system. Similarly, Alce et al. [1] demonstrate how body-tracking tools and VR can be combined to prototype wearable AR systems—highlighting the role of embodied interaction in iterative design. Game-based environments also offer insight. Rooney et al. [30] present "Serious Gordon", a VR kitchen simulation used to teach hygiene standards to food industry workers. While their study suggests that real-world prototypes still hold advantages for procedural learning, others like Pettersson et al. [39] report no significant difference in user responses between field and VR-based evaluations. Hoggemueller et al. [22] further nuance this view, comparing 2D video, 360° VR, and fully immersive CG environments—each proving valuable for different UX research goals.

Collectively, these studies reveal the richness of multimodal and context-aware VR for simulating use scenarios and collecting layered user feedback. However, few explicitly examine how designers can make use of rich qualitative, quantitative and behavioural data collected through XR prototyping and insight collection tools. That is, they rarely ask how these tools serve designers' insight-generation needs, or how immersive platforms might be configured to support iterative workflows. Our work addresses this by repositioning designers—not users or engineers—as core users of XR tools, investigating how they interpret, evaluate, and act on the insights gathered in immersive contexts like *KitchXR*. This perspective reframes the design process itself as a form of inquiry, where XR serves both as medium and method.

3 METHODOLOGY

Here in this section, we explain the methodological approach in detail, study procedure we followed (illustrated in Fig. 2) and how we developed the VR prototype.

3.1 Design and Prototyping of the VR Kitchen Experience

We designed and developed a room-scale immersive VR application for the VR Kitchen experience using Unity. Scenes and assets were modeled in Blender, with its plugin named Archimesh, and the interactive elements were implemented in Unity. The application runs on a VR-capable laptop connected to an HTC Cosmos headset. The experience consists of four scenes (Figure 3). The first is a welcoming scene, designed to onboard users to both the VR equipment and the overall experience. In this scene, users are guided by a voice-over, accompanied by visual cues placed throughout the environment. They are encouraged to practice basic VR interactions, such as teleporting and interacting with virtual objects and appliances. The remaining three scenes present alternative kitchen

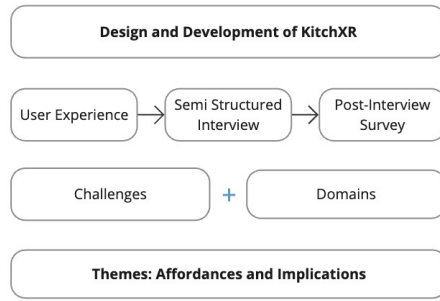


Figure 2: Schema of the methodology. Overview of the study process from prototype development to data collection and thematic analysis.

layouts, which users can switch between by interacting (grabbing or touching) with miniature models representing each scene. These three layouts—a compact kitchen, a large kitchen, and an American-style kitchen connected to a living room—were designed to offer designers a range of spatial experiences and user scenarios to explore. Each kitchen includes a variety of functioning appliance models with basic, natural interaction possibilities (Figure 4). These include a tea maker, coffee machine, toaster, robot vacuum cleaner, refrigerator, stove, and oven. On the countertops, we also placed food packages and fruits, which users can interact with freely. Only the third kitchen—the American-style layout—includes enhanced interactive features; a set of 2D UI menus, predefined tasks, and a laser pointer integrated into the user’s right hand to enable interaction with the UI. Through these elements, users can toggle between various visualization functions for selected appliances, such as viewing an exploded technical diagram of a washing machine, changing appliance models and colors, and seeing inside the devices. These menus (Figure 4) are positioned near the corresponding appliances—refrigerator, hood, and dishwasher—and operate semi-dynamically through custom Unity scripts.

3.2 Expert Evaluation Study

We used the VR application described above, KitchXR, to explore with designers the potential of employing VR for evaluating kitchen applications. We conducted an expert evaluation study with 17 participants. As noted by Lanius et al. expert review is widely adopted in UX practice, being ranked as the most common research method and the third most common evaluation method among practitioners [28]. Building on this recognition of its relevance, we complemented the evaluation with follow-up interviews and a post-experience survey. This study has been exempted from ethical approval by the Research Ethics Committee (Ref. No: 2025.362.IRB3.135), as it involves an expert evaluation.

3.2.1 Participants. 17 professionals from different departments at the headquarters of a global home appliance company in Türkiye, were purposefully recruited through an expertise sampling method. We reached participants through an open call within the company, targeting various departments involved in product development and design processes. We conducted the study in a dedicated office

space at the company. Participants’ age ranged from 22 to 48 (median 29). Only two of the participants have backgrounds in business administration while the others (15 out of 17) held a bachelor’s or higher degree in product design or visual media arts. Their professional experience ranged 1 to 24 years (median of 7 years); notably, 7 participants had less than 4 years of experience. As shown in Table 1, 8 participants held roles within UX/UI design departments, while the remaining 9 were involved in industrial design, consumer insights, R&D, or usability testing. Most participants (10 out of 17) indicated that this was their first experience using immersive VR.

Table 1: Table of participants. SKA: Small Kitchen Appliances

Part.	Age	Gender	Profession and Department	Exp. (Year)	First VR
P1	37	M	Technology Lead - Usability	12	No
P2	48	M	Global Consumer Insights	24	Yes
P3	27	F	UX Designer	3	Yes
P4	40	F	Lead Prod. Designer – SKA	16	Yes
P5	40	M	Lead Prod. Designer - Cooling	21	Yes
P6	39	M	Senior Product Designer – Cooking	15	Yes
P7	23	F	UI/UX Designer (UX/UI)	2	Yes
P8	36	F	Color/Material/Finish Designer	15	No
P9	25	F	Assistant UX/UI Designer	1	Yes
P10	25	F	Product Designer	3	No
P11	23	F	Project Assistant	2	Yes
P12	22	F	UX/UI Design Assistant	1	Yes
P13	29	M	Product Designer - SKA	7	Yes
P14	26	M	UI/UX Designer	6	No
P15	39	M	UI/UX Designer	16	No
P16	29	F	UX/UI Designer	1	No
P17	31	M	Product Designer, SKA	10	No

3.2.2 Expert Evaluation Study Procedure. The expert evaluation study began with a short briefing session explaining the VR Kitchen application and how to use the controllers. After the briefing, participants wore the headset and started freely exploring the VR Kitchen. The first scene was designed specifically to onboard participants with little or no VR experience. During this onboarding phase, users were guided to perform basic interactions such as teleporting, grabbing objects, and opening doors. Once they demonstrated familiarity with these actions, they were invited to explore three different kitchen scenes, each offering distinct spatial layouts. Participants navigated these environments freely—either by walking or teleporting—and interacted with various objects in all four scenes. During this phase, they were encouraged to think aloud, sharing their immediate impressions, emotional responses, and reflections on the visual and interactive aspects of the experience. After this free exploration, participants were guided via external voice instructions to complete four specific tasks in the third kitchen scene. These tasks were: (1) customizing the refrigerator, (2) placing packages into shelves and cupboards, (3) turning the hood on and off, (4) filling the tea maker with water from the faucet. These tasks were designed to demonstrate how VR can support scenario-based UX research. They allowed designers to experience how user behaviors can be observed during goal-driven interactions, how 2D UI menus can be operated with a laser pointer, and how differences in appliance size or interface design may influence user experience.

Semi-Structured Interview. Following the VR experience, each participant engaged in a semi-structured interview guided by 14 core questions (see Table 2). The conversational format allowed us to probe deeper into participants’ professional perspectives and

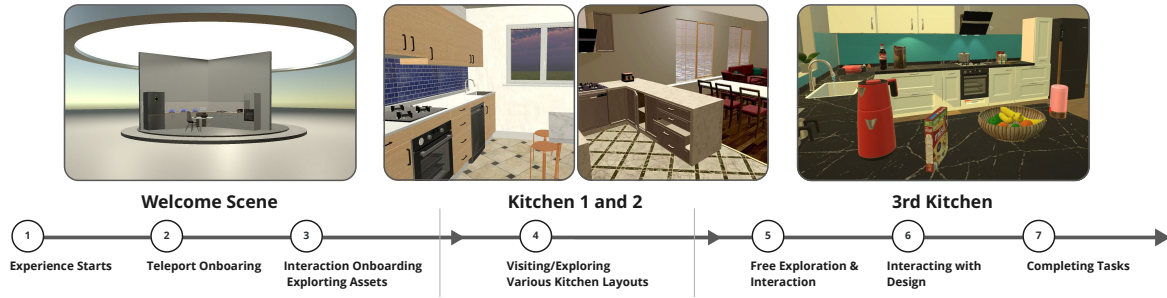


Figure 3: User journey map of the VR Kitchen experience, outlining key stages from onboarding to exploratory interaction and task completion across three virtual kitchen layouts.

explore emergent themes relevant to XR integration in home appliance design processes. Interview questions focused on three areas: (1) participants' current UX research practices and challenges in home appliance design, (2) their experiential evaluation of the VR Kitchen prototype compared to existing research tools and real kitchen environments, and (3) their envisioned applications of XR technology for user insight collection—including product conceptualizing, interface testing, and iterative feedback gathering. Specific probes examined the design effectiveness of the VR interface, the role of task scenarios in eliciting authentic user behaviors, and participants' professional assessment of how virtual prototype fidelity and interactivity levels might impact the quality of user insights collected through such tools. All interviews were audio-recorded and transcribed with the informed consent of participants. To complement our qualitative findings, we administered two standardized usability measures: the System Usability Scale (SUS) [24] and the After Scenario Questionnaire (ASQ) [29].

System Usability Scale (SUS). We used the SUS to assess the overall perceived usability of the KitchXR. The SUS is a widely recognized 10-item scale that provides a quick and reliable measure of system usability through a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). The SUS allowed us to examine whether the prototype achieved a level of usability that would enable designers to focus on their UX research practices and its objectives rather than overcoming interface barriers.

After Scenario Questionnaire (ASQ). To evaluate task experience, we employed a 3-item ASQ measuring perceptions of task difficulty, time adequacy, and system feedback on a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree) [29]. The ASQ offered more granular insights than the SUS, particularly on whether task scenarios supported meaningful interactions by evaluating ease, convenience, and guidance. This is critical, as scenario design directly affects the quality of user insights in XR-based research tools. These quantitative benchmarks indicated whether the prototype met usability thresholds for professional insight collection, clarifying which aspects aligned with designer expectations and which required improvement. Complementing this, our qualitative analysis examined how XR could reshape UX research practices and identified the most valuable capabilities from the perspective of home appliance design professionals.

Table 2: Questions of semi-structured interview.

Interview Questions	Context
1 Have you experienced a VR application before? What kind of application did you try? Could you compare this experience with Virtual Kitchen?	Experience
2 Which prototyping methods do you find more efficient in the design process? Could you compare your VR experience with other prototyping methods?	Experience
3 Would conducting user experience studies and usability tests in a VR environment contribute to the process? What kind of contributions would it make?	Experience - Goal
4 Considering your current task, if you had such a tool, how frequently would you use it in the design process?	Experience - Scenario
5 What information would you like to gather from users while collecting insights in a VR experience?	Experience - Scenario
6 What kind of interactions and features could be added to make the tests more efficient?	Experience - Scenario
7 How could the kitchen product experience be enhanced with the addition of other types of functions to the Virtual Kitchen?	Experience - Func.
8 How would you prefer to ask questions to the user during the experience?	UI
9 When evaluating the overall appearance of the Virtual Kitchen, how did you find the size of the environment, the number/placement of products, the space allocated to you, etc.?	Model
10 Was the ambient light sufficient and realistic in the kitchens you experienced in the VR environment? Would you need different lighting to try out products?	Model
11 Would experiencing products in different types of kitchens contribute to the design process?	Experience
12 Were you able to interact easily with the products in the Virtual Kitchen? For example, how was opening the refrigerator door, moving products, in terms of the functions you used? How could it be improved?	Interaction
13 What are your thoughts on using a laser pointer?	Interaction
14 What are your thoughts on AR menu design? How were the choices in terms of comprehensibility and applicability?	UI

3.3 Data Analysis

In our analysis, we complimented qualitative thematic analysis of interview responses with quantitative assessment of usability metrics. This dual approach allowed us to evaluate both the practical usability of our VR prototype and gain deeper insights into designers' experiences, needs, and expectations regarding the use of VR tools for UX research.

3.3.1 Qualitative Data Analysis. The qualitative analysis was conducted using the six-phase approach to thematic analysis proposed by Braun and Clarke [7], which is well-suited for exploratory research aimed at examining user experiences and professional practices in order to identify patterns within qualitative data. Two researchers independently reviewed the dataset, noting observations and identifying potential patterns. We employed an inductive approach for systematic coding, allowing the codes to emerge directly. Codes were generated based on participants' current UX research practices, their experiences and reflections on the VR prototype,

and their suggestions for potential improvements or additional features. The final analysis identified two main problem areas in existing practices, directly addressing the first research question (RQ1), and three key themes regarding the potential of VR to support user data collection and facilitate its integration into existing design processes, addressing the second research question (RQ2). This structure allowed us to clearly delineate between the current challenges faced by designers in UX research and the envisioned future roles of XR technologies. The thematic analysis revealed three major themes corresponding to design features that could be integrated into VR tools for UX research: (1) enriching user experience design with complex scenarios and alternative testing environments, (2) integration of bio-sensory measures to enable richer, multi-modal data collection, and (3) enabling revisitation of the design process by capturing data that can inform future iterations. Each theme was supported by sub-themes that reflected how participants envisioned VR integration into their workflows. In our reporting, we also present the frequency of similar responses among participants ($n/N = 17$), offering insight into the consistency and variability of perspectives within our study cohort.

3.3.2 Quantitative Data Analysis. SUS scores were calculated following standard protocols [24]. This recoding produced values on a 0–4 scale, with 4 representing the most positive response. The recoded scores were summed and multiplied by 2.5 to yield a final score on a 0–100 scale. Scores above 68 generally indicate above-average usability [24]. In our study, individual participant scores ranged from 47.5 to 82.5 (mean of 62.5). While the overall SUS score summarizes general usability, Table 3 presents the mean raw Likert ratings (1–5) for each item, highlighting specific aspects of the VR Kitchen that participants rated more positively or negatively and providing insight into usability strengths and areas for improvement. ASQ responses were analyzed to examine task-specific interactions participants engaged in the VR kitchen experience. For each of the three ASQ items, we report the frequency of participant ratings on the 1–7 scale, providing a descriptive summary of responses (Figure 5). These quantitative measures from the SUS and ASQ complemented our thematic analysis, contributing to the discussion on practicality of use under the discussion on commonly targeted in appliance UX research user insight domains (SUS), and to the theme of complex scenarios and alternative testing environments, where we discuss participants' responses to the ASQ.

4 FINDINGS

This section presents findings from our mixed-methods expert evaluation study assessing the utility and perceived value of the VR-based prototype KitchXR for collecting user insights in home appliance design. The study addressed two questions: (RQ1) What challenges and limitations exist in current UX research workflows for home appliances? and (RQ2) How might immersive VR extend or enhance these workflows for iterative testing?

The findings are organized into three main parts. First, we examine current UX research practices in product design and the emerging opportunities for VR integration, identifying three key limitations: access, timing, and workflow integration. Second, we analyze user insight domains commonly targeted in appliance UX research—practicality, aesthetics, and emotional connection—and

how VR can support them. Third, we present three overarching themes derived from qualitative analysis: (1) enriching user experience through complex and contextual scenarios, (2) integration of biosensory measures for deeper insight, and (3) enabling revisitation of the design process across multiple stages. Together, these findings provide a nuanced understanding of where immersive VR tools may meaningfully augment existing design research practices.

4.1 Opportunities and Challenges for Integration

Participants, working for the same global home appliance company, described their current UX research practices as geographically distributed across national and international locations. These practices typically involve using 3D-printed or foam-core prototypes at full scale in dedicated testing studios. UI interactions are often simulated through embedded tablets or smartphones, enabling users to engage with interfaces as if integrated into the physical product. During sessions, facilitators guide participants through predefined scenarios, while UX researchers observe behaviors, collect verbal feedback, and sometimes record physiological data such as eye tracking or skin conductance. Our analysis identified three critical limitations within this workflow that highlight the potential value and challenges of immersive VR tools: (1) restricted access to geographically diverse and demographically varied user data, (2) time constraints that delay or limit user feedback in early development stages, and (3) challenges in integrating VR into existing digital design and engineering pipelines.

4.1.1 Access: Expanding User Data Collection. The first challenge centers on geographical and logistical constraints. Participants (10/17) described the difficulty of transporting large physical prototypes across locations and the complications involved in setting up consistent testing environments abroad. Many believed that VR could allow UX teams to conduct more frequent and remotely distributed studies, increasing both the variety and number of users involved (12/17), while enabling faster data collection (10/17). Several participants (7/17) noted that such studies could also be more easily automated in VR environments, offering the possibility of collecting rich data asynchronously and at scale. A few participants (2/17) emphasized the added benefit of standardizing study conditions across locations via controlled VR environments. Participants also pointed out that VR could better simulate product variants and materials that are difficult to prototype physically. For example, 3D-printed models are usually monochrome and made of materials that differ significantly from the final product. VR could address this by allowing users to preview different appliance models, color variations, and finish types more accurately (14/17). Moreover, immersive environments can simulate diverse kitchen settings (6/17), and even replicate individual users' home environments (2/17), offering more contextually relevant scenarios for evaluation.

4.1.2 Timing: Enabling Earlier and More Frequent Feedback. The second issue relates to the temporal distance between product prototyping and user testing. Participants (14/17) reported that current processes often delay user testing until late in the development cycle, limiting the ability to incorporate feedback into the final product. Three participants explicitly mentioned that they are not



Figure 4: Screenshots from the application. Left: Oven is interactable in exploded-perspective mode. Middle: Tea-maker as a small home appliance. Right: 2D Menu presents options.

involved in the final stages of the design process, and therefore cannot respond to final usability issues or design revisions. Participants (9/17) suggested that VR could help fill this temporal gap by enabling earlier and more frequent user testing sessions. They emphasized that integrating VR into the prototyping process could foster stronger collaboration across teams and ensure that changes based on user feedback are validated more quickly. Some participants (7/17) highlighted VR's potential in collecting insights even in the concept development phase (2/17), helping designers *"better grasp how the finished version looks and works"* (P17), and potentially improving design quality through earlier validation loops.

4.1.3 Integration Challenges. Despite its potential, the reason of not adapting VR into existing workflows, mostly relies on integration challenges identified by our participants. One significant barrier that we also confront in our study, lies in the incompatibility between design software formats. Product designers and engineers commonly use CAD tools which are based on NURBS (Non-Uniform Rational B-Splines) to present 3D models with high geometric precision. By contrast, VR environments and game engines such as Unity primarily rely on polygon-based 3D models, necessitating an often time-consuming conversion process to adapt CAD-based product model for use in immersive experiences (P6). This conversion introduces friction into fast-paced design cycles and may deter adoption without automated translation pipelines. Additionally, participants noted that virtual environments still fall short in delivering certain physical feedback cues—particularly those related to ergonomics and haptics—which are critical in evaluating physical appliances (8/17). This limitation was especially noted by industrial designers, who emphasized the continued need for physical prototyping when evaluating aspects such as button resistance, weight, or tactile feedback.

Nonetheless, most participants viewed VR not as a replacement but as a valuable addition to their existing UX research toolkit. It was considered a complementary method for expanding the scope of insight collection, particularly in areas where physical prototyping is difficult, expensive, or logistically impractical. As one UX/UI designer (P2) put it: *"As long as you expand the toolbox, you will have new areas that you need that tool for."*

4.2 User Insight Domains in UX Research

Analysis revealed three primary areas that participants address in UX research can be meaningful in VR: (1) practicality of use, (2) material quality and aesthetics, and (3) connection between user

and product. UX designers focused primarily on user experience when interacting with different features, while product designers emphasized physical qualities such as ergonomics, material, and color.

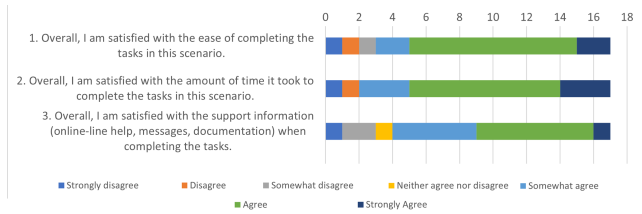
4.2.1 Practicality of Use. Participants defined practicality in terms of ease of use and functionality, emphasizing the first encounter phase and initial learning process (2/17). Both UX/UI designers (P2, P12) and product designers (P10) identified the learning phase—when users are first introduced to the product and learn to use its UI or features—as crucial for UX research. As one participant explained: *"When you are learning how to use a tool for the first time, it takes some effort and time to adopt, and then it gets automated. This phase gives us crucial insights into how to guide the user experience in a way it is more practical to use the product"* (P2). The learning curve provides essential information about user experience and product design. Questions sought by designers in this domain include: *"Can the user feel comfortable and confident while using the product? How will they experience the first learning process, where will they press, will they press accurately?"* (P12). Two participants (P2, P10) suggested that instructing users on product operation while observing their performance, as well as understanding how users naturally operate the product, provides valuable information that could be implemented in VR applications.

The System Usability Scale (SUS) yielded a score of 62.3, below the average benchmark of 68. Table 3 summarizes mean scores for each item. Interviews indicated that lower scores were largely attributable to participants' limited VR experience, which made adapting to the controllers and virtual environment dynamics challenging. Strengths included low perceived complexity ($M = 2.14$ for unnecessary complexity, inconsistency, and cumbersomeness) and strong functional integration ($M = 3.57$), suggesting that the prototype's features felt cohesive and conceptually clear. In contrast, middling ratings for ease of use, confidence, and learnability ($M = 2.85$ across related items) highlight friction in onboarding and first-time use. Addressing these challenges—particularly through enhanced guidance, feedback, and control affordances—could substantially improve both user confidence and overall usability.

4.2.2 Material Quality and Aesthetics. Participants (11/17) focused on how users respond to physical properties and visual attributes such as material type, weight, volume, and visual design. They noted that current rapid prototyping methods have limitations: foam prototypes are faster to produce but lack detail in visual and material

Table 3: System Usability Scale (SUS) responses of 17 participants. 1-Strongly Disagree / 5-Strongly Agree

SUS Items	Mean
1. I think that I would like to use this system frequently.	3.57
2. I found the system unnecessarily complex.	2.14
3. I thought the system was easy to use.	2.85
4. I think that I would need the support of a technical person to be able to use this system.	2.85
5. I found that the various functions in this system were well integrated.	3.57
6. I thought that there was too much inconsistency in this system.	2.14
7. I would imagine that most people would learn to use this system very quickly.	2.85
8. I found the system very cumbersome to use.	2.14
9. I felt very confident using the system.	2.85
10. I needed to learn a lot of things before I could get going with this system.	2.14

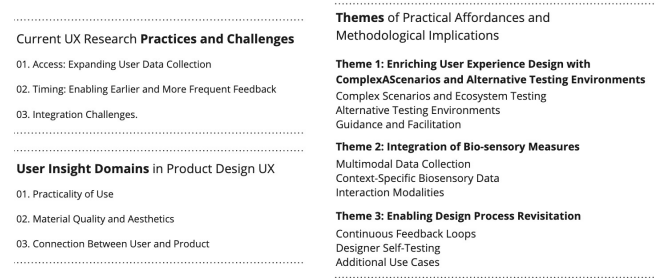
**Figure 5: After Scenario Questionnaire (ASQ) responses of 17 participants.**

quality, providing only basic information about form, mechanics, and ergonomics (4/17). Virtual prototypes such as rendered images present visual details but lack functional characteristics (5/17). One product designer (P8) explained how VR could support insights into user preferences and aesthetic perception: *"I would like to learn about users' love and hate perception"* regarding various models presented with visual detail alongside functional features.

4.2.3 Connection Between User and Product. The third domain concerns the emotional bond users develop with products (3/17). One UX designer (P14) emphasized that establishing emotional bonds with physical products requires users to *"see, touch, and feel"* the product, highlighting the importance of embodied interaction. Another participant (P2) noted that this connection is influenced by ease of use, requiring intuitive interaction that aligns with users' previous experiences with similar products. The connection domain appears to be shaped by the other two domains—practicality of use and material quality and aesthetics—which together provide essential information about user-product relationships.

4.3 Themes of Practical Affordances and Methodological Implications

Building on the previously identified challenges in current UX research practices and the user insight domains observed across design roles, this subsection presents three themes that emerged from our qualitative analysis (see Fig. 6). These themes illuminate how immersive VR tools may enhance user insight generation in product design workflows. Each theme corresponds to a distinct area where VR introduces new capacities or reconfigures existing ones: (1) enriching user experience design through complex scenarios and flexible testing environments, (2) integrating biosensory measures for deeper behavioral and emotional analysis, and

**Figure 6: Overview of key findings from the expert evaluation study. Left: Current UX research practices and user insight domains in product design; Right: 3 thematic areas that highlight the practical affordances and methodological implications of integrating VR into UX research.**

(3) enabling iterative revisitation and reflection during early-stage design. The findings highlight both the practical affordances and methodological implications of embedding immersive tools in the UX research process.

4.3.1 Theme 1: Enriching User Experience Design with Complex Scenarios and Alternative Testing Environments. Participants gave high ratings on the After Scenario Questionnaire (ASQ) for evaluating tasks, scenarios, and activities in the VR experience (Fig. 5). Across the three ASQ items—ease of task completion, time required, and adequacy of system guidance—12 of the 17 participants responded *"somewhat agree"* or higher, indicating that most found the tasks straightforward and well supported. However, while task execution was perceived as easy from an interaction standpoint, participants questioned the depth of user engagement the current scenarios could support. They recommended incorporating more complex, goal-oriented sequences (e.g., preparing breakfast) that better reflect everyday kitchen activities (4/17) and ensuring functional interactions with model appliances match real-world expectations (3/17). As one participant (P3) explained: *"I want to learn how many modes the user is employing in real life. We know that users don't employ all the 14 different modes in their washing machine. But the question is, how many modes they are using—2 or 4? And what if they only see these 4 modes instead of 14? How would they react? We cannot get this kind of insights since we are not able to produce many alternatives of a physical prototype."*

Complex Scenarios and Ecosystem Testing. UX designers sought sophisticated user insights in complex situations, such as using multiple connected devices, sometimes including mobile devices (4/17). Participants suggested that VR provides a suitable environment for connecting different prototypes in working relationships within larger use contexts. One participant (P16) noted: *"I would like to see how you feel in such an environment as the washing machine rang, buy clothes, the tea maker rang, the doorbell rang, etc. We always look at the products individually, but when they come together and create an ecosystem."*

Alternative Testing Environments. Based on their VR Kitchen experience, participants suggested novel ways of designing test

environments and interaction possibilities (3/17). One UX/UI designer (P15) proposed that unlike real-life testing with mock-up kitchens, VR could create environments exclusively for specific cases, allocating more time for product-specific tasks and scenarios: *"For some products, it may be more accurate to see the prototype alone, not in the environment. There are specific tasks related to that product, especially on the UI/UX side, where the environment the user is in does not have much effect."* Another participant (P7), a product designer, agreed that using alternative user interaction techniques for situations where it is unnecessary to build an environment testing setting for all models, but capture user interaction solely on one model with specific and detailed user insights. One participant (P5) gave positive feedback on the functionality where the VR-kitchen application displayed the internal working, such as air movement in a fridge model, and shared that these types of interactions, impossible to convey with physical prototypes, can be used as a general functionality in the VR application. P1 (UX) suggested that VR could enrich the user experience by adding more product-specific use cases to get more user feedback.

Guidance and Facilitation. Participants (10/17) emphasized the need for guidance in VR user experience studies, similar to existing protocols where UX researchers are present during testing. This could involve UX researchers being virtually and/or physically present, either embodied as avatars (4/17) or through voice-based interaction without visual representation (4/17). Participants (3/17) suggested incorporating visual, auditory, and spatial cues to help users navigate through steps and tasks.

4.3.2 Theme 2: Integration of Bio-sensory Measures. Participants (9/17) endorsed VR's potential to provide broader user observation capabilities through in-situ assessments of different user experience aspects. While bio-tracking techniques like eye-tracking are already used in user insight studies (3/17), they are not collected systematically or automatically (1/17). Embedded bio-tracking sensors in VR headsets (6/17) and aligned study scenarios (3/17) could address deeper and broader user insight requirements.

Multimodal Data Collection. More in-depth user experience information could be supported through physiological measures (heart rate, galvanic skin response) (4/17), motion tracking (6/17), and eye tracking (8/17). One participant (P1) highlighted the advantages of coupling eye-tracking and movement tracking: *"What the user is looking at and how the user views the interaction is important for us. Also, the movement; what kind of movements does the user make when using the prototype. This can be recorded and analyzed to give better insight into the process of interaction."* Another participant (p12) suggested in understanding how users interact with the UI features, *"the hand movements are important. And it's good to know how you feel"*. Eye and hand tracking coupled is mentioned by three participants (p12, p16, p11) while the finger sensitivity put forwards as a necessary data to complement insights for small appliances (p12).

Context-Specific Biosensory Data. Participants suggested that different biosensory data should be selected according to specific insights being collected. For instance they suggested that skin conductance and heart rate might be suitable for connection insights

(P2), eye-tracking coupled with hand-tracking for practicality insights (P10, P12), and eye-tracking coupled with skin conductance for aesthetic insights (P8).

Interaction Modalities. Three participants found that controller-based interaction broke the feeling of natural manual interaction, especially with small appliances having multiple parts. Hand gloves with vibro-tactile feedback were recommended as better alternatives to standard VR controllers (P15). This was less problematic for UI feature interactions (3/17). One participant (P2) suggested VR could be particularly suitable for UI research *"as we are not concerned with measuring finger sensitivity or other ergonomic issues there. We are interested in the flow of menus, whether it follows a certain logic and it is understandable."*

4.3.3 Theme 3: Enabling Design Process Revisitation. Participants suggested that VR can be used in early design stages (P11, P5) to support designers' decision-making processes (P17, P11). The early concept development phase could particularly benefit from VR-collected user data to establish design directions (3/17). One UI designer working on novel interface design and interaction techniques (P12) noted it would benefit stages where imagination is primarily used.

Continuous Feedback Loops. Different design alternatives can be tested across various design stages, providing continuous feedback loops (P16, P17). One UX designer suggested VR could be useful for preparing user experience scenarios for formal physical tests (P15) and for pilot testing to identify potential problems and revise scenarios before formal physical studies (P12, P15).

Designer Self-Testing. Five participants—two UX/UI designers (P16, P12) and three product designers (P7, P11, P17)—suggested VR could enable designers to experience prototypes themselves and foresee problems typically found later in UX testing. As one participant (P11) noted: *"The biggest advantage would be to try it ourselves. We could have found the bugs that will be found in some future tests early and here. In fact, we could perform these tests ourselves."* VR could also bring together teams of designers and engineers to test prototypes collaboratively (P8, P14).

Additional Use Cases. Participants identified additional potential application areas: remote collaboration (2/17), decision-making (4/17), co-design (5/17), prototyping across various design and development stages (6/17), scenario-testing for UX studies (2/17), benchmarking and market research (2/17), training and professional development (1/17), ergonomic research (1/17), and color material finish (CMF) design (2/17).

5 DISCUSSION

This study explored two interrelated research questions: (RQ1) What challenges and limitations exist in current UX research workflows for home appliances? and (RQ2) How might immersive VR extend or enhance these workflows for iterative testing? The discussion below engages with these questions by synthesizing our findings with relevant literature in product design, UX, HCI, Spatial Computing, and XR.

Across product design industries such as home appliances, UX research traditionally relies on physical prototyping—yet participants revealed persistent frictions in these practices. These include logistical constraints in transporting large or fragile models, difficulties replicating contextual qualities, and the late-stage timing of UX tests. Our findings echo prior work on inefficiencies in physical prototyping workflows, especially in globally distributed teams and high-cost industries [8, 11, 13]. These challenges inform the growing interest in alternative research environments. While Living Lab methodologies enable in-situ testing, they introduce complexity in setup, lack standardization, and often fail to meet industry demands for rapid, repeatable, low-cost methods. In contrast, immersive VR systems like our KitchXR prototype offer scalable, remote, and controlled UX evaluations [4, 42]. This study contributes by examining how immersive tools might realistically be integrated into existing design workflows. Rather than focusing on end-user acceptance, we foreground the perspectives of design professionals—UX/UI designers, product designers, and engineers—revealing how they assess the value, limitations, and applicability of VR in their own practices. This practitioner-centered view remains underexplored in XR adoption research. However, it also limits our findings to projected use rather than observed long-term integration.

Participants emphasized VR's ability to simulate richer scenarios and alternative contexts. This supports Bruno et al.'s argument that realistic context fidelity is key for meaningful UX insights [8]. However, several participant suggestions—such as integrating haptic gloves—require critical consideration. While richer interaction fidelity could enhance realism, these technologies also risk reducing accessibility and scalability, undermining one of VR's key advantages: low-cost, distributed testing. Thus, design teams must weigh realism against inclusivity and workflow feasibility when envisioning adoption. Another opportunity lies in VR's multimodal data collection. Eye tracking, motion tracking, and biosensory measures (e.g., galvanic skin response) embedded in headsets can deepen understanding of user emotion, cognition, and behavior. These findings build on work by Krauß et al. [26], though participants stressed the importance of aligning data types with research goals—e.g., combining hand tracking with motion data for ergonomics, or eye tracking with skin conductance for aesthetic evaluation. This suggests that while VR affords richer data capture, its true methodological contribution lies in selectively integrating these measures where they extend—not duplicate—existing UX research practices. Participants also envisioned VR supporting early-stage design work. Designers described using immersive environments for self-testing, bug detection, and rapid prototyping—resonating with speculative and generative XR practices advocated by Höggenmüller et al. [22]. VR becomes not just a testing platform but a creative and reflexive tool in iterative workflows. The semi-structured interviews show that KitchXR functioned not only as an evaluation setup but also as a probe for eliciting designers' expectations. Immersive VR also enables asynchronous collaboration among globally distributed teams. Echoing prior studies [11, 13], participants imagined Metaverse-style environments facilitating UX research and coordination when physical access to users or teammates is constrained.

Despite these benefits, adoption barriers remain. Participants noted the friction between high-fidelity CAD models and game engine formats, echoing Deininger et al. [14]. Time-consuming mesh

conversions hinder rapid iteration. Concerns also surfaced about the lack of tactile realism and ergonomic fidelity—particularly for evaluating physical properties crucial to product design. These limitations confirm that VR is best positioned as an augmenting layer, complementing physical prototyping and traditional UX research methods, rather than replacing them.

Together, these insights contribute to HCI and design by mapping how spatial computing and immersive VR might integrate into product development processes. We offer an empirically grounded view of how design teams imagine the affordances and limits of VR in practice. Our study also contributes to XR literature by identifying domain-specific insight domains—such as practicality, material perception, and emotional connection—that immersive tools like KitchXR can uniquely support. While our study focused on kitchen appliances and a single prototype, the findings have broader relevance across product domains where spatial interaction, physical form, and context are key. In relation to RQ2, KitchXR illustrates how immersive environments can act both as research instruments and as generative probes, shaping the types of insights available to design professionals. Yet their adoption depends not only on technical advances but on organizational readiness and willingness to reshape workflows—an ongoing challenge for future research.

6 CONCLUSION

This study examined how immersive VR tools might be integrated into UX research within the product design processes of home appliance industries. We developed KitchXR, a scenario-based VR kitchen prototype, and conducted expert evaluation study with 17 professionals from a global home appliance company. Through interviews and usability evaluations, we identified key workflow challenges and explored the design affordances VR can offer for user-centered research. The original contribution of this study lies in foregrounding the perspectives of product team members—UX/UI designers, product designers, and decision-makers—on how immersive tools could realistically fit into their existing practices. Rather than focusing on end-user interaction, our approach centers on how design professionals evaluate the potential, limitations, and adoption of XR tools in early-stage development.

Our findings show that current UX practices, heavily reliant on physical prototyping, face constraints in access, timing, and iteration. KitchXR was perceived as a complementary tool that enables remote, immersive, and scenario-rich user testing. We identified three key affordances for XR-based UX tools: enabling complex, contextual scenarios; supporting biosensory integration for deeper insight; and fostering iterative, revisitable design processes. These address RQ1 by highlighting persistent gaps in current UX research practices and RQ2 by illustrating how VR can extend methodological options to fill them. Beyond the immediate case of kitchen appliances, these findings have broader implications for design and HCI. They suggest that VR's role in UX research is not merely to replicate physical prototyping but to create new conditions for insight generation—especially where context, iteration, and multimodality matter. For industry, this points toward a shift in how design teams conceive user research infrastructures. For research, it underscores the value of VR as both a methodological probe and a platform for integrating qualitative and biosensory data.

At the same time, the study's contributions are bounded. Its original focus on speculative perspectives—rather than long-term deployments—limits claims about sustained integration. Additional limitations include the domain-specific context and reliance on a single VR prototype. Future work should examine longitudinal use across full design cycles, investigate integration of tactile and ergonomic evaluation, and develop frameworks for merging immersive prototyping with engineering pipelines. By doing so, we can better understand not only how VR responds to today's UX research challenges but also how it may reshape the methodological landscape of design research in the future.

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4.4 Paper IV - Evaluating Phygital Artifacts with Designers

Preface

This paper provides an integrative study that brings together the thesis’s preceding contributions and examines designer insights through an expert evaluation. Building on the problem space defined in Paper I, adopting the artifact concept developed in Paper II, and informed by methodological insights from Paper III, the study introduces three speculative XR-based prototypes embedded in an immersive VR kitchen: (1) *Ambient Cooking* for subtle, ambient awareness and affective connection; (2) a *Portal* for synchronous, immersive communication; and (3) *MagMemo* for asynchronous three-dimensional messaging. Using a VR-based evaluation with 14 professional designers, the paper examines how these artifacts may reconfigure social interaction in hybrid connected kitchens and what design trade-offs they surface in practice.

Within the thesis, this paper serves as the central *capstone prototype-based evaluation*, contributing to both RQ1 and RQ2. For **RQ1**, it operationalizes earlier-identified interaction challenges (e.g., visibility, audio, spatial awareness, and the need for hands-free and potentially multi-sensory communication) into concrete phygital communication modalities. Rather than treating these prototypes as finalized solutions, the study investigates how designers interpret their potential to support communication, coordination, and social connection in kitchen contexts, including the conditions under which such support may break down.

Findings position hybrid connected kitchens as *phygital communication hubs* that can support synchronous, ambient, and asynchronous forms of social presence. Designers’ reflections articulate opportunities for togetherness alongside concerns around privacy, control, household rhythms, inclusivity, and the trade-offs between intimacy and intrusiveness. For **RQ2**, the study demonstrates how immersive VR can function as a design research method: professional designers explore speculative kitchen futures within an experiential testbed, using the prototypes as probes to reason about domestic workflows, appropriation, and longer-term integration.

This paper therefore bridges user needs and designer practice by translating the challenges from Paper I into functioning XR prototypes and critically evaluating them through a designer-centered perspective. I led the design and implementation of the VR environment and artifacts, the study design, data collection, analysis, and primary writing. Co-authors contributed to supervision, methodological refinement, and positioning the contributions within social XR and domestic HCI. For the thesis version, I expanded the connections to both research questions

and clarified how each artifact addresses specific aspects of communication and design practice.

A version of the following section has been conditionally accepted for publication as: O. E. Öğretmen et al., “Hybrid Connected Kitchens: Phygital Social XR for Domestic Communication and Collaboration”, Virtual Reality (Springer), special collection Connected Realities: Exploring Collaboration and Social Interaction in eXtended Reality (under revision).

4.4.1 Manuscript 4: Evaluation Phygital Artifacts with Designers

Hybrid Connected Kitchens: Phygital Social XR for Domestic Communication and Collaboration

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Abstract

Extended Reality (XR) technologies are increasingly used to support communication, collaboration, and social connection across distance. While social XR research has primarily focused on workplaces, education, and entertainment, far less is known about how hybrid physical–virtual communication might unfold within everyday domestic environments—particularly kitchens, where cooking, coordination, and care shape social life. This paper investigates how design practitioners envision social interaction and collaboration if kitchens become hybrid connected spaces augmented through phygital artifacts. We conducted a designer-focused evaluation using three speculative prototypes—an ambient cooking companion, a synchronous communication portal, and an asynchronous three-dimensional messaging system—embedded within an immersive VR kitchen environment. Through situated tasks, think-aloud VR sessions, semi-structured interviews, and a structured design-evaluation survey, we examined how designers conceptualize communicative and relational possibilities, how they imagine practices emerging around hybrid kitchen artifacts, and what tensions arise when XR enters domestic life. Findings show how connected kitchens can function as phygital communication hubs that support synchronous, ambient, and asynchronous social presence; how designers balance opportunities for togetherness with concerns around privacy, household rhythms, and inclusivity; and how immersive speculative prototyping enables early-stage reasoning about domestic XR futures. We conclude by outlining design implications for socially meaningful, equitable, and ethically grounded connected kitchen ecologies and identify directions for future work on long-term domestic appropriation, cross-household interaction, and broader spatial ecologies of hybrid home life.

Keywords: Social XR, Domestic XR, Connected kitchens, Phygital communication, Hybrid interaction, Speculative design, Collaboration

1 Introduction

As wearable computing, generative AI, and connected devices converge with Extended Reality (XR) technologies—including Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), they are reshaping how people connect, collaborate, and sustain social relationships across distance. By blending digital content with physical environments or immersing users in fully virtual spaces, state-of-the-art XR systems promise alternative forms of communication enriched by embodied cues, shared spatial attention, and immersive social presence [1–3]. Yet as these technologies evolve, concerns remain about whether they can genuinely support meaningful social interaction, or whether they risk reproducing the limitations of screen-based media in three dimensions. While research on social XR has grown rapidly, it has predominantly examined work-oriented collaboration, education, entertainment, and user behavior in large-scale social platforms [2]. Far less attention has been paid to domestic environments, where communication and coordination intertwine with routines of care, intimacy, and shared living. Kitchens are particularly socially dense domestic settings: they are sites of teaching and learning, hosting, improvisation, negotiation, and emotional connection. In many families, cooking together or sharing food is a cornerstone of social life, one that increasingly extends to remote encounters as people cook together over the phone, video chat or send photos and messages about meals.

HCI research has begun exploring technologically mediated cooking and dining. Telematic dinner parties use multimodal setups to create playful, performative forms of togetherness [4], while virtual cook-along experiences using videoconferencing illustrate the opportunities and challenges of remote collaboration around food [5]. Yet such systems largely remain tied to flat screens and limited camera views, offering only partial support for the embodied, spatial, and sensory cues that shape cooking as a social practice. In parallel, AR and MR work has examined situated cooking assistance. *Shadow Cooking* demonstrated how spatial projection of instructions onto utensils and ingredients can enable fluid, embodied task performance [6]; inter-generational AR cooking tools show how overlays and quantity visualizations can reduce confusion and strengthen emotional connection between remote family members [7]. More recent work such as “Cookie” [8] highlights how multimodal smart kitchen assistants can support cooking while preserving user autonomy, balancing proactive system support with human agency. However, these systems primarily emphasize instruction, safety, and task performance rather than broader questions of social presence, asynchronous communication, ambient connection, or long-term household integration.

Beyond food contexts, domestic HCI research has explored how networked and ambient technologies can shape family life and co-presence. Situated displays that visualize physiological data support coordination and reflection in households with ADHD children [9]. Research-through-design explorations of passive co-presence—such as the *There Chair* and *Fragrance Frame*—illustrate how subtle tactile or olfactory cues support emotional connection without demanding attention [10]. Moreover, technology-probe studies of AR in the home show that users value AR not only for utility but also for its capacity to support social activity, awareness, and playful shared experiences in domestic life [11]. Collectively, these works suggest a design space of

phygital domestic experiences in which ambient, spatial, and embodied cues create a sense of “being together apart” that differs from traditional communication methods.

Additional research in smart homes and hybrid-reality systems highlights the value of coupling physical and virtual spaces. Hybrid smart home prototypes [12] and methodological platforms like AR-in-VR simulators [13] show how immersive environments can enable early-stage evaluation of domestic technologies. Recent work on VR-based kitchen prototyping further demonstrates how designers use immersive scenes to reflect on appliance interactions, workflows, and user experience in context-rich domestic environments [14]. Yet across these domains, the question of how people—especially designers—conceptualize *social* interaction, collaboration, and relational practices in hybrid kitchens remains largely unexplored. Despite growing interest in social XR, phygital interfaces, and domestic computing, we still lack an in-depth understanding of how designers envision the communicative and emotional possibilities of hybrid connected kitchens.

While prior smart-home and kitchen XR research has largely focused on task assistance, safety, interaction techniques, or procedural guidance, much less is known about how future hybrid kitchen systems might mediate social presence, everyday communication, and the subtle rhythms of domestic life. Existing systems tend to treat the kitchen as a technical workspace rather than a socially charged environment where routines of care, coordination, and shared meaning unfold. Our goal is therefore not to extend smart-kitchen automation, but to examine how *hybrid spatial communication* tools may reshape relational and communicative practices in domestic settings. To investigate these questions, we developed three speculative phygital artifacts; an ambient cooking companion (ambient channel), a synchronous communication portal (real-time channel), and an asynchronous 3D messaging system (temporal channel), embedded within an immersive VR kitchen environment. Together, these artifacts sketch a three-channel communication model for hybrid connected kitchens, representing different temporal and sensory modalities of domestic communication: ambient, synchronous, and asynchronous. To address this gap, we investigate: **(RQ1)** how professional designers interpret and appropriate speculative phygital artifacts within an immersive VR-connected kitchen as supports for synchronous, ambient, and asynchronous social interaction; and **(RQ2)** what opportunities and concerns they identify regarding social presence, household rhythms, inclusivity, privacy, and the broader socio-emotional dynamics of XR-enabled domestic ecologies.

To explore these questions, we conducted a designer-focused evaluation study featuring three speculative phygital artifacts—an ambient cooking companion, a synchronous communication portal, and an asynchronous three-dimensional messaging system—embedded within an immersive VR kitchen environment. Through situated VR tasks, think-aloud walkthroughs, semi-structured interviews, and a structured design-evaluation survey, we analyze how designers imagine everyday domestic practices emerging around these artifacts, where they see value and potential, and where tensions and ethical considerations surface. We focus intentionally on professional designers rather than end-users. Designers in home-appliance contexts act as *proxy futurists—experts* who routinely anticipate emerging technologies, interpret future domestic scenarios, and translate speculative ideas into product trajectories. Their

conceptualisations reveal early signals about how hybrid connected kitchens may be framed, appropriated, and problematised before such systems ever enter people’s homes. Studying designers therefore provides insight into the *future-making practices* that shape domestic XR long before widespread adoption.

Our investigation contributes a conceptual, empirical, and methodological foundation for understanding hybrid connected kitchens as socially meaningful XR environments. We advance the notion of kitchens as *phygital communication hubs*; provide insights into how designers reason about social presence, co-presence, ambient connection, accessibility, and privacy in domestic XR; and demonstrate how speculative VR-based prototyping can serve as an effective design-research method for probing future domestic technologies.

2 Background

Our work sits at the intersection of domestic XR, phygital interfaces, and social presence. We review how XR has been explored in kitchens, dining, and domestic contexts as sites of collaboration; how phygital and ambient interfaces connect physical artifacts, sensing infrastructures, and hybrid representations; how social presence and co-presence shape interaction in XR; and how inclusivity, accessibility, and ethics inform emerging domestic hybrid environments.

2.1 Social XR in Domestic Life and Connected Kitchens

XR has been widely explored for cooking guidance, food experiences, and domestic social encounters. Early projection-based systems such as Shadow Cooking coupled instructions with physical action [6], while recent reviews show how VR/AR support food preparation, dining, and affective manipulation [15]. Kitchen-focused XR further includes hands-free collaborative cooking systems like HoloCook [16], AR guidance tools that use multimodal prompts and context-aware overlays [17], and AR-based appliance support leveraging virtual assistants and 3D overlays [18].

Cooking has also become a site for empathy and sensory experimentation. *Cooking in the Dark* uses spatial audio, passive haptics, and voice interaction to simulate blindness and foreground accessibility [19]. VR co-cooking platforms explore social cohesion through synchronized cooking and shared gestural interaction [20]. Beyond kitchens, remote dining systems like the Telematic Dinner Party [4] and Mixed Dining for older adults [21] highlight hybrid togetherness around meals. Creative and design-oriented food XR extends these ideas. CakeVR supports remote cake co-design through shared 3D artifacts [22, 23], while KitchXR positions VR as a context-rich environment for understanding kitchen workflows and appliance use [14]. AR-enhanced intergenerational cooking reduces confusion and strengthens emotional connection between dispersed relatives [24]. More recent smart-kitchen work such as “Cookie” [8] explores multimodal assisted cooking that preserves human agency, balancing proactive system behavior with user control through Wizard-of-Oz prototyping. Virtual cook-along studies map the design space of remote cooking across planning, enabling, co-presence, and food interaction [5].

Existing work establishes kitchens and dining rooms as promising XR sites for guidance, empathy, co-creation, and remote togetherness [5, 6, 14, 15, 22, 24]. Yet most systems prioritize task assistance, entertainment, or single-purpose interactions, treating XR as a layer rather than as part of a broader hybrid communication ecology. We instead frame the *connected kitchen* as a social XR hub where phygital artifacts and an immersive VR prototype probe how domestic communication and cohesion might shift when physical and virtual layers become tightly interwoven.

2.2 Phygital Domestic Interfaces: Ambient Displays, Smart Homes, and Situated Guidance

Phygital and ambient domestic interfaces investigate how physical environments can host hybrid representations and subtle communication cues. Early ambient displays like Hubscape visualized home sensor data across surfaces [25], and implicit-interaction systems adapted content through proximity or identity sensing [26]. Hybrid smart homes use immersive models to evaluate context-aware services [27], while IoT Avatars link physical devices with manipulable mixed-reality representations [28]. Ambient artifacts increasingly support reflection and social awareness. Situated displays for ADHD co-regulation [9], family negotiation tools like Schlusslicht [29], and workspace awareness systems such as LiteCo [30] illustrate how subtle cues and continuous feedback shape routines. Always-on in-home energy displays reveal how ambient representations influence everyday decision-making [31]. Multi-sensory systems like LiveNature integrate visuals, audio, and tactile cues to support calm engagement for people with dementia [32]. Passive co-presence systems—There Chair, Fragrance Frame—highlight how peripheral tactile or olfactory traces sustain distant connection [10, 33]. Technology-probe studies of domestic AR reveal that users value augmented information not only for practical support but also for enhancing social activity, awareness, and playful shared experiences in the home [11]. This suggests that domestic mixed-reality systems must account for household rhythms, interpersonal dynamics, and the expressive dimensions of everyday communication.

Phygital smart-home control expands interaction through tangible and spatial interfaces. Everyday objects used as TUIs demonstrate how physical metaphors shape control literacy [34]. ARKitchen combines magnetic markers and iron workspaces for tangible AR cabinet authoring [35, 36]. MR interfaces explore hierarchical UI frameworks and near-object interactions [37], while comparisons of AR, MR, and VR highlight trade-offs in appliance control [38]. Systems like HoloHome integrate XR with Ambient Assisted Living for older adults [39], and AR-in-VR toolboxes enable prototyping domestic AR concepts under ecologically valid conditions [13]. Phygital domestic systems—ambient displays, IoT avatars, tangible controllers—show how hybrid cues can be woven into physical environments [25, 27, 28, 35, 38, 39]. Yet few explicitly target kitchens as sites of *social* communication or examine how such artifacts are experienced in synchronous social XR environments. We address this gap by embedding speculative phygital artifacts in a VR-connected kitchen and treating them as communicative media for awareness, togetherness, and domestic coordination.

2.3 Social Presence, Co-presence, and Collaboration in XR

Social presence and co-presence provide key lenses for understanding social interaction in XR. Foundational work highlights the illusion of non-mediation and sense of community as central to multi-user VR [1]. Reviews emphasize how visual feedback, embodied interaction, and motion fidelity shape emotional involvement and collaboration [3, 40]. Empirical studies examine how visual and embodied cues influence coordination. Avatar visualisation affects attention and interaction quality [41]. Behaviourally realistic bidirectional gaze reduces coordination overhead [42]. Eye tracking in XR maps gaze as input, communicative cue, and shared attention signal [43]. Movement fidelity enhances presence and influences interpersonal distance [44]. Avatar realism can strengthen trust and social presence [45, 46], though effects vary across contexts [47, 48].

Collaboration studies compare symmetric and asymmetric media configurations, showing benefits for hybrid PC–VR collaboration [49]. Co-located VR game collaborations reveal how roles, task structures, and communication channels shape collective efficacy [50]. In remote XR collaboration, 3D reconstruction views often outperform 2D or 360° video [51]. SocialVR platforms using volumetric reconstruction support emotion perception and social presence [52]. Multimodal cues—vibrotactile, haptic, physiological—further modulate connectedness and coordination [53–55]. Several studies complicate simple presence–performance relationships. AR-based remote guidance shows social presence is not always predictive of collaboration quality [56]. Accounts of coextensive space highlight how digital and physical layers blur in XR [57], while social exchange frameworks link avatars, objects, and cues to trust and reciprocity [58].

Work on social presence and collaboration provides rich insight into how gaze, embodiment, and multimodality shape shared experience [1–3, 42, 55]. Yet these studies rarely consider kitchens as materially dense and socially charged environments. Our study brings these insights into connected kitchens by analysing how speculative phygital artifacts shape presence, coordination, and cohesion within domestic routines.

2.4 Inclusivity, Accessibility, and Ethical Considerations in Connected Domestic XR

Inclusive design plays a growing role in domestic XR and smart home research. MR and VR-based AAL systems support older adults and users with mobility or cognitive challenges [39, 59]. Digital twin–MR collaborations help dispersed families assist elders remotely [60]. Social VR exergames and virtual cultural experiences for older adults show how shared embodied tasks strengthen connection for specific user groups [61, 62]. Mixed Dining foregrounds social interaction among elderly individuals eating alone [21]. Domestic XR has also been used to explore sensory difference and family dynamics. *Cooking in the Dark* embodies blindness to foreground sensory-motor accessibility challenges [19]. ADHD-oriented co-regulation displays [9] and negotiation-oriented ambient systems like *Schlusslicht* [29] demonstrate how domestic technologies scaffold regulation and shared awareness. Passive co-presence research highlights how ambient sound, tactile cues, and object traces sustain family connection across distance [10, 33].

Ethical concerns are emerging prominently. Gaze-based identification research reveals privacy risks in social VR [63]. Telepresence robot design emphasises balancing embodiment, control, and acceptability in intimate domestic settings [64]. Social exchange research reframes XR interactions through trust, agency, and long-term relational dynamics [58], while presence research warns against assuming presence is universally beneficial [56]. Domestic XR research increasingly attends to accessibility, care, and ethical risk [10, 29, 33, 59, 60, 63]. Yet kitchens remain under-explored as active domestic workspaces where inclusivity, privacy, and social communication intersect. Our work examines speculative phygital artifacts as relational and ethical actors in connected kitchens, highlighting how such systems may support—or complicate—equitable and sustainable forms of domestic social interaction.

3 Methods

We designed our methodological approach to understand how professional designers interpret speculative phygital artifacts and imagine social interaction in hybrid connected kitchens. To address our research questions, we employed the quantitative and qualitative methods. We conducted an immersive VR-based evaluation study followed by semi-structured interviews and a design-evaluation survey. The quantitative results are derived from the analysis from the survey as a control/validation mechanism of the qualitative results which are the results of analysis of outcomes from interview and open ended questions from survey.

3.1 Methodological Approach

Our study followed a research-through-design-inspired evaluation protocol, in which interactive speculative artifacts were embedded into an immersive VR kitchen environment and experienced by participants through situated tasks. This combination of experiential exploration, reflective interviewing, and structured survey responses allowed us to investigate how designers make sense of synchronous and asynchronous communication concepts and how they envision everyday practices emerging around them. The methodological structure was deliberately aligned with our research questions: the VR sessions offered first-person experiential grounding for RQ1 and RQ2, the interviews elicited deeper reflections on social presence, collaboration, and concerns, and the survey provided structured comparative insights across design, interaction, usability, and contextual fit.

3.2 Participants

Fourteen professionals from multiple departments at the headquarters of a global home appliance company in [blind] took part in the study. Participants were recruited using purposeful expert sampling through an internal open call targeting teams engaged in product development, innovation, and design. Sessions took place in a dedicated testing room within the company. Participants ranged in age from 26 to 45 (median: 38). All participants provided informed consent.

Table 1 Participant IDs, demographics, educational background, field of expertise, years of experience, and their self-reported familiarity with kitchen and home-appliance design and VR/MR experience level (7-point Likert scale).

ID	G	Age	Edu	Field	Exp	Prod. Design Exp.	VR-MR Exp.
P1	M	37	Bachelor	Product Design	11	5	5
P2	M	40	Master	Engineering	15	7	5
P3	M	45	Master	Product Design	15	7	4
P4	M	40	Bachelor	Product Design	14	7	4
P5	F	31	Master	Product Design	6	5	3
P6	M	41	Bachelor	Product Design	15	5	5
P7	F	34	PhD	HCI	6	7	3
P8	F	26	Bachelor	Product Design	3	7	1
P9	M	34	Master	HCI	10	7	7
P10	F	28	Bachelor	Product Design	4	5	4
P11	M	43	Master	Product Design	15	5	4
P12	F	39	Master	Engineering	15	4	4
P13	F	32	Bachelor	Product Design	10	5	5
P14	M	39	Master	Product Design	15	7	4

3.3 VR Experience Procedure

The study consisted of three sequential phases: a VR experience phase, a semi-structured interview, and a post-study survey. After a brief introduction to the purpose, equipment, and procedure, participants provided informed consent and were guided into the VR environment. Each participant interacted with three speculative communication artifacts using a think-aloud protocol, while the facilitator observed from the physical environment and simultaneously joined the VR scene through a multiplayer setup. The facilitator prompted participants to verbalize thoughts throughout the session as a part of the thinking aloud protocol. All sessions were recorded using screen capture and external audio/video.

VR Kitchen Environment

For this study, we developed a dedicated VR prototyping environment inspired by the structure of our prior tool (blinded for review), but rebuilt from the ground up to avoid identity disclosure. The environment featured a fully interactive U-shaped kitchen with functional cabinets, appliances, a sink, utensils, and a variety of food items. Scenes were modeled using Blender3D, implemented in Unity, and scripted using MetaXR SDK components combined with custom interaction logic. Multiplayer functionality was enabled through Photon Fusion and Meta Avatar SDK. This environment allowed us to embed the three artifacts, Ambient Cooking, Portal, and Magmemo, into a coherent spatial and interaction-rich domestic setting.

Artifact Selection Rationale

The three artifacts examined in this study, Ambient Cooking, Portal, and MagMemo, were selected through an exploratory concept development phase in which we elaborated on common communication challenges in domestic kitchen environments on cooking, remote togetherness, and ambient domestic technologies. From a broader set of early concepts, we intentionally chose three artifacts that represent complementary modalities of hybrid communication: (1) an *ambient*: *Ambient Cooking* modality that foregrounds mood, atmosphere, and low-demand co-presence; (2) a *synchronous*: *Portal* modality that supports high-bandwidth, real-time social interaction; and (3) an *asynchronous*: *MagMemo* modality that enables expressive, persistent messaging embedded in the physical environment. This trio allowed us to probe a diverse spectrum of spatial and temporal communication practices while remaining feasible to prototype within a shared VR kitchen environment.



Fig. 1 The VR kitchen environment used in the study. Participants interacted with a fully functional U-shaped kitchen featuring food items, appliances, and spatially embedded prototypes.

Ambient Cooking

In the first stage, participants engaged in an interactive pizza-garnishing task designed to foreground ambient, atmospheric qualities. The kitchen lighting was adjusted to evening conditions, and participants used bowls containing ingredients such as mozzarella, olives, peppers, corn, ham, sausage, and pepperoni. Ingredients were poured or placed onto dough via shaking or grasping interactions; the system instantiated corresponding ingredient meshes dynamically. As the task began, the Ambient Cooking prototype activated: an AI-generated animated visual of a Mediterranean seaside town (integrated as a subtle moving backdrop) and ambient soundscape played, complemented by soft colored countertop reflections. The goal was to simulate how ambient and aesthetic cues might shape social and emotional experiences during cooking.



Fig. 2 The default Portal design viewed by participants. The circular frame includes ambient blue backlighting and subtle visual distortion, offering life-size synchronous video and audio from another kitchen space.

The Portal

The second stage introduced the Portal—a synchronous communication artifact inspired by ancient architectural motifs and their popular reinterpretations in science-fiction media. The Portal enabled real-time audio-visual communication and provided a parallax effect, allowing participants to shift viewing angles through embodied movement [65, 66]. To simulate communication across two distinct households, a second kitchen was added to the scene. The facilitator moved into this secondary kitchen in VR, leaving the participant alone to explore portal behaviors. Participants were then guided through a scenario where they described their previous pizza creation using hand gestures and explored how portals might support everyday communication. Portals were implemented using the *Portals for VR* Unity asset (limited to video and audio transmission). Multiple form-factor variations—including circular, compact, square, and rectangular portals—could be toggled by the participant.

MagMemo

In the final stage, participants explored MagMemo, an asynchronous communication artifact inspired by refrigerator magnets used for notes and reminders in everyday homes. Six 3D+audio messages were recorded by designers (and in one case, a child), each presenting a distinct domestic scenario such as forgotten items, grocery requests, recipe sharing, or dinner invitations. Messages were captured using depth-enabled mobile-phone cameras and reconstructed as animated 3D mesh sequences. In VR, each MagMemo appeared as a framed object pinned to the refrigerator; participants activated them using point-and-click interaction to trigger playback.



Fig. 3 A participant (shown as a Meta Avatar) interacting with Magmemo. Each message is represented as a framed 3D animation with accompanying audio, activated through point-and-click interaction.

3.4 Semi-Structured Interview

Following the VR experience, participants took part in a semi-structured interview consisting of three parts. First, they were asked about VR sickness symptoms and general impressions of the VR environment. Second, for each artifact, three guiding questions prompted evaluative reflection, imagined use cases, and broader design considerations. Finally, participants identified which artifact they felt held the most value for further development and expressed any concerns—including privacy, usability, and ethical issues. Each interview held one by one and recorded in audio, which are then transcribed from the analysis.

3.5 Survey

To complement the qualitative insights, a survey was administered including 18 design-evaluation items per artifact and six demographic questions. Sixteen items of artifact questions used a 7-point Likert scale; two were open-ended. Items were grouped into four constructs—Design, Interaction, Usability, and Contextual Fit—allowing comparison across conceptual dimensions and across artifacts.

3.6 Data Analysis

In our analysis, we integrated a qualitative thematic examination of notes taken during the VR exploration, interview responses and open-ended survey items with a quantitative evaluation of usability metrics. This mixed-method approach strengthened the validity of the designers’ feedback and provided more comprehensive insights into the overall experience and the proposed artifacts. In qualitative analysis we followed the six-phase reflexive thematic analysis framework by Braun and Clarke [67], an approach well suited to exploratory studies seeking to understand user experiences and professional practices. Two researchers independently reviewed the dataset, documenting observations and identifying preliminary patterns. Using an inductive coding strategy,

No.	Statement	Category
1	The Artifact is well designed in outlook (form, color, material, size, etc).	Design
2	The Artifact is well designed in terms of interactions.	Interaction
3	The Artifact needs a redesign in terms of usability, security, privacy, or other aspects.	Design
4	The Artifact uses intuitive interactions.	Interaction
5	The Artifact needs a redesign for better interaction.	Interaction
6	The Artifact should provide more ways to interact.	Interaction
7	The Artifact is designed flexibly and provides sufficient control to users.	Design
8	The Artifact fits well into the kitchen context.	Context
9	The Artifact is well positioned in the room.	Context
10	The Artifact is user-friendly and accessible to most users.	Usability
11	The structure, technology, and techniques of The Artifact are clear to understand and imagine.	Usability
12	The Artifact has potential for future contexts such as cooking, communication, smart home, or others.	Context
13	The Artifact has user potential in today’s context.	Context
14	There are competitive tools/technologies available today that can replace The Artifact.	Context
15	The Artifact fits well into domestic environments outside the kitchen.	Context
16	The Artifact fits better into non-domestic spaces (public, work, meeting rooms, etc.).	Context

Table 2 The 7-point Likert survey items used to evaluate each artifact, categorized into Design, Interaction, Usability, and Contextual Fit.

we allowed codes to emerge directly from the data. These codes reflected participants’ reflections on social presence, collaboration, and concerns.

4 Findings

This section presents the results of our mixed-methods study. The survey data provide a structured evaluation of each artifact’s design, interaction qualities, usability, and contextual fit, while the interviews and think-aloud observations offer deeper insight into how designers interpreted the prototypes and imagined their place in everyday kitchen life. Together, these complementary perspectives reveal both how the artifacts were assessed and how they were envisioned as components of future hybrid connected kitchens.

4.1 Survey Findings: Artifact Evaluation

Across all three prototypes, participants rated contextual fit in the kitchen and perceived future potential particularly positively, indicating that the overall concept of hybrid connected kitchen artifacts was well received. Portal and MagMemo tended to receive more consistently positive scores on appearance and interaction, whereas the Ambient Cooking Experience elicited more polarized ratings, especially around interaction intuitiveness and redesign needs. At the same time, all three artifacts were seen as candidates for further refinement, with respondents frequently highlighting the need for richer interaction methods and improved usability, privacy, or safety.

4.1.1 Portal

Overall, participants evaluated the Portal positively across design and contextual factors. Appearance (Median = 5, SD $\approx$ 1.26) and interaction design (Median = 5, SD $\approx$ 1.52) received consistently high ratings, and participants agreed that interactions were intuitive (Median = 5, SD $\approx$ 1.56). Contextual integration within the kitchen was among the strongest results (Median = 6, SD $\approx$ 1.12), alongside clarity of structure and technology (Median = 6, SD $\approx$ 1.41). The highest-rated dimension was perceived future potential across cooking, communication, and smart-home scenarios (Median = 7, SD $\approx$ 0.96).

At the same time, several items highlighted clear areas for refinement. Participants indicated a strong need for additional interaction methods (Median = 6, SD $\approx$ 1.50) and moderate agreement regarding redesign in terms of usability, privacy, or interaction improvements (Median = 5, SD $\approx$ 1.58). While competitive technologies were acknowledged (Median = 4, SD $\approx$ 1.70), participants still viewed the artifact as adaptable beyond the kitchen into broader domestic and, to a lesser extent, public contexts.

4.1.2 MagMemo

Participants also evaluated MagMemo favorably across several core dimensions. Appearance (Median = 4, SD $\approx$ 1.15) and interaction design (Median = 5, SD $\approx$ 1.10) were rated positively, and participants generally agreed that the artifact enables intuitive interactions (Median = 5, SD $\approx$ 1.60). Contextual fit in the kitchen showed consistently strong endorsement (Median = 6, SD $\approx$ 1.18), and the clarity of the artifact’s structure and technology was rated above the midpoint (Median = 6, SD $\approx$ 1.39). As with Portal, a strong signal emerged around future potential, with high agreement regarding its suitability for cooking, communication, or smart-home scenarios (Median = 6, SD $\approx$ 1.00) and solid perceived usefulness in current contexts (Median = 5, SD $\approx$ 1.24).

MagMemo likewise revealed room for improvement. The need for additional interaction methods showed strong agreement (Median = 6, SD $\approx$ 1.67), and moderate support emerged for redesign related to usability, privacy, or safety (Median = 5, SD $\approx$ 1.41). Some respondents acknowledged the presence of competitive technologies (Median = 5, SD $\approx$ 1.71). Adaptability beyond the kitchen received generally positive but more variable ratings (domestic non-kitchen spaces: Median = 6, SD $\approx$ 1.46; out-of-home contexts: Median = 5, SD $\approx$ 1.62), suggesting that participants viewed the concept as extendable but increasingly context-dependent outside domestic environments.

4.1.3 Ambient Cooking Experience

Evaluations of the Ambient Cooking Experience were more polarized than for the other two artifacts. Appearance (Median = 5, SD $\approx$ 1.33) and interaction design (Median = 5, SD $\approx$ 1.86) received moderate-to-positive ratings, yet intuitive interaction was endorsed less consistently (Median = 5, SD $\approx$ 2.12), indicating greater variability in perceived ease of use. Contextual fit nonetheless remained strong overall: participants

agreed that the artifact integrates well within the kitchen (Median = 6, SD $\approx$ 1.44), is appropriately positioned (Median = 6, SD $\approx$ 1.60), and is generally understandable in terms of structure and technology (Median = 6, SD $\approx$ 1.67). Future potential again emerged as a highly rated dimension, with strong scores for suitability across cooking, communication, and smart-home scenarios (Median = 6, SD $\approx$ 1.33), as well as for current user applicability (Median = 6, SD $\approx$ 1.15).

At the same time, critical perspectives were more pronounced for this artifact. Participants gave more mixed responses regarding the need for redesign in usability, privacy, and safety (Median = 3, SD $\approx$ 2.13), suggesting that some saw a clearer need for change than others. The need for additional interaction methods was strongly endorsed (Median = 6, SD $\approx$ 1.86), and competitive technologies were acknowledged at similar or higher levels than for the other artifacts (Median = 5, SD $\approx$ 1.41). Adaptability outside the kitchen showed mixed support: while participants considered the artifact suitable for other home contexts (Median = 6, SD $\approx$ 1.33), ratings for applicability in public or professional settings exhibited wide variance (Median = 5, SD $\approx$ 2.10). Overall, the results suggest that although the Ambient Cooking Experience is viewed as contextually meaningful and future-ready, it raises clearer concerns around interaction adequacy and redesign requirements.

4.2 Interview and Observation Findings: Designer Reflections

Designers' reflections, drawn from think-aloud sessions, interviews, and open-ended survey responses, converged around four recurring areas that shaped how they imagined these speculative artifacts becoming part of everyday kitchen life. The themes of Design and Flexibility, Privacy and Security, and Cooperation and Communication capture how participants envisioned the artifacts adapting to different routines, protecting the intimacy of the home, and enabling new forms of synchronous and asynchronous social connection. Complementing these, the Use Cases theme highlights the alternative scenarios designers proposed.

4.2.1 Design & Flexibility

For the interviews, we prepared a set of prompting questions aimed at sparking discussion around design decisions such as form, size, interaction, placement, and flexibility. These questions were asked separately for each artifact. In addition, each participant was asked how they would design the related artifact or what kinds of changes they would make. Flexibility emerged as one of the shared and significant outcomes across participants. All participants emphasized flexibility in the design; under this concept, they referred primarily to the mobility and personalization of the artifacts. This may be related to the different placement options of the Portal that participants tried during the experience.

"In its current design, the portal feels like a kind of video-chat tool. If I were to design it, the first thing I would do is free the portal from fixed forms and fixed positions. In my opinion, if the portal were something that the user could create in any position and in any form using the controller, the potential interactions would increase significantly. For example, in the kitchen context, the user could place the portal on the countertop or under

the upper cabinets if needed. This would allow the interactions enabled by the technology to expand.” (P7)

Feedback for the Portal included comments in a similar direction, such as: *“I would prefer it to be flexible in terms of placement—being able to move it to a different location” (P10)*, and *“... it could even have a feature that follows me as I move around the house.” (P1)*. Comparable observations appeared in interviews: *“Sometimes I cook here, then I move to the table. The portal should come along too.” (P7, P13)*. Participant 7 also suggested more imaginative options: *“I would like to be able to draw/place that portal wherever I want... like with a laser or something.”* While these quotes clearly point to mobility, some participants suggested partial flexibility for the Portal: *“Being able to adjust the portal’s position depending on where the person spends most of their time... there should be 2–3 placement options.” (P4)*. Beyond placement, others highlighted interaction flexibility: *“...I might even want to move the camera around—to show what’s on the stove, inside the oven, or something on the countertop.” (P13)*

Insights about the Ambient Cooking Experience focused more on personalization. During the interview, P6 noted: *“If I can choose the music or the ambiance myself... that would be great. It helps with motivation.”* Similar opinions appeared in the survey: *“It would be good if the content were under my control whenever I want, or if I could use AI.”* and *“Changing the music, increasing/decreasing the volume, changing the light color (colors adjusted according to context and time of day, or selected from predefined favorites).”*

Flexibility suggestions were less frequent for the MagMemo artifact, even though some participants questioned its placement: *“...I should be able to see it wherever I want... otherwise it would make me question why I’m forced to look there.” (P3)*. Flexibility in design was articulated more concretely as: *“Being able to change the size. I would also want features such as start, stop, add/remove, hide, and volume on/off.” (survey)*

Beyond flexibility and personalization, participants also shared a range of additional design suggestions. One commonly recurring idea was expressed as follows: *“Instead of appearing like a new product in an already cluttered kitchen, it should have a built-in, concealed design.” (P4)*. This feedback highlights an important point about how designers conceive kitchen technologies—even virtual ones—as ideally integrated rather than additive. Another participant suggested reinforcing functionality and coverage by changing the form: *“Rather than being planar, the portal could have been a concave surface, creating a more immersive perception through its form.” (survey)*

Design suggestions for the Ambient Cooking Experience often focused on the content itself. Participants proposed ideas such as: *“If there were something that could create an ambiance like an actual Chinese kitchen... an American kitchen... or the kitchen of some famous cook in France, it would really put you in the mood.” (P4)*, and *“...it could include live views of certain cities or places.” (survey)*. One survey response also pointed to a perceived lack of interaction, noting: *“There was no interaction space for the user in this concept; this aspect could have been addressed more comprehensively.”* For the MagMemo artifact, most feedback emphasized noise and disruption in the 3D visual quality. Additionally, two different preferences for visual

representation emerged: *“Its bust-like appearance feels unpleasant to me... it might be better if it looked more integrated as a whole.”* and *“It doesn’t have to be a direct video recording. The avatars would already be loaded, so when the other person sends a written message, you could see their avatar and hear its voice.”*

4.2.2 Privacy & Security

Given the communicative nature of the artifacts, privacy emerged as a central concern for both participants and researchers. Several comments about the Portal shared similar worries; for example: *“In terms of security and privacy, it should not open directly; instead, it should have a structure that can be opened or closed only with the user’s approval.”* (survey), and *“I think the kitchen is a private space. I would use it to communicate with my spouse, friends, or mother, but you don’t always want it to be open at home. The person on the other side may not want to turn on their camera...”*. Three different comments on MagMemo proposed ways to address privacy. Two participants suggested access control mechanisms: *“Only my messages should be visible. Like Face ID.”* (P2), and *“The sender should be able to choose the visibility of the message.”* (P14). In contrast, P3 considered that, since these messages are sent to a household, they are inherently common messages intended for all occupants.

The integration of XR technologies into the kitchen also prompted designers to reflect on safety. Concerns about this issue concentrated largely on the Ambient Cooking Experience. One survey comment reflects this tension: *“I thought it might not help me focus better on what I am doing in the kitchen, but rather could be distracting. For me, good lighting and proper ergonomics are more important.”* In contrast, some participants suggested enhancing functionality to support cooking and safety, such as: *“It could warn me if the kettle is hot.”* (P6)

4.2.3 Cooperation & Communication

The communication artifacts, Portal and MagMemo, were frequently interpreted as tools for cooperation. In the communication dimension, the Portal was mostly associated with cooking and co-cooking scenarios, with feedback such as: *“You talk to your mother... and at the same time, you cook together with her.”* (P3) and *“It can be used for a synchronous cooking experience, allowing someone with limited cooking experience to learn by following a more experienced person.”* (survey). Additionally, this form of communication was imagined for assisting others, as expressed by P6: *“It could also be used for helping an elderly person.”* Further scenarios clustered around workshop-like uses, for example: *“A famous person giving a live cooking demonstration.”* (survey). Additional comments on cooperation also appeared under Section 4.2.4. Beyond cooking, lightweight social communication was discussed for both Portal and MagMemo by P6, P4, and P14, for example: *“Family far away... it’s good for easing that longing.”* (P6). P7 similarly described: *“My aunt once gave me a cornbread recipe... that paper has been with me for 8 years. It feels inherited—valuable—so I keep it. It would have been nice to record it in my aunt’s real voice, or even have a 3D model of the final bread, a 3D cue showing a critical step in the recipe, or maybe an image of her eating it or of the dinner table.”* (P7)

In line with this quote, 4/14 participants imagined MagMemo as an asynchronous communication tool for cooperation. Comments such as the following underline collaboration through annotation-like usage: *“I should even be able to create a shopping list; when it’s placed on the fridge, it can offer more interaction and usefulness, and it can be combined with different user needs.”* In the discussion of the Ambient Cooking Experience, the first participant imagined blending playfulness and cooperation, commenting: *“It could even offer a new, concept-driven cooking recipe experience... like a gamified family cooking activity.”* (P1)

4.2.4 Use Cases

The thematic analysis revealed various use cases from the survey and interview data. We divided these use cases into cooking and non-cooking categories. Although most cooking-related use cases appeared in earlier sections under the themes (Section 4.2), we also cover additional examples here. Non-cooking use cases include both in-kitchen and out-of-kitchen scenarios, as well as entertainment-oriented applications.

Cooking By the nature of the artifact context, all participants initially envisioned the use cases of Portal and Ambient Cooking within cooking scenarios. Additional examples were also mentioned, such as: *“It could be used to make the environment more enjoyable in amateur cooking classes.”* (P11) As discussed under cooperation, adding playfulness to cooking was also mentioned for both Ambient Cooking and MagMemo, with comments such as: *“It could even go into gamification—you’re chopping something and it guides you by projecting the cutting lines, it augments the sound of cutting, and it could even show how much 300 grams of cheese would be by projecting a guideline.”* (P14)

Non-Cooking Similar to P7’s comment about using the Portal artifact during dinner, P11 shared that *“A more holistic experience could have been combined with a kitchen scenario. It could be integrated into the meal-preparation process.”* Most participants (10/14) also agreed that the Portal has the potential to enhance communication between people beyond the kitchen, as expressed in comments such as: *“It could also be used while exercising or playing games.”* (P7). Similar ideas were shared for the Ambient Cooking artifact, including comments such as: *“It would be a technology that could be integrated into everyday life—it could be used for anything, even making cleaning a more enjoyable experience.”* (P14) Although some participants questioned anchoring MagMemos to the fridge, one survey comment supported this decision: *“Considering that the kitchen is the shared hub of the home, scenarios such as household members leaving notes for one another come to mind.”* Beyond communication and annotation use, participants also commented that MagMemo has strong potential as an augmented picture frame: *“I would want to collect content here related to memories I’d like to keep. That way, I could also place it in different locations around the house.”* (P7) and *“On the fridge... I would put the ‘We love you very much’ message my mom and sister recorded. I’d look at it whenever I feel like it.”* (P8)

5 Discussion

This study examines how design practitioners conceptually frame communication, collaboration, and social presence when kitchens become hybrid connected environments. The following sections discuss our insights across five thematic areas that respond to our research questions and situate our findings within broader scholarship on social XR, domestic HCI, and phygital interaction design. Each theme reflects a conceptual lens shaped by existing literature and the design space of domestic social technologies, while also pointing toward implications for future design research.

5.1 Reframing the Kitchen as a Hybrid Connected Space

Kitchens have long been recognized as socially charged domestic environments where coordination, care work, and interpersonal routines unfold. Prior HCI research on cooking and dining technologies has largely emphasized guidance, safety, and efficiency [6, 15], with some work beginning to foreground co-cooking and remote togetherness [4, 5]. Our findings extend this trajectory by framing the kitchen not only as a task-oriented workspace but as a *hybrid connected space* that can function as a social XR hub. Designer feedback highlights two intertwined aspects of this reframing. First, participants resisted fixed spatial configurations for XR artifacts in kitchens. As reflected in Section 4.2.1, the Portal was repeatedly described as something that should “come along too” when moving between preparation areas and the dining table, or be drawable “wherever I want” on demand. This insistence on mobility and reconfigurability suggests that hybrid connected kitchens should be conceived less as static installations and more as *reconfigurable communication zones* that flex with domestic rhythms. Second, designers framed the kitchen as a broader household hub. Under the non-cooking use cases (Section 4.2.4), participants imagined the artifacts supporting activities such as exercising, playing games, cleaning, or simply spending time together, aligning with domestic AR findings that highlight “social activity” and shared awareness as key values of home AR systems [11].

These insights suggest several implications for future design research. First, connected kitchen systems should explicitly support *spatial fluidity*: portals and ambient surfaces that can be repositioned, resized, or re-anchored across countertops, cabinets, and adjacent spaces rather than being bound to a single wall or appliance. Second, designers and researchers should treat the kitchen as part of a larger domestic network, designing for cross-room and cross-activity continuity instead of isolated cooking scenarios. Finally, the strong desire for concealed or built-in forms (e.g., designs that do not appear as “a new product in an already cluttered kitchen”) points to an agenda of *architectural integration* in which social XR functions are blended into cabinetry, lighting, and surfaces rather than introduced as standalone gadgets.

5.2 Social Presence and Co-Presence in Domestic XR Contexts

Social presence research typically focuses on work, play, or entertainment contexts, emphasizing immersion, non-verbal cues, embodiment, and shared attention as core drivers of presence and collaboration [1–3]. Our study brings these concerns into a domestic setting and shows that in kitchens, presence is evaluated not only in terms of

intensity or realism but also in relation to privacy, attentional demand, and household rhythms.

The three artifacts embody distinct modalities of presence. The Portal offers high-bandwidth synchronous presence through life-size audio-visual communication; Ambient Cooking foregrounds low-demand, atmospheric co-presence; and MagMemo supports asynchronous, memory-like presence via 3D+audio messages. Designers recognized the value of these different modalities but were cautious about when and how each should be invoked. For example, the Portal was associated with rich co-cooking and learning scenarios (e.g., cooking together with a parent, or receiving step-by-step guidance from a more experienced cook), yet participants also worried about having such a strong visual presence open “all the time” in a private kitchen. Similarly, Ambient Cooking was praised for its motivational, mood-setting qualities but flagged as potentially distracting or even unsafe if it drew attention away from heat sources or cutting tasks. These tensions echo broader findings that more presence is not always better [56] and that domestic technologies must adapt to complex temporal and social rhythms. Our results suggest treating presence in hybrid kitchens as a *tunable gradient* rather than a fixed design goal: households may need to shift between intense, synchronous presence (e.g., co-cooking, workshops), gentle ambient accompaniment (e.g., shared music and atmosphere), and lightweight asynchronous traces (e.g., messages, reminders) depending on time of day, task, and emotional state. Design research might therefore explore interaction mechanisms and policies that allow dynamic modulation of presence intensity, including “do-not-disturb” modes, scheduled or context-aware activations, and subtle transitions between presence modalities.

Moreover, the emphasis on supporting users’ competence and autonomy resonates with work on smart kitchen assistants such as “Cookie” [8], which balance proactivity with human agency. Future domestic XR systems should similarly ensure that social presence augments users’ sense of control and confidence rather than overwhelming them with constant connectivity or performative pressure.

5.3 Phygital Artifacts as Communicative Media in Everyday Life

Phygital systems—hybrid combinations of physical form, spatial interaction, and digital content—have been explored in smart homes, tangible interfaces, and ambient displays as expressive media for coordination, reminders, and emotional connection [10, 25, 28]. Our three artifacts can be read as such phygital communicative media: they are not simply tools for task completion but *spatial communication channels* embedded in the kitchen.

Designer reflections in Section 4.2.3 show how participants imagined these artifacts as supports for both everyday coordination (e.g., shopping lists, reminders, safety warnings) and more affective or playful exchanges (e.g., a cherished recipe recorded in a loved one’s voice, gamified family cooking activities, or ambient scenes that transport the household to another place). In these scenarios, the kitchen becomes a site where communication is enacted through light, sound, spatial framing, and embodied gestures—not just through text or voice. MagMemo, in particular, highlights the potential of phygital artifacts as *memory surfaces*. Participants envisioned it as an augmented

picture frame or a repository for important family messages that could be revisited over time, aligning with prior work on passive co-presence and situated traces in domestic environments [33]. At the same time, the critiques of its visual form (e.g., discomfort with “bust-like” representations, or a desire for more stylized or avatar-based renderings) underscore the importance of carefully chosen representational strategies when embedding personal content into shared spaces.

For future design research, these insights point to at least three directions. First, there is value in designing *families of phygital artifacts* that together span synchronous, ambient, and asynchronous communication, allowing households to compose their own communication ecologies rather than relying on a single monolithic system. Second, materiality and integration matter: artifacts that appear as lightweight additions or overlays may be perceived differently from those that are structurally embedded into furniture, walls, or appliances. Third, the expressive range of phygital artifacts could be expanded to include additional modalities (e.g., haptics, scent, or subtle motion), building on prior multi-sensory work in domestic care contexts [32] while remaining sensitive to clutter and overload.

5.4 Tensions Around Privacy, Boundaries, and Domestic Inclusivity

Domestic technologies inevitably raise questions around privacy, agency, inclusivity, and the negotiation of shared space. Prior research on smart homes, telepresence, and social VR has highlighted concerns about surveillance, control, and the long-term implications of always-on sensing [60, 63, 64]. Our findings show how these concerns play out specifically in the context of hybrid connected kitchens.

Participants articulated clear expectations for *boundary mechanisms*. For the Portal, they emphasized that it “should not open directly” and should only be activated with explicit user approval, reflecting a desire to avoid accidental exposure in a private setting such as the kitchen. For MagMemo, designers proposed granular visibility controls (e.g., FaceID-style access or sender-defined visibility), while others argued that messages posted in a shared kitchen should be treated as inherently collective. These perspectives echo ongoing debates in domestic HCI about shared versus individual ownership of data and displays in multi-person households. Concerns about safety and cognitive load surfaced most strongly around the Ambient Cooking artifact, where some participants worried that immersive ambience might distract from core cooking tasks and undermine ergonomics. At the same time, designers suggested safety-enhancing extensions such as warnings about hot kettles or other hazards, indicating that hybrid kitchen systems must carefully balance ambience and assistance.

Inclusivity emerged more implicitly, through comments about placement flexibility, clutter, and integration into existing kitchen layouts. The desire to reposition artifacts and to avoid additional “products” in already crowded workspaces suggests that future connected kitchen systems must account for diverse body sizes, mobility constraints, and sensory sensitivities. These concerns resonate with work on ambient displays for families with ADHD children [9], Ambient Assisted Living systems [39], and negotiation-oriented domestic artifacts such as Schlusslicht [29], all of which highlight the need for configurable, transparent, and respectful domestic technologies.

For design research, this points to a clear agenda: hybrid connected kitchens should be designed with *privacy-by-default*, explicit opt-in mechanisms for high-bandwidth communication, fine-grained message visibility and retention settings, and accessible controls for muting, relocating, or reshaping artifacts. Future studies might investigate how these controls are negotiated within households over time, and how hybrid kitchen systems can support vulnerable or marginalized household members (e.g., children, elders, or people with sensory differences) without reinforcing existing power imbalances.

5.5 Methodological Reflections: Speculative VR Probes for Domestic Futures

Our methodological approach combines speculative artifact design with an immersive VR prototyping environment, extending prior work on hybrid smart homes [27], AR-in-VR simulation platforms [13], and VR-based kitchen tools such as KitchXR [14]. By embedding Ambient Cooking, Portal, and MagMemo in a coherent, interactive VR kitchen, we were able to give designers a first-person experience of not-yet-available domestic technologies and prompt situated reflection on their communicative, emotional, and ethical implications.

Participants used this environment to experiment with placement, scale, and interaction styles, and to imagine use cases that extended beyond the scripted tasks. The think-aloud protocol and subsequent interviews revealed how speculative VR probes can surface both enthusiasm and skepticism: designers could articulate nuanced scenarios of togetherness, play, and support, while also identifying privacy concerns, aesthetic misalignments, and potential safety issues. This aligns with broader arguments in design research that speculative and immersive methods are well suited to probing emerging socio-technical configurations and making tensions more tangible.

At the same time, our approach has limitations. As a lab-based, time-bounded study with single participants, it cannot capture long-term appropriation, the dynamics of multi-person households, or the material constraints of real kitchens. The prototypes, though interactive, were not fully functional networked systems; this may have shaped participants' expectations and the granularity of their critiques. Future methodological work could therefore combine VR-based speculative probes with longitudinal in-home deployments of lower-fidelity artifacts (e.g., screen-based proxies, tangible mock-ups, or AR overlays), as well as cross-cultural studies that explore how different domestic norms and housing configurations shape the reception of hybrid kitchen technologies. For design research, a promising direction lies in developing *iterative hybrid pipelines* where concepts are first explored in VR (for spatial and experiential reasoning), then materialized as partial physical prototypes, and finally re-embedded into extended domestic trials. Such pipelines could also integrate participatory methods with non-designer household members, broadening the perspectives beyond professional designers while preserving the critical and imaginative potential of speculative VR.

6 Conclusion

This study explored how professional designers envision social interaction, collaboration, and communication in hybrid connected kitchens by experiencing three speculative phygital artifacts within an immersive VR domestic environment. Through situated VR tasks, reflective interviews, and structured design-evaluation surveys, we examined how designers interpret synchronous and asynchronous communication tools, how they imagine everyday practices emerging around them, and what opportunities and tensions they see when domestic spaces become XR-enabled communication hubs.

Our findings contribute to the growing body of VR and HCI scholarship on domestic XR, phygital communication, and socially meaningful hybrid environments. First, we introduce the notion of *hybrid connected kitchens* as a conceptual frame that positions kitchens not merely as task-oriented spaces but as socially charged, relational sites where ambient, synchronous, and asynchronous cues can meaningfully support togetherness. Second, we provide empirical insights into how design practitioners make sense of speculative phygital artifacts—highlighting perceptions of social presence, emotional comfort, contextual appropriateness, and concerns around privacy, accessibility, and household rhythms. Third, we demonstrate how immersive speculative prototyping can serve as an effective research probe for eliciting designer interpretations of domestic XR futures, offering methodological value for design-led research on emerging technologies.

This work also carries limitations. Our participant group consisted exclusively of designers from a single global home appliance company, which may limit the diversity of perspectives and restrict generalizability to other user groups, households, or cultural contexts. The VR prototypes were speculative and not functionally complete, which may have influenced how participants imagined long-term use. Furthermore, the study focused on designers rather than everyday household members; future work should expand to families, children, older adults, and multi-person households to better understand lived domestic dynamics.

Future research could explore long-term appropriation of hybrid kitchen artifacts in real homes through longitudinal deployments, examine cross-household interactions among family members with varied cultural and sensory practices, and investigate how AI-driven adaptation, multimodal cues, or shared embodiment can support richer domestic communication. Extending this work beyond kitchens to other domestic spaces—or to inter-home ecologies that link multiple rooms, households, and contexts—offers fertile ground for understanding how XR might shape new forms of togetherness, support, and coordination in everyday life. Ultimately, this study contributes an initial but substantive step toward imagining socially meaningful, inclusive, and ethically grounded XR experiences for domestic environments, pointing toward a broader vision of connected realities woven into the textures of everyday home life.

Declarations

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this study did not require formal ethical review or approval. **Consent to participate.** All participants were informed about the purpose and procedures of the study and provided written informed consent prior to participation. **Consent for publication.** Participants consented to the use of anonymized data and study outcomes in academic publications.

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Chapter 5

Findings & Discussions

I examine how Extended Reality (XR) technologies can support mediated spatial interaction in domestic kitchens, and how XR can function as a design instrument for envisioning, prototyping, and evaluating future domestic experiences. This discussion chapter brings together insights from the empirical and design-oriented studies presented throughout the thesis and interprets them in direct relation to the research questions introduced earlier.

In this chapter, I will be referring to my findings as reflections, suggestions, designer constraints, and frictions. I use *reflections* as overall observations and lessons I drawn from the findings across the studies over the course of the PhD as a researcher. *Suggestions* are findings that can be used in further design and design research studies by others. *Designer constraints* are practice-level limitations on the design process that obstruct designers and researchers from conducting broader and deeper studies within this domain. On the other hand *frictions* are recurring points of resistance that surface in everyday kitchen practice when communication, coordination, attention, and care intersect with technological mediation.

In recent HCI literature, "friction" has come to denote intentionally designed resistance introduced into otherwise smooth interactions to provoke reflection, deliberation, or mindful engagement. In origin, this term refers to any point of resistance or effort encountered during interaction with technology that interrupts the flow of a user's activity. Drawing on its physical metaphor, the term refers to a force that acts upon movement through an interaction sequence. As Cox et al. [144] articulate, design frictions are points of difficulty that can disrupt automatic, mindless engagement and instead prompt moments of reflection, while Mejtoft et al. [145] define friction as interaction that hinders people from painlessly achieving their goals and argues if a

more mindful interaction induced with friction by design will increase the satisfaction of users experience.

In this thesis, frictions refer to empirically observed interactional tensions, (recurring breakdowns in visibility, audibility, spatial reference, sensory cues, adaptability, and control) that surfaced across the studies and contribute to the structuring of the design space of mediated kitchen interaction. These are not interventions but programmatic constraints in the sense of Redström and Binder’s programmatic design research [23, 24]: persistent concerns that motivate, shape, and delimit successive design moves. These frictions are not treated as anomalies to be eliminated, but as productive tensions that expose the limits of existing interaction paradigms and point toward alternative spatial, embodied, and phygital design possibilities.

Across the studies, I repeatedly revisit the same interactional tensions through different XR configurations, methodological lenses, and design artifacts. In each case, a friction is translated into a concrete design intervention, such as alternative strategies for visibility, spatial anchoring, hands-busy interaction, or hybrid communication. They are examined through use, critique, or evaluation, and then folded back into subsequent iterations. Rather than converging on a single solution to kitchen interaction challenges, the thesis progressively charts a design space, making explicit its possibilities, limits, trade-offs, and boundary conditions.

Accordingly, I present the thesis’s central contributions not as prescriptive solutions but as a set of design suggestions that articulate intermediate-level design knowledge [146], specifying where and how XR can productively intervene in kitchen interaction. I developed suggestions and reflections across the two research questions. The first four (Suggestions 1–4) address RQ1 and concern user-facing interaction: spatial flow across zones and task transitions, multi-modal and hybrid communication, phygital anchoring, and tunable presence. The remaining four contributions of this chapter take a different register. Because they concern what I learned about XR as a method rather than what future designs should do, I present them not as suggestions but as reflections.

A single reframing spans all suggestions and reflections and summarizes the contribution of this thesis: In the domestic kitchen, social interaction is a primary design concern which is to be designed for, not treated as a byproduct of task support, and XR can strengthen it through spatial flow and multimodality without disrupting the cooking it accompanies.

This discussion chapter synthesizes the insights generated through this iterative process. I organize the discussion around the thesis’s research questions and reflect on what the studies

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collectively reveal about mediated spatial interaction in kitchens, as well as about XR's role as a methodological and design resource. In the sections that follow, I discuss findings in relation to the following research questions:

- **RQ0.** How can Extended Reality (XR) technologies support mediated spatial interaction in domestic kitchens, addressing both everyday communication challenges and the design of meaningful domestic experiences?¹
- **RQ1.** In what ways can state-of-the-art technologies address the interaction challenges that emerge in domestic kitchen environments, particularly during hybrid or remote engagements?
 - **RQ1a.** What communicative, spatial, and sensory breakdowns do users encounter in kitchen contexts?
 - **RQ1b.** How can XR mediate presence, awareness, and coordination in the kitchen environment, and with what implications for collaboration, safety, and togetherness?
- **RQ2.** How can designers meaningfully employ XR as a tool for envisioning, prototyping, and evaluating domestic experiences in kitchen contexts?
 - **RQ2a.** How do designers use XR as a methodological tool to reason about spatial and interactional dynamics in kitchens?
 - **RQ2b.** What unique design insights arise from immersive and speculative XR prototyping?

Taken together, this discussion reflects on how these questions are addressed across the studies and how the findings collectively reposition the kitchen as a dynamic, socially charged environment, one in which spatially mediated interaction and spatially grounded design practice can be meaningfully explored through XR.

5.1 Addressing RQ1: How XR Mediates Interaction Frictions in Kitchens

The first research question (RQ1) addresses the experiential dimensions of mediated spatial interaction in domestic kitchens. It asks: *In what ways can XR technologies address the interaction challenges that emerge in domestic kitchen environments, particularly during hybrid*

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or remote engagements? To operationalize this inquiry, I examine how everyday kitchen interaction is shaped by recurring communicative, spatial, and sensory frictions, and how these frictions influence presence, awareness, and coordination under hands-busy and safety-critical conditions. Together, these perspectives structure an investigation into how XR might intervene in domestic kitchen interaction without displacing existing practices.

Across Paper1, *Kitchen Talks*(4.1) and the subsequent studies, I identify recurring breakdowns that shape everyday kitchen interaction during hybrid or remote engagement. These frictions reveal where communication, spatial reference, and sensory cues fail, and why those failures matter for coordination, safety, and togetherness in a dynamic domestic workspace. Rather than treating these issues as isolated usability problems, I approach them as *programmatic tensions* within a Programmatic Design Research framework [23]. Each friction is translated into a design intervention, examined through successive XR implementations and evaluations, and then revisited as part of an evolving design space shaped by trade-offs and boundary conditions.

The following sections present the insights related to RQ1 in detail. I first articulate the key interaction frictions that shape kitchen communication and coordination (RQ1a). I then discuss design implications for how XR can mediate presence, awareness, and coordination in response to these frictions (RQ1b).

5.1.1 (RQ1a) Communicative, Spatial, and Sensory Breakdowns in the Domestic Kitchen

This subsection synthesizes the core frictions reported across the studies as recurring breakdowns in mediated kitchen interaction. These frictions reflect fundamental tensions between the kitchen’s dynamic, hands-busy nature and the assumptions embedded in screen-based communication systems. In this thesis, I foreground kitchen dynamics, movement, and task zones as primary design considerations rather than treating communication as a stationary, screen-centered activity.

Friction 1: Visibility Breakdowns Paper 1(4.1 surfaced a recurring visibility problem; sustaining mutual visual access through a single, fixed viewpoint during synchronous remote interaction in a dynamic, hands-occupied workspace. As users move between kitchen zones and rapidly shift attention, anchored camera views or screens routinely obscure relevant actions,

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tools, and changes in workspace state.

Why it matters. In cooperative cooking moments, visibility is not primarily about seeing faces, but about seeing action context at the right moment. The coordination burden increases precisely when users need to share situational information opportunistically, for example during a quick check, an improvised step, or a safety-critical moment. Visibility therefore becomes foundational for extending smart-kitchen interaction beyond task efficiency toward social coordination, learning, and family bonding. Prior systems (e.g., *Living Cookbook*, *Personal Chef*) propose partial remedies through camera configurations, overlays, or alternative workspace representations. However, across my studies, visibility emerges as an *interactional requirement* under movement, time pressure, and object-centered reference, not simply a matter of “better video.” In response, this thesis introduces the *Portal* concept as an exemplar design direction, aiming to reduce visibility friction by re-establishing shared spatial reference and action context through an XR-mediated, spatially grounded communication threshold. The Portal is revisited as a speculative probe in Paper II and IV (4.2-4.4) to examine how anchoring, spatial representation, and alignment can reduce the coordination burden imposed by fixed-camera video.

Friction 2: Disruptions in Audibility Alongside visibility, Paper I 4.1 revealed breakdowns in *audibility*: the ability to maintain intelligible, well-timed conversation while moving through space and while the environment produces fluctuating noise. Appliance sounds, running water, and ventilation reduce audio clarity, while movement and divided attention further undermine turn-taking and conversational flow.

Why it matters. As audio quality degrades or latency increases, coordination and conversational rhythm deteriorate rapidly. In hands-busy conditions, users cannot reliably compensate for these disruptions; interruptions become costly, and misunderstandings can escalate into safety-relevant issues. These findings suggest that “good audio” is necessary but insufficient. Mitigating audibility friction requires treating communication as a *spatial and multi-channel system*, in which voice is complemented by additional cues that convey attention, state, and reference without demanding continuous verbal exchange, especially during moments of high activity and divided attention.

Friction 3: Spatial Awareness Deficits A persistent breakdown concerns shared spatial awareness. Conventional smart devices and screen-centric interfaces convey limited spatial cues

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and contextual detail, and they do not adapt well to the kitchen’s distributed zones and rapidly shifting activities. As users move between zones, the absence of a shared spatial reference frame disrupts coordination and mutual understanding.

Why it matters. This deficit helps explain why remote cooperation differs so markedly from co-present interaction. Without stable spatial reference, users experience increased cognitive load and greater communication overhead as they attempt to reconstruct context verbally or visually. This thesis argues that supporting mutual spatial awareness is essential for hybrid and remote kitchen interaction. XR offers a promising pathway; through mediated spatial interaction and spatial anchoring, it can restore reference, reduce re-orientation work, and improve coordination.

Friction 4: Multi-Sensory Communication Needs Across the studies, participants highlighted a gap beyond vision and hearing: remote cooking experiences neglect other sensory cues that shape kitchen work and social experience.

Why it matters. Smell, taste, and thermal sensations contribute to how people experience communication and coordination. When these cues are absent, the sense of *togetherness* is not fully supported. Users may rely on heavier verbal explanations or uncertain visual inferences, which can reduce confidence and shared understanding while increasing coordination effort. Fully simulating these senses remains an open scientific and technical challenge and is not yet practical at scale. However, prior work suggests XR systems can elicit or approximate aspects of such sensations even without corresponding physical stimuli. In this thesis, this friction operates as a boundary condition: it clarifies where mediated experience can be strengthened today, and where meaningful equivalence to co-present cooking remains constrained.

Friction 5: Spatial Adaptability and Flexibility Domestic kitchens vary widely in layout, circulation, sightlines, device placement, and ambient noise. This heterogeneity complicates the design of appliances and smart-kitchen interfaces intended for broad adoption.

Why it matters. A placement strategy that works in one kitchen may fail in another. In the Paper I(4.1 co-design sessions, participant feedback did not converge on a single “best” placement. At the same time, consistent with the spatial awareness deficit, participants strongly wanted systems that support tasks across multiple work zones. This friction motivates a requirement for flexible, reconfigurable interaction that can adapt across layouts and shifting

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zones of activity. It frames generalization as an interaction-design problem such as supporting movement, zone transitions, and context change, rather than only a deployment problem.

Friction 6: Control, Privacy, and Identity Under Hands-Busy Conditions This friction comprises two coupled breakdowns in user control. **(1) Control over interaction during hands-busy work.** Cooking often produces hands-wet or hands-dirty conditions, making touch-based input impractical. This interrupts interaction and undermines reliable device control at precisely the moments when attention and safety demands are high. **(2) Control over boundaries, privacy, and identity.** Beyond operational control, Paper II 4.2 surfaced recurring concerns about privacy and selective identity management in cooperative and collaborative settings. Participants described a need to protect privacy, manage information boundaries, and selectively regulate anonymity and identity. At the same time, they noted that accountability and continuity of identity can be necessary for trust and coordination.

Why it matters. Taken together, these breakdowns show that mediated kitchen interaction is not only constrained by sensory and spatial gaps, but also by users' ability to regulate exposure and control interaction while remaining engaged in hands-busy activity. In this thesis, Friction 6 functions as a programmatic constraint that shapes which mediation strategies remain viable in domestic settings.

5.1.2 (RQ1b) Mediating Presence, Awareness, and Coordination Through XR

Building on the frictions synthesized above, this subsection discusses how XR can mediate presence, awareness, and coordination in domestic kitchens, and what implications follow for collaboration, safety, and togetherness. Rather than presenting these implications as universal prescriptions, I position them as design-oriented suggestions that emerged across the studies as plausible responses to recurring breakdowns, and as directions that were further examined through speculative artifacts and critique in Papers II and IV (4.2-4.4).

Suggestion 1: Enabling Spatial Flow and Interaction Building on the communication frictions identified in RQ1a, this thesis proposes *spatial flow and interaction* as a guiding concept for XR-mediated kitchen communication. Rather than treating interaction as anchored to a single interface or viewpoint, this concept foregrounds continuity across movement, task shifts,

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and changing attention, the conditions that define everyday activity in domestic kitchens.

Core claim. Spatial flow and interaction prioritize a spatial mindset in interaction design: core interactions should remain usable and legible across distinct kitchen zones, rather than being optimized for a single location or moment. Kitchen zones are not defined solely by physical location. They also reflect shifts in task, posture, focus, and attentional demand, for example moving from preparation to cooking, from observation to action, or from individual work to shared coordination. Designing for zone transitions therefore requires interaction concepts that remain stable under movement, changing bodily orientation, and hands-busy conditions.

Link to identified frictions. This design direction addresses visibility breakdowns, audibility disruptions, and spatial awareness deficits by maintaining interaction continuity across zones and by enabling spatial reference that does not rely on fixed camera viewpoints. It also motivates interaction approaches that do not depend on buttons or touchscreens, instead favoring gestural, spatially anchored, and context-aware cues that align with hands-occupied work (see Friction 6).

Positioning in the literature. This concept parallels arguments in spatial computing research that emphasize architectural and spatial logics over screen-centric interface metaphors. Zhang [68] argues that spatial computing requires adopting architectural design principles, foregrounding movement, adjacency, and spatial relationships while exposing conflicts with conventional interface paradigms. Related work similarly highlights the need for intuitive interaction techniques that leverage spatial computing capabilities rather than reproducing two-dimensional interaction patterns in three-dimensional space [69–71]. Kumari et al. further stress that user-friendly spatial interfaces must align with embodied action and environmental context [70]. By foregrounding spatial flow and the mechanics of kitchen work zones, this thesis extends these theoretical arguments into a more situated, domestic context. The complexity of kitchen environments becomes more tractable when analyzed through kitchen and task zones derived from users’ movement patterns, task sequences, and contextual shifts. These zones, identified through empirical analysis, provide a practical structure for designing XR interactions that remain coherent, legible, and supportive across the rhythms of everyday kitchen life.

Suggestion 2: Fostering Communication with Multi-Modal and Hybrid Modes A further implication of treating the kitchen as a communication and coordination hub is the need to support interaction that distributes communicative effort across modalities and time.

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Kitchens produce fluctuating noise, interrupted attention, and hands-busy conditions; under these dynamics, communication cannot depend on continuous conversation or uninterrupted visual attention.

Core claim. XR-mediated kitchen communication should be designed as multi-modal and hybrid: it should combine modalities (e.g., spatial annotations, lightweight audio, embodied cues) and support multiple temporalities of engagement, including moments of live exchange and moments where awareness or coordination must persist without sustained attention.

Link to identified frictions. This direction responds directly to audibility disruptions (Friction 2) and multi-sensory gaps (Friction 4) by reducing reliance on a single channel and by distributing coordination work across complementary cues. It also supports spatial awareness (Friction 3) by enabling referential cues to remain anchored to the environment rather than re-established verbally each time users move across zones.

Positioning in the literature. Prior work suggests both a gap and an opportunity. In the article of Kim et al. [91], emphasize the effectiveness of combining several cues rather than relying on a single cue.. Reviews such as Ens et al. [82] indicate that multimodal and temporally flexible XR-based collaboration remains comparatively underexplored within CSCW, despite its relevance for distributed coordination. RStudies of channel transitions and hybrid workflows, reviewed in Section 2.2.4, argue that cooperation improves when systems support switching, partial overlap, and continuity of context across modes rather than enforcing a single channel at a time. In this thesis, this implication is explored through the lens of domestic rhythms: communication must accommodate interruption management, safety, attention budget, and household boundaries. These constraints shape what forms of multi-modal and hybrid mediation remain viable without drifting into overload or surveillance.

Suggestion 3: Strengthening Interaction with Phygital Artifacts A third implication concerns the role of *phygital artifacts* in mediating presence and coordination. Across the studies, participants consistently reasoned about mediated interaction through placement, adjacency, workflow integration, and the material logic of the kitchen. This suggests that effective mediation requires interaction resources that operate as part of the kitchen’s spatial and material ecology rather than as external interfaces layered onto it.

Core claim. Phygital artifacts can operate as interaction anchors that support presence and coordination while remaining legible within everyday domestic practice, moving beyond

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context-agnostic “smart” devices toward situated interaction resources.

What counts as “phygital” in this thesis. Phygital artifacts refer to: (1) digital entities designed with object-like, tangible qualities that support embodied interaction and respond credibly to physical-world constraints; and (2) physical objects augmented with sensing, computation, and connectivity such that they operate as interactive digital media. Conceptualizing these artifacts as cross-realm interaction resources clarifies how they can reduce breakdowns in visibility, coordination, and control while supporting more natural, context-aware interaction.

Link to identified frictions. This direction directly addresses Frictions 3 and 5, and it also bears on Friction 6. Papers II and IV (4.2-4.4) investigate this direction through speculative probes. Across these studies, participants evaluated artifacts in architectural terms (where they belong, what they disrupt, and what they reveal) suggesting that interaction quality depends not only on interface mechanics but also on spatial embedding. In this respect, phygital artifacts are positioned to address adaptability (Friction 5), spatial awareness deficits (Friction 3), and control constraints (Friction 6) by distributing mediation into situated objects rather than concentrating it in screens or dashboards.

Positioning in the literature. This framing aligns with Zhang’s argument that spatial computing should be approached through an architectural lens [68]. Phygital design supports ambient continuity by fitting form factors to domestic objects and mapping cues onto architectural elements rather than centralizing interaction in display surfaces. It also provides a practical path toward making system inferences legible and easy to act upon through well-placed, contextually meaningful cues that can support trust and coordination in everyday use.

Suggestion 4: Tunable Presence for Privacy and Boundary Control Finally, the studies suggest that mediated presence must remain tunable rather than absolute. Tunable refers to continuously adjustable rather than fixed or binary. Domestic kitchens are socially charged environments where people coordinate care work, share routines, and manage household boundaries. In such settings, the same interaction that supports *togetherness* can also create unwanted exposure.

Core claim. XR mediation should treat presence as a user-tunable property, so that inhabitants can regulate privacy and exposure as interaction moves across moments, zones, and social situations. The studies indicate that this regulation runs along three axes: space (control over where the connection is positioned among the kitchen’s zones rather than bound to a single

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surface), time (control over whether an exchange is live or time-shifted), and identity (control over self-presentation and what is disclosed about one’s appearance, activity, and recording status).

Link to identified frictions. This implication responds directly to Friction 6. It recognizes that hands-busy interaction constraints intersect with privacy and identity concerns: increasing awareness without boundary control can become intrusive, and increasing automation without user control can feel destabilizing. A tunable model of presence supports mediated coordination while preserving domestic agency, allowing systems to selectively reveal cues, modulate interruption, and protect household rhythms.

5.1.3 Synthesis: XR Mediation as Situated, Spatial, and Bounded Support for Domestic Interaction

Taken together, the findings addressing RQ1 position mediated kitchen interaction as a fundamentally spatial and rhythmic coordination problem rather than a narrow communication deficit. Across the studies, breakdowns in visibility, audibility, shared spatial awareness, and multi-sensory experience reveal that screen-centered paradigms misalign with how kitchen life actually unfolds: through movement across zones, hands-busy work, fluctuating noise, interruption, and care. In response, this thesis argues that XR’s value is not in “adding immersion”, but in restoring and reorganizing *shared reference*, which makes action context, attention, and coordination cues perceptible in ways that remain legible under domestic constraints. The suggested directions (spatial flow and interaction, multi-modal and hybrid communication, and phygital artifacts) collectively propose how mediated presence can be made more continuous across zones and moments without demanding continuous visual attention or uninterrupted conversation. At the same time, the studies clarify that stronger mediation introduces its own risks: increasing awareness can become intrusive, and richer presence can destabilize household boundaries unless presence remains tunable and controllable. In this sense, RQ1 is answered not by a single solution, but by mapping a design space in which XR can meaningfully support collaboration, safety, and togetherness while making explicit the boundary conditions (privacy, identity, sensory limits, and spatial heterogeneity) that shape what forms of mediated connection remain viable in everyday kitchen life.

5.2 Addressing RQ2: XR as a Design Instrument for Kitchen Experiences

The second research question (RQ2) focuses on methodological practices for employing XR technologies within design work. It asks: *How can designers meaningfully employ XR as a tool for envisioning, prototyping, and evaluating domestic experiences in kitchen contexts?* To address RQ2, I examine design practice through the development and use of *KitchXR* as an experiential design and evaluation tool, alongside studies conducted with designers engaging with it. These investigations surface recurrent designer constraints and unmet needs in prevailing design workflows, particularly in relation to understanding situated behavior, reasoning about spatial and embodied constraints, and iterating on interaction ideas within realistic domestic contexts. Building on these findings, the same tool, which iteratively refined through successive improvements, is later reused to evaluate speculative design artifacts within an immersive, kitchen-like setting. Taken together, these perspectives position XR as both a medium for design exploration (envisioning and prototyping) and a substrate for situated evaluation, enabling an evidence based account of how XR can extend and strengthen iterative design workflows for domestic kitchen experiences.

RQ2 shifts the focus of the thesis from *interaction outcomes* to *design practice*. Rather than treating XR solely as a medium for end-user interaction, this research positions XR as a designer-facing methodological and epistemic resource. XR becomes a means through which designers can reason about space, enact domestic scenarios, and explore interaction constraints that are difficult to access through conventional representations, mock-ups, or screen-based prototypes. In this framing, XR supports designers not only in visualizing future systems, but in thinking with spatial configurations, bodily movement, and domestic rhythms as active components of the design process.

Within the programmatic logic of this thesis, XR enables iterative cycles of exploration in which spatial layout, movement, attention, and social coordination can be examined *in situ* without requiring fully deployed smart-home infrastructures. By supporting immersive and speculative prototyping, XR reshapes how designers investigate domestic interaction, allowing breakdowns, trade-offs, and design possibilities to be surfaced early and revisited across iterations. The following sections discuss how designers engage with XR as a methodological tool for reasoning about kitchen interaction (RQ2a), and what kinds of unique design insights emerge

through immersive and speculative XR prototyping (RQ2b).

5.2.1 (RQ2a) Understanding the Kitchen Through XR as a Methodological Tool

The first sub-question of RQ2 asks: *How do designers use XR as a methodological tool to reason about spatial and interactional dynamics in kitchens?* This question foregrounds design practice itself, examining how XR enters designers' existing workflows and what constraints emerge when designers attempt to understand, prototype, and evaluate interaction in a spatially complex, socially embedded domestic environment.

Across studies with designers using KitchXR, this thesis surfaced recurring limitations in prevailing design and evaluation practices, particularly in relation to prototyping cost, access to realistic contexts, and the difficulty of reasoning about embodied, multi-actor interaction in kitchens. Two of these limitations are articulated below as designer constraints. The remainder of this subsection then answers RQ2a in a reflective register: rather than prescribing how designers should use XR, I reflect on what I observed when XR entered design work through KitchXR, and on how my own evaluative stance shifted in the process.

Designer Constraint 1: Resource Demands of Prototyping and User Testing Developing sufficiently functional prototypes and conducting user studies often require substantial time, budget, and coordination, which slows iteration and limits how broadly concepts can be explored. These burdens intensify when studies involve geographically distributed user groups, where physical prototypes introduce additional logistics, constraints, and costs for deployment, travel, or facilitation.

Why it matters. Prototype development and user testing are central to producing credible design knowledge, yet they are frequently constrained by time, budget, and coordination overhead. When prototypes must be sufficiently functional to support realistic interaction, iteration slows and the breadth of concepts that can be explored narrows. These constraints become even more pronounced when evaluating with geographically distributed user groups, where physical prototypes introduce additional logistics, deployment challenges, and financial costs. As a result, designers often rely on shorter sessions, fewer design cycles, or simplified scenarios, which limits geographic reach and reduces the depth and ecological validity of the evidence available to inform design decisions. Over time, these limitations increase the likelihood that designs

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progress with unresolved usability issues, leading to less usable outcomes and, in practice, a higher risk of unsuccessful products.

Designer Constraint 2: Barriers to In-Depth User Studies of Prototypes In many design contexts, user studies struggle to examine prototypes in depth, particularly when the interaction depends on situated routines, spatial constraints, and multi-actor coordination, as in domestic kitchens. Practical constraints (limited access to real contexts, short study durations, low prototype fidelity, and safety or installation overhead) often lead to evaluations that capture surface-level impressions rather than sustained use and emergent breakdowns. This limitation motivates XR-based workflows as a way to simulate context, increase evaluative control, and iteratively test interactions with greater methodological depth.

Why it matters. Envisioning usability and user needs in dynamic, high-interruption spaces such as kitchens is essential for effective design. Everyday kitchen activity is shaped by disrupted awareness, concurrent tasks, shifting attention across distinct zones, and interactions with devices, other people, and digital content. Designers therefore need ways to examine design decisions and user interactions holistically to develop solutions that remain robust under the kitchen’s inherent dynamism. As domestic technologies increasingly move toward spatial and immersive computing, these constraints become more salient, increasing the risk that designs perform well in controlled settings while failing in everyday use.

Reflection 1: From Efficiency Metrics to Household Rhythms I entered this research with the evaluative apparatus that dominates interaction design close at hand: task completion, error, efficiency. What participants in *Kitchen Talks* described as mattering, and what the designer sessions kept returning to, were not throughput problems. They were questions of awareness, togetherness, and coordination unfolding amid movement, interruption, and household rhythms. Holding on to efficiency as the criterion would have meant measuring the kitchen against the wrong thing.

The reflective outcome is the shift in evaluative stance that runs through the later studies and is consolidated as a contribution in Section 5.3 and Chapter 6: treating the kitchen as a domestic communication and coordination hub rather than solely a task-oriented workspace. Under this stance, the frictions identified across Papers I and II (Sections 4.1 and 4.2) stop reading as usability defects to be eliminated and start reading as programmatic tensions. They both characterize breakdowns, in visibility, audibility, spatial awareness, and boundary control,

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and provide concrete entry points for redesign. For design practice, the implication is that evaluation criteria are themselves a design decision. What a study can find out about a kitchen technology depends on whether the kitchen is framed as a workspace or as a hub.

Reflected onto XR as a tool for design studies, this framing determined what I asked the simulated environment to reproduce: movement across zones, interruption, co-presence, and divided attention, the conditions under which awareness and coordination become observable at all. It also set what I treated as missing when the environment fell short, a thread the next reflection picks up. The stance itself did not originate in XR. It was grounded in the inhabitant fieldwork of Kitchen Talks. The empirical understanding of the kitchen disciplined the configuration of the instrument, rather than the instrument defining what the kitchen was taken to be.

This stance is not idiosyncratic. Prior work already documents that kitchen activity extends beyond cooking and that home design shapes the social interaction it hosts [29], and that the kitchen concentrates domestic coordination work and embeds cultural practice in routine activity [16, 61, 147]. The vocabulary I use to name those modes, ambient co-presence, synchronous exchange, and asynchronous communication, is the thesis’s own rather than these authors’. The same rhythm-sensitive stance plausibly transfers to other active workspaces [55] and to hybrid spatial collaboration settings where coordination likewise depends on shifting attention and spatial reference. I note this transferability as an expectation rather than a demonstrated result.

Reflection 2: KitchXR in Designers’ Workflows The second reflection concerns what I observed when designers worked inside *KitchXR*. Frictions 7 and 8 name the constraints these designers operate under. Functional prototypes are costly, realistic contexts are hard to access, and study formats rarely sustain the depth that situated, multi-actor interaction demands. *KitchXR* was, among other things, a response to those constraints. Watching designers work inside it clarified where a simulated kitchen earns its place in the design process, and where it reaches its limit.

It’s clearest contribution was architectural anchoring. Inside the simulated kitchen, designers placed, scaled, and tested interaction concepts against layout, sightlines, adjacency, reach, and movement rather than against abstracted screen-based flows. This grounding let them reason early about whether cues would be noticed, whether interactions would disrupt task flow,

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and whether placement decisions would support or undermine coordination during hands-busy activity. The designers themselves emphasized this in the interviews: a simulated kitchen of this kind would make it possible to situate interaction concepts within holistic usage scenarios rather than evaluating them as isolated features detached from kitchen activity. The same argument plausibly extends to MR variants of such an environment, a direction I return to in Reflection 4. In this respect, the environment responds to a gap identified in Section 2.3.4, where point solutions exist but holistic toolchains spanning needfinding, ideation, prototyping, and insight collection remain underdeveloped.

This mode of reasoning also ties the approach back to the ubiquitous and invisible computing paradigms reviewed in Section 2.1.2, which judge a domestic system by how well it accords with context, position, and flow rather than by the attention it commands [30, 40]. Anchored prototyping makes that criterion examinable: whether a concept recedes into the room’s layout and the user’s movement can be tested inside the simulation, rather than asserted as a design intention.

The limit lay not in what *KitchXR* could simulate but in what it could measure. Across the studies, evaluation relied on the Continuous Fuzzy Kano Model (CFKM) and on observation. Both capture participants’ assessments from outside the medium and largely after the experience. Yet the environment itself continuously produces behavioral data that these instruments do not reach. Extending the testbed to capture such data, and to render it analyzable within the design process, would considerably strengthen XR’s role in design studies: gaze heat maps over the kitchen layout, spatial flow graphs of participant movement across work zones, and logs of interactions with individual design elements. Eye tracking is available on current headsets; richer physiological signals would require additional instrumentation. Instrumentation of this kind would allow designers to triangulate what participants report with what they attend to and what they do, a distinction that matters precisely in the hands-busy, attention-divided conditions this thesis has argued are central to kitchen interaction. This possibility was also raised independently by the designers in the *KitchXR* study (Section 4.3). I record it as a development direction rather than a claim, since the studies reported here did not implement such capture. Section 6.5 returns to this as a limitation.

5.2.2 (RQ2b) Methodological Reflections on XR as a Design Instrument

While RQ2a focuses on XR as a methodological tool within design workflows, RQ2b examines the kinds of design insights that become possible specifically through immersive and speculative XR prototyping. This question addresses what designers are able to see, anticipate, and reason about when interaction concepts are explored within embodied, spatially situated scenarios rather than through abstract representations or screen-based prototypes.

Reflection 3: Speculative Prototypes as Instruments for Productive Failure When I introduced speculative artifacts into *KitchXR* (Papers III and IV), I expected the sessions to function primarily as concept validation. Designers would enact scenarios and confirm whether interaction concepts worked. This expectation reflected a validation-oriented habit carried over from conventional usability practice. What the sessions produced instead was a catalog of conditions under which concepts stopped working. Ambient Cooking drew its most substantive commentary not on whether the modality was acceptable but on what content should appear and when, once attention was already divided across zones (Section 4.4). Portal’s form factor, which had seemed a matter of visual style, became a question of where in the kitchen a life-size opening could sit without occupying a work surface, and participants moved between circular, compact, square, and rectangular variants accordingly. MagMemo attracted markedly fewer flexibility suggestions than the other two artifacts, a silence that turned out to be diagnostic: the refrigerator door was already so strongly specified as a message surface that participants had little to reconfigure. The prototypes did not tell me whether a concept worked. They told me under what conditions it stopped working.

In one sense, this outcome should not be surprising. Exposing assumptions and provoking breakdowns is speculative design’s default function, and prior work reports that evaluating immersive prototypes can generate high-quality design insight and accelerate decision-making in product design contexts [113,114]. My aim here is to carry this repositioning beyond my own process and state it as transferable knowledge. I entered the studies treating XR prototypes as evaluation instruments and left treating them as instruments for productive failure in the sense of programmatic design research [23,24]: devices that reveal the boundary conditions of a design move before implementation commitments are made. In a spatially complex, high-interruption setting such as the kitchen, this reframing mattered because the most consequential design risks were embodied and temporal, and they surfaced only when concepts were enacted

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under movement, noise, and interruption. Stated as intermediate-level knowledge [146], the implication for researchers adopting similar workflows is to design speculative XR sessions around eliciting failure rather than confirming function, and to treat the resulting breakdowns as findings rather than as flaws in the prototype.

A further, more tentative observation came from facilitation. Participants with design or creative backgrounds tended to treat the speculative artifacts as provocations to push on, while the few without such backgrounds more often read them as finished products to be judged, and the speculative framing took longer to land. I note this as an impression rather than a finding, with one practical implication: in mixed-expertise sessions, the speculative stance may need to be scaffolded rather than assumed.

Reflection 4: KitchXR as an Intermediate Methodological Testbed *KitchXR* did not begin as a methodological position. It began as a response to constraint. In the early years of this PhD, developing interactions and experiences for MR headsets was technically demanding, and the immaturity of the available development ecosystem threatened to consume time and resources that the research program could not spare. I therefore envisioned *KitchXR* as a test environment for MR design ideas, one that would fit within the practical limits of a doctoral project. The premise was that it could serve as an intermediate methodological space between abstract prototyping and deployment in real homes, an approach that shares the Living Lab commitment to studying technology in context and sustaining design and evaluation cycles (Section 2.3.3) while substituting simulation for instrumentation. It is this position that the thesis names an intermediate living testbed. Rather than restating the premise as a finding, this reflection examines where it held, where it strained, and what the difference taught me.

The premise held on iteration, cost, and control. *KitchXR* allowed me to test and revise designs rapidly across studies, and its reuse made it a connective tissue of the research program. Working in a simulated environment detached from any physical kitchen also made study scheduling and participant recruitment considerably more manageable. The environment proved readily extensible as well. When the evaluation study required a second, adjoining kitchen scene so that a remote interlocutor could occupy the far side of Portal, and playback of volumetric message assets captured outside the simulation for MagMemo, the environment could be adapted to those scenarios without rebuilding the setup.

The premise strained on fidelity. The same detachment that made *KitchXR* flexible con-

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strained its ability to reproduce the accumulated domestic texture of a lived-in kitchen, and technical limits on haptic feedback and fine manipulation remained strong drawbacks throughout. Some of this strain can be mitigated, for example through extended onboarding, longer exposure time, and richer scenario embedding, but it cannot be removed. It marks the boundary of what a simulated testbed can access, and Section 6.5 treats it as such.

This trade-off is also why I named the environment an XR kitchen rather than a VR kitchen. The studies reported here used fully virtual configurations, and Paper III describes its medium accordingly. The name marks an intention rather than a description: to treat the degree of reality as a design variable, so that future configurations of the testbed can shift along the reality-virtuality continuum [21] depending on the design question, the study scenario, and the limitations at hand, rather than being fixed to a fully virtual setup.

Fidelity was not the only strain. The ecosystem immaturity that first motivated *KitchXR* did not resolve over the course of this PhD. It changed shape. More than a decade after modern consumer headsets arrived, XR development still rests on a fast-moving base of devices, runtimes, and toolchains. Hardware generations, SDK revisions, and platform requirements turn over faster than a single research program can absorb, and each turnover carried a maintenance cost for the testbed. The lesson I draw is that an environment like *KitchXR* is not a finished artifact but a maintained instrument. Its value depends on continued upkeep, and researchers or designers adopting a similar approach should budget for that upkeep as part of the method rather than treat it as incidental overhead. The same instability tempers any broader claim about XR as a design tool. What the field currently offers is a powerful but unsettled instrument, not yet the stable, long-lived infrastructure that design practice can take for granted.

Designer constraints place this account in a wider frame. The limitations that shaped *KitchXR* at the start of this PhD about time, budget, and access to participants for broad and in-depth user studies. These limitations are not peculiar to doctoral research. The two design constraints record the same conditions in design practice, where prototyping and user testing compete for time, budget, and coordination, and where studies of situated domestic interaction rarely reach sustained use. If anything, these pressures bear harder on industrial projects, since commercial timelines leave even less room for sustained user engagement than a doctoral program does. The first design constraint notes, they raise the risk that designs progress with unresolved issues and fail in everyday use. Reading my own early limitations through this lens

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reframed them. What I first treated as a limitation of one doctoral project turned out to be a structural condition of designing for domestic XR. *KitchXR* stands as one deliberate response to that condition rather than a workaround particular to this thesis.

5.2.3 Synthesis: XR as a Design Instrument

Together, the findings addressing RQ2 position XR not merely as a medium for representing future domestic systems, but as an active design instrument that reshapes how designers reason about space, interaction, and everyday practice. Two design constraints frame the problem: prototyping and testing are resource-bound, and study formats rarely sustain the depth that situated interaction demands. Four reflections trace what followed. Evaluation criteria proved to be a design decision rather than a neutral instrument. Simulation contributed architectural anchoring while leaving a measurement gap unaddressed. Speculative prototypes proved more useful as instruments for eliciting failure than as instruments for confirming function. And the testbed itself held on iteration and control while straining on fidelity and on the stability of its own toolchain. These strains mark the limits of the approach without invalidating it, and they recast it as a maintained instrument rather than a finished artifact. The design constraints that open the section thus return at its close as structural conditions of the practice itself, and *KitchXR* stands as one deliberate response to them. What XR extends is not the designer's confidence in a concept but the designer's access to the conditions under which that concept is answerable: movement, attention, interruption, and the rhythms of a room that was never only a workspace.

5.3 Addressing RQ0: What the Studies Reveal Together

RQ0 asks how XR technologies can support mediated spatial interaction in domestic kitchens, addressing both everyday communication challenges and the design of meaningful domestic experiences. Its answer lies not in either strand alone but in how the experiential and methodological findings constitute one another. Read together, the studies show that XR's contribution is not to add immersion or improve video, but to redistribute presence, awareness, and coordination across the spatial and temporal rhythms of kitchen life. In Verbeek's terms, XR operates here as a *mediator* that co-shapes how cohabitants perceive and engage one another rather than as a neutral channel between them [148, 149]; what it redistributes is precisely the substrate

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of everyday coordination that awareness and presence research identify, the peripheral understanding of others' activity that contextualizes one's own [150, 151], and the felt salience of a remote other [152, 153].

This redistribution becomes intelligible through what the studies reveal together. First, mediated kitchen interaction does not break down at random but along recurring, structured points of resistance; visibility, audibility, shared spatial awareness, multi-sensory experience, spatial adaptability, and control over privacy and identity. These are not isolated usability costs to be engineered away, but persistent tensions between the kitchen's dynamic, hands-busy character and the assumptions embedded in screen-based systems; taken together they form a structured phenomenon rather than a list of failures. Second, the support that inhabitants will accept is one they govern: control over identity and privacy registered across the studies as near-essential rather than merely desirable, and asynchronous, time-shifted communication was readily accepted for the kitchen's social functions. Effective mediation therefore cannot rest on fixed system defaults but must remain adjustable by the inhabitant.

Third, the studies reveal a specific character to the kitchen that conventional smart-kitchen framings overlook. Across the inhabitant-grounded fieldwork, the kitchen consistently emerged as a socially central, shared space: a site of collaboration, learning, and interpersonal connection rather than a private cooking facility. The speculative studies then showed that remote, hybrid, and asynchronous forms of connection are readily taken up within it. At the same time, it is spatially heterogeneous: not a uniform room but a cluster of specialized workzones between which inhabitants continually move, so that mediated interaction must be anchored to this distributed structure rather than to a single screen or surface. Its social centrality and everyday ubiquity together make the kitchen a plausible anchor for connection that reaches across, and beyond, the home.

Taken together, these findings answer RQ0 in substantive terms. *XR supports mediated spatial interaction in the domestic kitchen by redistributing presence, awareness, and coordination across the kitchen's spatial and temporal rhythms, acting as a mediator rather than a neutral channel between cohabitants.* The consequence is twofold: **(1)** it reframes social interaction as a core function of the kitchen alongside cooking, to be designed for in its own right rather than treated as a byproduct of task support; and **(2)** it positions the domestic kitchen as a candidate hub for spatial, hybrid, and remote connection within the home. This answer is also methodologically constituted. It became reachable because XR served throughout as both the

medium under design and the instrument of designing: the frictions were grounded in inhabitant fieldwork, elaborated through speculative prototypes, and examined under the controlled conditions of an intermediate living testbed. The answer to RQ2 is, in this sense, internal to the answer to RQ0. It is this integrated reframing, rather than any single study, that the concluding chapter abstracts into the thesis’s conceptual contributions.

5.4 Summary

In summary, the thesis answers RQ1 by identifying six recurring interactional frictions in mediated kitchen interaction and by articulating four design suggestions that show how XR can reintroduce shared spatial reference, awareness cues, and hybrid communication across ambient, synchronous, and asynchronous modes, while also exposing implications and constraints related to safety, attention, and boundary control. It answers RQ2 by identifying two designer constraints in how designers prototype and study domestic interaction, and by positioning XR not only as an interaction medium but also as a designer-facing instrument: an intermediate living testbed that supports speculative prototyping, situated evaluation, and iterative decision-making under domestic and practical constraints. Four reflections trace where this instrumental stance held and where it strained.

Across RQ1 and RQ2, several themes operate as boundary conditions that shape the viability of XR-mediated domestic communication and the feasibility of designer-facing XR workflows. The thesis’s framing of the kitchen as a dynamic social and cultural hub highlights that domestic coordination unfolds across routines, zones, and attention shifts, which affects what becomes meaningful, desirable, or problematic in mediated connection. In parallel, the recurring concerns around privacy, identity, and boundary management underline that strengthening presence and awareness can also intensify intrusion unless disclosure is tunable. Spatial heterogeneity and multi-sensory limits constrain how broadly interaction concepts can generalize and what aspects of co-present experience can realistically be mediated, simulated, or evaluated. Finally, the conditions of design practice constrain the workflow side directly: prototyping and user studies remain limited by time, budget, and access, as Frictions 7 and 8 record, and the toolchain on which XR workflows rest is still fast-moving rather than settled, so that a testbed built on it remains a maintained instrument rather than a finished artifact.

Chapter 6

Conclusion

This thesis sits at the intersection of Interaction Design, HCI, and Extended Reality (XR), and focuses on mediated spatial interaction in domestic kitchens. It examines the kitchen as a materially dense and socially meaningful environment where communication, coordination, and shared presence unfold during everyday cooking as well as during hybrid and remote engagements. Conceptually, the work investigates how presence, awareness, and togetherness can be mediated through ambient, synchronous, and asynchronous XR-supported interactions, engaging smart home and spatial computing discourses through the concrete lens of kitchen practices. Methodologically, the thesis follows a design-oriented, programmatic approach that advances situated, intermediate-level insights through a sequence of interrelated artifacts and studies, treating XR primarily as an exploratory and epistemic instrument rather than an end in itself.

Three terms recur throughout this chapter. In this thesis, frictions are the recurring interactional tensions observed across the studies that structure the design space. Reflections are my own observations and lessons as a researcher, drawn from the findings across the studies over the course of the PhD. Designer constraints are practice-level limitations on the design process that obstruct designers and researchers from conducting broader and deeper studies.

The motivation behind this thesis is to bridge task-oriented HCI studies of cooking with the social, relational, and cultural dimensions of the domestic kitchen. To do so, the work pursues three intertwined aims: first, identifying and characterizing the distinctive interaction challenges that emerge in kitchen environments; second, conceptualizing XR-enabled interventions that can mediate presence and togetherness, enable richer communication, and support multiple

modes of interaction; and third, investigating XR as a methodological and creative resource for designers envisioning, prototyping, and evaluating domestic experiences. The domestic kitchen matters as a setting for mediated spatial interaction because it is not merely a task environment but a domestic coordination hub where social, practical, and cultural activities overlap. It has played a central role in shaping modern domestic life, maintaining its distinctive personal and cultural significance, and it continues to host everyday forms of collaboration, learning, and care that extend beyond cooking itself.

At the same time, kitchens are boundary-intensive spaces. Everyday routines, household rhythms, and moments of care become visible and audible, making privacy, identity, and boundary management central design concerns, particularly during hybrid or remote engagements. These characteristics make the kitchen a challenging yet generative context for exploring the capabilities, constraints, and boundary conditions of XR technologies. Accordingly, this thesis conceptualizes the domestic kitchen as a hybrid environment that simultaneously supports productive doing and meaningful relating. To address these aims, the thesis is structured around the following research questions:

1. **RQ0.** How can Extended Reality (XR) technologies support mediated spatial interaction in domestic kitchens, addressing both everyday communication challenges and the design of meaningful domestic experiences?
2. **RQ1.** In what ways can XR technologies address the interaction challenges that emerge in domestic kitchen environments, particularly during hybrid or remote engagements?
3. **RQ2.** How can designers meaningfully employ XR as a tool for envisioning, prototyping, and evaluating domestic experiences in kitchen contexts?

6.1 The Shift to Spatial and Phygital Ecologies

Regarding RQ1, this thesis concludes that the prevailing “Smart Kitchen” paradigm must evolve beyond fixed screens toward a spatially aware, phygital ecology. The investigation of interaction breakdowns shows that domestic frictions, such as limited visibility, audio interference, and hands-busy constraints that cannot be resolved by simply improving video quality or adding more screens. Instead, the findings point to three fundamental shifts in how domestic interaction should be designed.

From Stationary Screens to Spatial Flow The research establishes that effective XR interventions must align with the *spatial flow* of the kitchen. Interactions should remain legible and accessible as users transition between zones (e.g., from preparation to cooking), rather than being anchored to a single stationary interface. In this way, technology supports the fluidity of kitchen work instead of constraining it.

From Single-Channel to Multi-Modal, Hybrid Communication Kitchen communication cannot rest on a single channel of continuous conversation or uninterrupted visual attention, which the kitchen’s fluctuating noise, interruptions, and hands-busy conditions routinely defeat. Instead, mediation should combine modalities, such as spatial annotations, lightweight audio, and embodied cues, and span multiple temporalities, from live exchange to awareness that persists without sustained attention. Distributing communicative effort across complementary cues and across synchronous, asynchronous, and ambient registers allows coordination to continue even when any one channel is unavailable.

From Interfaces to Phygital Artifacts Successful mediation relies on *phygital artifacts*: digital entities with tangible qualities or augmented physical objects embedded in the kitchen’s material ecology. By distributing interaction across space and objects, designers can support coordination without disrupting physical workflows, moving away from generic devices that compete for attention.

From Binary to Tunable Presence The thesis argues that social presence in domestic settings cannot be binary (online/offline). Drawing on privacy as a dynamic boundary-regulation process [154, 155], it must instead function as a tunable property that users can regulate along three axes: (1) space (where a connection is placed in the kitchen), (2) time (whether an exchange is synchronous or asynchronous), and (3) identity (how much of one’s appearance, activity, and recording status is disclosed)..

Designing presence as tunable thus means giving households continuous control along these axes, managing the tension between coordination and the preservation of domestic boundaries.

Answer to RQ1 Overall, the thesis argues that XR technologies can support domestic interaction only when they respect the kitchen as a rhythmic, socially charged environment.

The goal of design is not to eliminate friction entirely, but to work with it in creating hybrid interactions that support care, safety, and togetherness within everyday domestic life.

6.2 XR as a “Living Testbed” for Situated Design

Regarding RQ2, this thesis establishes XR as a critical methodological instrument for design research. In this respect, the thesis makes a foundational contribution by reframing the kitchen itself from a sanitary food preparation facility to a dynamic, rhythmic social and cultural hub.

From Efficiency to Social Rhythms By foregrounding the kitchen’s social missions, this work shifts evaluation away from narrow efficiency metrics toward understanding how awareness, togetherness, and coordination unfold amid household rhythms. This reframing reveals that conventional user studies often fall short, as they isolate usability from the social dynamics of domestic life.

The Necessity of a “Living Testbed” To capture these dynamics, the thesis argues that XR must function as an *intermediate living testbed*. This approach allows designers to simulate architectural and contextual realities (sightlines, reach, and adjacency) without the logistical burden of deploying physical prototypes across multiple homes.

Toward Behavioral and Bio-Sensory Insight Immersive XR prototyping also opens a path toward capturing behavioral and bio-sensory data, such as gaze and movement traces, that could provide empirical access to the often invisible costs of coordination and attention. The studies reported here did not implement such capture; Chapter 5 records it as a development direction, and Section 6.5 treats its absence as a limitation. If realized, such instrumentation would complement qualitative methods by revealing interactional dynamics that are difficult to articulate retrospectively.

Answer to RQ2 Ultimately, this thesis argues that XR enables designers to meaningfully engage with domestic complexity by enacting and evaluating interaction in situ. By anchoring design exploration in spatial reality, XR supports a shift from treating kitchens as sites of labor toward recognizing them as sites of care, culture, and coordination.

6.3 XR as Design Material and Empirical Tool

RQ0 questions, at the most foundational level, what XR is to a design researcher working on domestic spatial interaction; Is it a communication channel to be optimized, or something with a more constitutive role in the work itself. Across the four studies, XR is examined on these suggestions. On one side, it is used as a design material: virtual assets, spatial layout, and embodied presence were not fixed givens but malleable elements that designers composed, configured, and aligned with the kitchen’s physical dynamics, which is consistent with the thesis’s premise that spatial technology and virtual assets must stay in touch with real-world physics, layout, and interaction.

On the other side, XR is proposed as an empirical tool: by allowing complicated interactions to be enacted and evaluated in situ rather than merely described, it functioned as a living testbed through which situated constraints, preferences, and breakdowns could be surfaced under realistic conditions. Treating XR in these two roles, is what allowed the thesis to move between design exploration and empirical evaluation within a single programmatic arc, and it underpins both the claim about supporting domestic interaction (RQ1) and the claim about engaging domestic complexity in situ (RQ2) developed above.

Answer to RQ0 Substantively, XR supports mediated spatial interaction in the domestic kitchen by redistributing presence, awareness, and coordination across the kitchen’s spatial and temporal rhythms, acting as a mediator rather than a neutral channel (Section 5.3). This thesis argues that XR delivers this support through two roles. As a design material, it is the flexible medium from which designers compose hybrid domestic interaction. As an empirical tool, it is the instrument through which that interaction is enacted and evaluated in situ.

6.4 Contributions

This thesis contributes to Interaction Design and HCI by articulating how XR can mediate everyday kitchen interaction and by examining how XR can function as a design instrument for domestic experiences. Its contributions rest on four pillars: the interactional frictions and user needs surfaced through Kitchen Talks (Paper I), the insights generated in the subsequent studies with designers, the speculative artifacts built to address those findings, and the methodological infrastructure consolidated in KitchXR. Rather than restating these as a taxonomy, this section

traces how they emerged from the research journey, because the conditions under which they emerged are part of what they claim. The friction lens that structures the empirical findings and the design construct of tunable presence are presented in full in Chapter 5 and are therefore not repeated here; this section concentrates on the two conceptual reframings, the design knowledge, the methodological contribution, and the artifacts.

6.4.1 Conditions that Shaped the Research

My doctoral research began a few months before the Covid-19 pandemic, and the restrictions that followed left a visible imprint on its trajectory. For a long time I read that imprint as a purely negative one. In retrospect, it was directive rather than merely disruptive: it relocated domestic sociality into mediated channels at exactly the moment I was beginning to study the kitchen, and it thereby made the phenomenon at the center of this thesis observable at scale. Not every study conducted in that period is reported here. An early large-scale survey, completed by more than seven hundred participants in the first weeks of the pandemic, examined kitchen use for cooking, recipe sharing, and social media practices. Its unusually wide participation likely reflected the circumstances of lockdown as much as the instrument itself, and its descriptive findings, while informative, did not yield patterns robust enough to carry analytical weight on their own. Its role in the programme was formative rather than evidential: it indicated that questionnaire-based accounts of kitchen practice would not reach the situated, embodied dynamics I was after, and it redirected the research toward participant observation.

That redirection produced the first anchor of the thesis. The participant observation within Kitchen Talks was itself made possible by the pandemic: online cooking workshops had emerged as one of the ways people countered loneliness and boredom under prolonged restrictions, and they offered a naturally occurring site of mediated kitchen sociality. Most of the interactional frictions analyzed in Chapter 5, breakdowns in visibility, audibility, device control, and the felt absence of togetherness, were first recorded in the observation notes from these workshops. Individually, such breakdowns are well documented in prior work on cooking and connecting at a distance, on the embodied coordination of family cooking, and in the CSCW literature on awareness and coordination in shared activity [17,18,78,150]. The contribution of this thesis is not their discovery but their consolidation into the friction lens developed in Chapter 5, which treats them as persistent, constitutive features of mediated kitchen interaction to be designed around rather than defects to be engineered away.

6.4.2 Framing Kitchen and Dynamics

The co-design sessions that followed the observations yielded the first of the two conceptual reframings this section advances. The sessions were structured toward a final group-ideation phase, with earlier activities intended as warm-ups; one of those warm-ups proved more consequential than the ideation itself. Participants drew the two-dimensional layout of the kitchen they used most, using a set of tagged, variably sized boxes I had prepared, in a deliberately imprecise, lego-like manner, and then marked where they placed their phone, where they worked most, where they took calls, and where tablets or laptops rested. I did not expect these sketches to yield a generalizable spatial pattern, and they did not; the insight came from the attempt to analyze them. Reading the layouts as wholes obscured more than it revealed, whereas dividing them into clusters of work-related activity made the participants' accounts legible. Beyond its methodological value, this exercise reshaped my conceptual understanding of everyday kitchen dynamics. The kitchen is not a homogeneous area of uniform layout but a cluster of spatially distributed workzones, each specialized for particular processes, between which the inhabitant continually moves. This structure is partly explained by the distribution of appliances and tools, which bind processes to locations. Building on Ju's account of active workspaces [55], I position the kitchen as an instance of a broader class of active working spaces and conjecture that this heterogeneous, workzone-cluster structure characterizes that wider class, while noting that the present evidence is drawn from kitchens alone.

This reframing carries direct implications for XR design in the kitchen. Although this thesis addresses the domestic kitchen as a whole, scoping design work to one or a few workzones would allow designers to engage the dynamics of those zones in depth and to propose more precise interventions, without foreclosing holistic tools for the kitchen as such. It also suggests that kitchen technology should adapt to the workzone and task content the user currently occupies or moves toward. This position is adjacent to, but distinct from, the ubiquitous and invisible computing paradigms [30, 40]: the tools I envision are not invisible so much as reachable and relevant, available in relation to the user's current work whenever needed, closer in spirit to Rogers's call for engaging rather than calm ubicomp experiences [33]. Prior augmented kitchen research has pursued safety and support through continuous analysis of the user's task, for instance through sensing and recognition of ongoing activity [55, 106, 156, 157]. I argue that, for many adaptation and safety purposes, such fine-grained task recognition may not be necessary: a workzone-level model, pairing each cluster with its characteristic tasks and adapting the

experience accordingly, may suffice, at considerably lower sensing and processing cost. Future spatial tools for the kitchen could follow this reframing to balance the lived kitchen experience with task augmentation.

6.4.3 Domestic Kitchens Beyond Cooking

The second conceptual reframing concerns the kitchen’s social function. A substantial strand of prior work resists reducing the kitchen to a site of functional cooking labor and foregrounds its social and cultural significance [15,19]. This thesis advances that view into a design position: the kitchen is a strong domestic candidate for a spatial connection hub, a site for telepresence, telecollaboration, and telecoordination, by virtue of its ubiquity, its shared use across inhabitants, its established social role, and its comparatively lower privacy stakes than more private rooms. The consequence is that social interaction becomes a primary design concern for domestic XR rather than a byproduct of task support. This does not displace cooking and other food-related activity; rather, I suggest the kitchen can hold both functions in balance by accommodating dedicated space for communication. Read together with the workzone-cluster reframing, the communicative function can be understood as one further zone within the cluster. I therefore suggest that interaction designers and architects alike might account for both working zones and a communication zone from the outset, rather than treating communication as an overlay added to a space planned solely around tasks.

6.4.4 Paradigms for Spatial Interactions

These reframings were not developed in the abstract; they were exercised through artifacts. Ambient Cooking, MagMemo, and Portal were built as speculative VR artifacts that bind the findings, design insights, and concepts of the preceding studies into forms that designers and experts could ideate with and evaluate. In designing them, I consolidated a concept I refer to as phygital artifacts. The term phygital, a contraction of physical and digital, originates in the marketing literature, where it describes blended retail experiences. In this thesis it designates digital artifacts, presented to users within an XR experience, that remain in constant communication with the real world and are bound by its conditions and rules. The design stance this implies is that such interfaces and artifacts should be designed as if they were real-life products, and the relation is bidirectional: physical objects in the kitchen should likewise communicate with the XR experience and may carry functions that shape it. This position is

closely related to Ishii and Ullmer’s Tangible Bits [41], but it is articulated here from within the kitchen’s specific spatial and social constraints.

6.4.5 Controlling Space, Time and Identity

The ideation sessions around these speculative artifacts also surfaced a finding I had not anticipated: the XR experts repeatedly returned to the question of presence control at the boundary between real and virtual worlds. That discussion is distilled in Chapter 5 as the design construct of tunable presence (Suggestion 4), and I do not restate the construct here; what belongs in this chapter is how it emerged and what it opens up. User-centered approaches to a spatial interaction tool would begin, reasonably, with the dynamics of use: how the inhabitant engages the tool, how they initiate and end a connection, where they stand in relation to it. What such an account risks missing is the inhabitant’s presence as a design object in its own right, their presence within the system and toward the remote other. Presence, on this view, is not a simple toggle in the manner of switching a camera on or off; it is a property to be regulated continuously along the axes of space, time, and identity. Spatially, an inhabitant might be fully present within a communication zone yet have their presence attenuated, reduced rather than removed, as they step out of it, or have audio channels dampened in a working zone where noisy equipment and safety demand undivided attention. In terms of identity, an inhabitant may want embodied presence while withholding who or where they are: an avatar can protect identity, the surroundings can be masked, and the same question extends to the auditory channel, since a voice can disclose identity as readily as a face.

The sessions also raised questions that reach beyond these axes and remain open. The first concerns the locus of control. An unannounced appearance beside someone absorbed in hands-busy work is rarely welcome, which suggests that presence is jointly governed: the sender specifies an upper bound on their own presence, and the receiver may attenuate below that ceiling according to their preferences and circumstances, but not exceed it. This immediately invites further questions, such as whether a receiver should be able to modulate the sender’s presence at all, and how far attenuation can go before presence becomes disappearance. Such questions exceed usability and enter the territory of privacy, safety, and ethics, in the sense in which Palen and Dourish characterize privacy as a continuous boundary negotiation rather than a fixed state [155]. The second concerns the frictions that regulation itself introduces. Masking identity may erode the credibility of the masked party in the eyes of the receiver,

and a counterpart who fades silently may leave the other misjudging whether anyone remains present at all. This points to a requirement of presence awareness: inhabitants should remain aware of how they are currently presented and whether the connection persists, echoing the long-standing CSCW concern with awareness in shared activity [150,151]. Presence regulation is well studied in CSCW; what this thesis adds is the argument that, for domestic spatial mediation, it must be treated holistically as a tunable, multi-parameter property, and that doing so is a load-bearing condition for the spatial connection hub envisioned for the kitchen.

6.4.6 Design Knowledge, Method, and Artifacts

Alongside these conceptual contributions, the thesis contributes design knowledge in the form of intermediate-level knowledge [146]: four design suggestions for XR-mediated kitchen interaction, developed in Chapter 5, addressing spatial flow across zones and task transitions, multi-modal and hybrid communication, phygital anchoring that distributes interaction across space, objects, and surfaces, and user-governed presence controls. Together they argue that low-friction mediation depends not on more information but on better-timed, better-placed cues aligned with hands-occupied work. Methodologically, the thesis shows how XR can function as an intermediate living testbed for domestic design research. KitchXR, developed and studied for this purpose, allows designers to prototype and evaluate interaction under realistic spatial and attentional constraints, surfacing failure modes and coordination costs earlier than screen-based methods typically permit; the boundaries and costs of this methodological role are examined reflectively in section 5.2. Finally, the thesis contributes the artifacts themselves: KitchXR as a maintained design and evaluation instrument, and the speculative probes Portal, MagMemo, and Ambient Cooking as exemplars that operationalize its programmatic concerns.

Taken together, these contributions can be stated as a single position: XR research for the kitchen should center phygital artifacts that serve as interfaces for both communication and cooking, flexible enough to adapt to workzone and task content across one or more zones rather than assuming a single, uniform site of use. I believe this repositions the construction of mediated spatial interaction tools for the kitchen from a computational problem to a design problem.

6.5 Limitations

This thesis is subject to several limitations that should be considered when interpreting its findings. First, the empirical studies were conducted with a limited number of participants and contexts, primarily within exploratory and design-oriented settings. While this aligns with the programmatic and qualitative nature of the work, it constrains the generalizability of the findings across diverse household types, cultural practices, and long-term domestic routines.

Second, the XR artifacts and prototypes explored in the studies were intentionally speculative and of intermediate fidelity. They were designed to surface interaction frictions and design implications rather than to function as deployable systems, which limits conclusions about long-term adoption, robustness, and everyday use. Third, the studies relied on current XR hardware and sensing capabilities, which impose constraints on comfort, field of view, input modalities, and sensing precision. These technical limitations shape both user experience and designer reasoning, and future advances may alter some of the observed dynamics. Finally, although the thesis emphasizes situated and embodied interaction, most evaluations were necessarily bounded in time and did not capture longitudinal patterns of appropriation, habituation, or fatigue that unfold in real domestic life.

This thesis carries epistemological and ecological limitations that should be acknowledged directly. Of the four studies, only Kitchen Talks engaged domestic kitchen users, through participant observation and ideation workshops; the interaction frictions that structure the thesis's findings are anchored in this inhabitant-grounded inquiry. The subsequent studies elaborated and examined these frictions with design practitioners rather than with household members. This scoping reflects the generative orientation of the research rather than an oversight: the artifacts developed here are speculative instruments for charting a possible design space, not deployable systems whose social outcomes could be meaningfully assessed in domestic use, and the practical constraints of virtual prototypes (reliant on head-mounted displays, simulated kitchens, and session-based engagement) precluded the kind of sustained, in-home deployment that an inhabitant evaluation would require. Within this scope, the thesis deliberately prioritized constructing artifacts and methodological infrastructure over extending empirical coverage. The consequence for interpretation is twofold. First, the findings of the designer-focused studies should be read as intermediate-level design knowledge about the articulability of this design space, not as evidence of how such systems would be appropriated, negotiated, or resisted in lived family kitchens. Second, as the history of ubiquitous computing

has repeatedly shown [158], design practitioners’ envisioned futures cannot be assumed to anticipate real domestic adoption. Evaluating the developed artifacts with inhabitants in everyday kitchen settings therefore remains a necessary next step, taken up in Section 6.6.

6.6 Future Research

The findings of this thesis open several directions for future research. First, longitudinal studies conducted in real homes could examine how XR-mediated kitchen interaction evolves over time, including how practices stabilize, adapt, or fade with repeated use. Second, privacy, identity, and boundary management merit deeper investigation, particularly as mediated presence becomes more ambient and persistent in domestic settings. Third, multi-party and asymmetric scenarios, such as interactions involving more than two participants, or differing levels of engagement and authority offer a rich space for extending the findings beyond dyadic collaboration.

Fourth, these inquiries would benefit from a stronger anthropological grounding of how households appropriate mediated kitchen technology. This thesis foregrounds presence and care but does not theorize the domestic power relations, authority structures, and gendered divisions of labor through which families would actually negotiate such systems. Future work should engage domestication theory and the moral economy of the household [159, 160], which treat adoption not as a designer’s forecast but as an ongoing process of appropriation, negotiation, and conversion shaped by domestic power relations and the gendered division of labor [161]. As the history of ubiquitous computing makes clear [158], envisioned futures are poor predictors of lived domestic practice; situating the developed artifacts within in-the-wild, longitudinal deployments in real family kitchens that analyzed through this anthropological frame, would test whether the frictions identified here persist, dissolve, or recombine once mediated presence is appropriated by households on their own terms.

A Vision for the Future Domestic Kitchen Revisiting the contributions and reflections of this thesis, I can articulate a broader vision for the future of domestic kitchens. Remote communication today remains dominated by mediated 2D video interfaces; the trajectory charted in this work points instead toward embodied, spatial, and multi-modal forms of interpersonal interaction. If such interactions become part of everyday domestic life, homes will require dedicated spaces that can host them while safeguarding safety and privacy. Building on the position

developed in Section 6.4.3, I argue that the domestic kitchen is a strong candidate for this role as a spatial connection hub.

The kitchen already operates as a social hub of the home, connecting inhabitants through cooking, dining, message-leaving, and the sharing of household tasks. Integrating mediated spatial interaction into this environment would not impose a new function upon it; rather, it would extend an established social role, while the presence of such interaction resources could, in turn, reinforce the kitchen's social value within the home.

The holistic vision that emerges can be summarized as follows: a smart kitchen in which appliances are centrally orchestrated through phygital artifacts and spatial interactions, and which supports togetherness and embodied presence among co-located and remote users through interaction, collaboration, and cooperation. Current smart-kitchen and XR technologies already bring aspects of this vision within reach, and emerging developments in spatial computing may progressively close the remaining distance, though, as argued in Section 6.5, whether and how households appropriate such futures remains an empirical question rather than a foregone conclusion.

6.7 Final Remarks

This thesis has approached the domestic kitchen not as a problem to be optimized, but as a lived space where interaction, care, and coordination intertwine. By treating frictions as productive sites of inquiry and XR as both an interaction medium and a design instrument, the work has argued for a more spatially grounded and socially aligned understanding of domestic technology. Rather than offering a single solution, it maps a design space, one that acknowledges limits, embraces complexity, and invites further exploration. In doing so, the thesis contributes to ongoing conversations in HCI and interaction design about how emerging technologies can support everyday life without flattening its social texture.

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Table .1: Prior kitchen HCI studies positioned along the task/social axis.

Author(s)	Title	Year
Task-oriented studies		
Ju	The Design of Active Workspaces	2001
Bonanni, Lee & Selker	Cooking with the Elements: Intuitive Immersive Interfaces for AR Environments	2005
Hamada et al.	Cooking Navi: Assistant for Daily Cooking in Kitchen	2005
Chia & Lee	Spatial User Interfaces: Augmenting Human Sensibilities in a Domestic Kitchen	2005
Patterson et al.	Fine-Grained Activity Recognition by Aggregating Abstract Object Usage	2005
Hashimoto et al.	Smart Kitchen: A User Centric Cooking Support System	2008
Olivier et al.	Ambient Kitchen: Designing Situated Services Using a High Fidelity Prototyping Environment	2009
Chen et al.	A Smart Kitchen for Nutrition-Aware Cooking	2010
Mennicken et al.	First-Person Cooking: A Dual-Perspective Interactive Kitchen Counter	2010
Pham, Plötz & Olivier	A Dynamic Time Warping Approach to Real-Time Activity Recognition for Food Preparation	2010
Chi et al.	Kitchen of the Future: Computing Innovations for Cooking	2012
Kosch et al.	The Digital Cooking Coach: Using Visual and Auditory In-Situ Instructions	2019
Eom et al.	TupperwareEarth: Bringing Intelligent User Assistance to the IoT Kitchen	2022
Social / ethnographic studies		
Warde	Consumption, Food and Taste	1997
Bell & Valentine	Consuming Geographies: We Are Where We Eat	1997
Miller et al.	Exploring the Role of Home Design in Fostering Family Interaction	2003
Grimes & Harper	Celebratory Technology: New Directions for Food Research in HCI	2008
Terrenghi	Computing Technologies in the Kitchen: The Living Cookbook (PhD Dissertation)	2009
Paay et al.	Cooking Together: A Digital Ethnography	2012

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Table .1 – *continued from previous page*

Author(s)	Title	Year
Wills et al.	Domestic Kitchen Practices: Findings from the Kitchen Life Study	2013
Nansen et al.	Kitchen Kinesics: Situating Gestural Interaction within the Social Contexts of Family Cooking	2014
Chai et al.	Cooking Together at a Distance: Sustaining Connectedness for Long Distance Families	2017
Bridging / critical perspectives		
Bell & Kaye	Designing Technology for Domestic Spaces: A Kitchen Manifesto	2002
Lee et al.	Smart Homes, Families, and Control	2006
Ferdous	Technology at Mealtime: Beyond the ‘Ordinary’	2015
Wozniak et al.	Inhabiting Interconnected Spaces: How Users Shape and Appropriate Their Smart Home Ecosystems	2023
Weber et al.	Bridging Distances through Virtual Cook-Along Sessions	2025